



Transportation Management (TM)

Generated on: 2026-09-01 12:17:38 GMT+0000

SAP S/4HANA | 2025 FPS01 (Feb 2026)

Public

Original content: https://help.sap.com/docs/SAP_S4HANA_ON-PREMISE/e3dc5400c1cc41d1bc0ae0e7fd9aa5a2?locale=en-US&state=PRODUCTION&version=2025.001

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Integration

You can integrate Transportation Management (TM) with the following applications:

Application	More Information
Logistics Integration	Integration of orders and deliveries: Logistics Integration Integration with settlement: Forwarding settlement Freight settlement
Advanced Shipping and Receiving	Advanced Shipping and Receiving
Extended Warehouse Management	Integration with Extended Warehouse Management
SAP Event Management	Integration with SAP Event Management
SAP Global Trade Services	Integration with SAP Global Trade Services

Advanced Shipping and Receiving

Advanced Shipping and Receiving is an integrated, end-to-end process that allows users to send, receive, and transport products. It supports mixed and multi-warehouse scenarios within one freight order. It helps you, to coordinate the process steps from arrival of trucks at the gate to the final goods receipt in the warehouse. A truck that has posted arrival at a checkpoint can carry out both loading and unloading activities at a location, or it can deliver goods at several warehouses at the same location. The process is also supported for rail freight orders.

Advanced Shipping and Receiving simplifies communication between the Transportation Management (TM), Extended Warehouse Management (EWM), Stock Room Management (STRM), Inventory Management and Physical Inventory (MM-IM), and Logistics Execution (LE) application components. You need Materials Management (MM) and Sales and Distribution (SD) to create transportation requirements.

Communication is simplified by using a harmonized data model, by applying the EDIFACT standard to EDI messages, and by providing dedicated apps and expanding existing apps and RF transactions for shipping and receiving processes. Information moves freely between the components and allows users to gain insights into the status of the products during the process.

Storage Location

The freight order can contain both EWM-managed and IM-managed storage locations, for example at a production location, and the two storage locations can share a loading point.

In Advanced Shipping and Receiving you can integrate TM with the following warehouse management applications in the same system:

- **Lean Warehouse Management**

You can use a very simple warehouse where inventory management takes place solely at storage location level. You need to create a place holder warehouse that ensures the document flow in TM.

- **Stock Room Management**

You can use Stock Room Management for small warehouse operations and for warehouse operations with low complexity.

- **Extended Warehouse Management**

You can use EWM for a flexible, automated support for processing various goods movements and for managing stocks in your warehouse complex.

Integration Based on Freight Order

The Advanced Shipping and Receiving process is **based on the freight order** and does not use the EWM transportation unit (TU) or the EWM vehicle.

Both integration processes, integration based on the freight order or integration based on the EWM transportation unit, can run in parallel. For warehouse-driven outbound processes, you start in EWM.

For information about **restrictions**, see SAP Note [3289977](#) .

For information about the integration of TM with EWM **based on the EWM TU** and about possible system combinations, see [Integration with Extended Warehouse Management](#) and [Integration Based on EWM Transportation Units](#). This integration is based on two objects: the TU on EWM-side and the freight order on TM-side.

Consignment Orders

In the Advanced Shipping and Receiving process you can use consignment orders to group transportation requirements based on certain criteria, such as the same source and destination location, the same delivery date/time, and the same business partner. You can create consignment orders using inbound messages as well as manually.

For more information, see [Consignment Order](#).

Detailed Organization of Loading and Unloading Activities

It is possible to deliver goods to or pick up goods from several warehouses at the same location.

If you have a large warehouse with multiple loading or unloading points and several doors, and you use more than one application component for warehousing, you can organize your loading and unloading activities in detail, and decide which door, loading or unloading point can be used by which application component. A loading point or unloading point is defined in master data as a location and represents a group of doors where goods are delivered or picked up. This allows greater control over where trucks load or unload cargo.

For more information, see [Delivering Goods to Several Warehouses Within a Location](#) and [Assignment of STRM Doors to EWM Doors or Loading Points](#).

Integration with decentralized Extended Warehouse Management

Decentralized EWM and TM enables you to conduct transportation logistics solely on the basis of an outbound delivery. Decentralized EWM and TM is embedded in SAP S/4HANA. The arrangement requires integration between TM and EWM, and EWM integration with an ERP system in turn. You have the options of carrying out the outbound delivery process as either transport-driven or warehouse-driven.

Customizing is required. For more information, see [Advanced SR for Decentralized EWM and TM](#).

More Information

- For a setup guide, see SAP note [3225241](#) .

- For best practice scenarios, see SAP note [3512117](#) .
- For a detailed **end-to-end example process**, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#).
- For an overview about available **features**, see [Overview of Features in Advanced Shipping and Receiving](#).
- For information about **prerequisites**, see [Prerequisites for Advanced Shipping and Receiving](#).
- For information about logistics integration (such as, integration of orders, deliveries or scheduling agreements), see [Logistics Integration](#).
- For information about **EWM-specific topics** in the **Advanced Shipping and Receiving** process, see [Advanced Shipping and Receiving for Embedded EWM](#).
- For information about decentralized EWM-specific topics in the **Advanced Shipping and Receiving** process, see [Advanced Shipping and Receiving for Decentralized EWM and TM](#).
- For information about **LE-specific topics** in the Advanced Shipping and Receiving process, see [Integration with Advanced Shipping and Receiving](#).
- For information about **master data replication**, see [Master Data Replication](#).
- For more information about the freight booking process with Advanced Shipping and Receiving, see [Ocean Freight Bookings for Shippers \(Outbound\)](#) and [Air Freight Bookings for Shippers \(Outbound\)](#).

Advanced SR for Decentralized EWM and TM

This deployment option is an outbound delivery-based arrangement in an Advanced Shipping and Receiving setup. The process is based on outbound deliveries that stem from sales orders.

Decentralized EWM and TM enables you to conduct transportation logistics solely on the basis of an outbound delivery. The arrangement requires integration with Extended Warehouse Management (EWM), and EWM integration with an ERP system in turn. You have the options of carrying out the outbound delivery process as either transport-driven or warehouse-driven.

Additionally, to manage foreign-trade activities, ensure ongoing trade compliance, and optimize the cross-border supply chain, you can integrate your decentralized EWM and TM system with SAP GTS.

Customizing is also required. For more information about Customizing, see [Prerequisites for Advanced Shipping and Receiving](#).

! Restriction

Decentralized EWM and TM enables you to execute the outbound process. Inbound features in Advanced Shipping and Receiving are not supported by decentralized EWM and TM.

In addition, integration of Dangerous Goods is not supported.

Supported Systems

- ERP System:
 - SAP ERP 6.0 (EhP7 or higher)
 - SAP S/4HANA

More Information

For more information about the end-to-end outbound process, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving \(Decentralized\)](#).

For more information about integration with SAP GTS, see [Integration with SAP Global Trade Services in Advanced Shipping and Receiving](#) and [SAP Global Trade Services](#).

Related Information

[SAP GTS Integration for the Outbound Process](#)

[Global Trade](#)

[Logistics Integration Profile for Decentralized EWM](#)

[SAP GTS Integration in Decentralized EWM and TM](#)

Prerequisites for Advanced Shipping and Receiving

To work with Advanced Shipping and Receiving you must configure some settings in the system.

Master Data

You have made the following settings in the location master data on the SAP Easy Access screen under **►►Logistics►►Transportation Management►►Master Data►►Transportation Network►►Location►►Location►►**:

- You have activated Advanced Shipping and Receiving.

You do this, by indicating in the master data for your source or destination location (location type 1003, shipping point), that the location is relevant for Advanced Shipping and Receiving. This setting is mandatory.

- You have defined a location with location type 1200 (loading point). The setting is mandatory.

Assignment of Doors

Assign Stock Room Management (STRM) doors or Lean Warehouse Management (Lean WM) doors to Extended Warehouse Management (EWM) doors or loading points. This setting is mandatory for all scenarios using shared doors and multiple warehousing components. The setting is not required for a scenario that only uses EWM.

To assign doors in scenarios that only use EWM, open the EWM app **Assign Storage Bin to Door** from the launchpad.

For more information about prerequisites in EWM, see [Prerequisites for Advanced Shipping and Receiving in EWM](#).

Customizing Activities

For more information about Customizing settings for Advanced Shipping and Receiving, see Customizing for **Transportation Management** under **►►Integration►►**.

For more information about Customizing settings for decentralized EWM and Transportation Management (TM), see Customizing for **Transportation Management** under **►►Integration►►Logistics Integration►►Decentralized EWM-TM Integration►►**.

For more information about Business Add-Ins for decentralized EWM and TM, see Customizing for **Transportation Management** under **►►Business Add-Ins (BADIs) for Transportation Management►►Integration►►Advanced Shipping and Receiving►►Decentralized EWM-TM Integration►►**.

More Information

- For a setup guide, see SAP note [3225241](#) .
- For information about LE-specific topics in the Advanced Shipping and Receiving process, see [Integration with Advanced Shipping and Receiving](#).
- For information about master data replication, see [Master Data Replication](#).

End-to-End Process (E2E) for Advanced Shipping and Receiving

This **example** describes an end-to-end process for Advanced Shipping and Receiving (Advanced SR) in which both the customer and the supplier use Transportation Management (TM). In case the systems are not integrated, you can use the inbound or outbound process separately.

Prerequisites

For more information, see [Prerequisites for Advanced Shipping and Receiving](#).

Customer Side: Inbound Process

1. Creating a purchasing document

Create a purchasing document in Materials Management (MM), for example, a purchase order or scheduling agreement (SA), to define which products have to be purchased.

You can use the [Create Scheduling Agreement](#) app from the launchpad, for this example process.

- Enter necessary data, such as agreement type, purchasing organization and purchasing group (all defined for integration with TM). Enter a supplier, define validity dates and insert materials that must be procured. In addition, it is important that you enter a confirmation control key.
- In the scheduling agreement item you can add a **TM loading or unloading point** as additional data. To define where exactly in the warehouse the goods shall be unloaded.

2. Creating schedule lines in the scheduling agreement

Create schedule lines in the scheduling agreement (usually done by a report).

To create them manually, open the [Manage SA Delivery Schedule](#) app from the launchpad.

For every schedule line, the system creates one or more **freight units** in TM.

3. Creating a scheduling agreement release

Create a scheduling agreement release (usually done by a report).

To create them manually, open the [Create Scheduling Agreement Releases](#) app from the launchpad.

The system creates intermediate documents (IDocs) and sends a **call off** (a release against the outline agreement) to the supplier's system.

Supplier Side: Outbound Process

With the call off, the supplier receives information about the products ordered. In the supplier's system you can display the ordered products in the corresponding scheduling agreement in Sales and Distribution (SD). In the forecast delivery schedule, the call off appears as a schedule line.

1. Creating outbound deliveries

Create manually or automatically an **outbound delivery** in Logistics Execution (LE) and Extended Warehouse Management (EWM).

You can use the [Create Outbound Deliveries](#) app from the launchpad.

a. Defining a warehouse

In the delivery, choose a **storage location** to define the kind of warehouse you want to use, for example, EWM-managed.

b. Displaying TM information in the delivery

You can use the [Manage Outbound Deliveries](#) app from the launchpad.

In the delivery, choose the **TM Status** tab to display the assigned freight units.

2. Performing warehouse activities

Picking (and possibly packing) is carried out by the warehouse workers. You create handling units (HU) or cross-delivery HUs.

You can use the [Load or Unload Freight Orders](#) app from the launchpad.

For more information about statuses, see [Statuses in the Outbound Process](#) and [Statuses in Advanced Shipping and Receiving](#).

For more information about cross-delivery HUs, see [Cross-Delivery Handling Units](#)

All warehouse-relevant tasks are now completed for the freight units. You can start with the TM process: You can plan by assigning the freight units to a consignment order, to a freight order and then start with the execution.

1. Creating a road freight order

Create a road freight order. You can use the [Create Road Freight Order](#) app or [Transportation Cockpit](#) app from the launchpad.

2. Creating consignment orders

Create **consignment orders** or have the system create them automatically. For more information, see [Consignment Order](#) and [Automatic Consignment Building](#).

You can use the [Create Consignment Order](#) app from the launchpad.

3. Check-In process at the gate

- o You can manage the check-in of the truck in the freight order.

To open the [Freight Order](#) app from the launchpad.

- o After the check-in, you can perform warehouse activities. Again, you can use the [Load or Unload Freight Orders](#) app or the [Freight Order](#) app from the launchpad. You can assign doors to the truck (assign loading point and warehouse door), load cargo, set the execution status accordingly, and post goods movement.

For more information about statuses, see [Statuses in the Outbound Process](#) and [Statuses in Advanced Shipping and Receiving](#).

- o Post goods movement to update the respective warehouse management system (independent of the kind of warehouse management you use). The system displays an **overall status overview** in the road freight order, under the [Advanced SR](#) tab page.

4. System communication using advanced shipping notifications (ASN)

The system sends an ASN from the supplier's to the customer's system. This ASN is a web service based on the consignment order to inform the customer about the expected cargo.

For more information, see [Creation or Update of Consignment or Freight Orders on Inbound Messages](#).

The supplier can also communicate with the carrier, so the carrier is aware of the consignment orders to be transported.

Customer Side: Inbound Process

1. Creating consignment orders using ASN

When the customer's system receives the ASN, it creates a **new consignment order** including a reference to the external order number that the supplier's system uses to identify the consignment. The relevant freight units are determined on the basis of criteria such as purchasing reference, supplier or material number, and assigned to the new consignment order. The ASN can also be used to update consignment orders.

2. Displaying unplanned freight units

You can display all order-based and unplanned freight units related to the scheduling agreement you created at the beginning of the process.

You can use the [Freight Units Worklist](#) app from the launchpad, and enter the scheduling agreement number under [Original Order](#).

Once the related ASN message arrives and the systems has assigned the freight units to the consignment order, the freight units are no longer displayed in your list.

3. Creating inbound deliveries

The system can automatically create an **inbound delivery** in LE and EWM. The deliveries are blocked.

To view the inbound deliveries, open the [Display Inbound Delivery](#) app or the [Maintain Inbound Delivery](#) app on the launchpad.

4. Displaying consignment orders

You can open the **document flow** of the consignment order to display further information, such as related purchasing document, assigned freight units and automatically created **inbound deliveries**. The **item structure** shows the package information that has been transferred.

To display information, open the [Consignment Orders](#) app from the launchpad.

5. Creating road freight order

Create a road freight order based on consignment order stage.

You can use the [Create Road Freight Order](#) or [Transportation Cockpit](#) app from the launchpad.

Alternatively, this can be an automatic step when the freight order is sent as load message from the carrier. For more information, see [Creation and Editing of a Consignment Order](#).

6. Set execution status to [Set to Ready for Transportation Execution](#).

7. Check-In process of trucks at the gate

- You can manage the check-in of the truck in the freight order.

To do this, open the [Freight Order](#) app from the launchpad.

- After the check-in, you can begin with warehouse activities.

You can use the [Load or Unload Freight Orders](#) app or [Freight Order](#) app from the launchpad.

- o When a truck arrives at the gate you can compare the transportation documents of the truck with the freight order in system. (Earlier, you might have received planning information via EDI message from the carrier.)

The packages in consignment orders and freight orders that are created using EDI messages are automatically **fixed**, to avoid accidental changes. The system automatically sets the load plan status of the packages to "Finalized". This means that you can only change the packages if you reset the status.

If the load of the truck differs from the freight order data in the system, you can directly change discrepancies in the business document. As an example, changing the data is necessary, if the truck contains more, less or other cargo than expected.

The system can automatically update the referenced predecessor documents of the freight order and keep them up-to-date. For more information, see [Synchronization of Predecessor Documents](#).

- o You can assign doors, assign unloading point for a location (with warehouse number and warehouse door) and guide the truck to the right place.
- o You can set different statuses for the freight order. Once the gate check-In process is finished, you can set the warehouse processing status to **Ready for Warehouse Processing**. The receiving process is now completed, so warehouse activities, such as unloading or posting goods movement can start. The EWM delivery is unblocked. The document flow now shows warehouse requests.

For more information about statuses, see [Statuses in the Inbound Process](#) and [Statuses in Advanced Shipping and Receiving](#).

Displaying Information

- o You can display the complete document flow in the inbound delivery.
- o The system displays a **status overview** in the road freight order, under the **Advanced SR** tab page. For more information, see [Delivering Goods to Several Warehouses Within a Location](#).

More Information

For more information, see the SAP Best Practices for solution scenario at https://me.sap.com/processnavigator/SolS/EARL_SolS-055 . Navigate to **Supply Chain**.

For more information about of EWM-specific topics in the Advanced Shipping and Receiving process, see [Advanced Shipping and Receiving for Embedded EWM](#).

For information about LE-specific topics in the Advanced Shipping and Receiving process, see [Integration with Advanced Shipping and Receiving](#).

For more information about processes in the integration with SAP Global Trade Services, see [Integration with SAP Global Trade Services](#).

Related Information

[Correction of Errors in Business Documents](#)

[Example Checks to Find Errors in Business Documents](#)

Transport-Driven Execution

A transport-driven process prioritizes transportation planning while warehouse activities are blocked until manually released.

A transport-driven outbound process takes place in the context of internal logistics integration between Transportation Management (TM) and Delivery Management as well as TM and Extended Warehouse Management (EWM), Lean Warehouse Management, and Stock Room Management. A transport-driven execution prioritizes transportation planning. After delivery creation, the delivery is blocked for warehouse execution until transportation preparations have begun and a block on warehouse activities has been removed. When you release the delivery for warehouse processing, it is then unblocked, and warehouse execution can start.

You must set the warehouse processing status to **Ready for Warehouse Processing** manually. Consequently, the warehouse receives the information needed to plan and execute its warehouse activities accordingly.

Prerequisites

You have defined the following criteria for the determination of outbound deliveries that are relevant for transport-driven processes:

- Warehouse Number
- Shipping Point
- Delivery Type

When a delivery meets the criteria set in the Customizing activity, the transport-driven process is automatically enabled.

For more information, see Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration > Determine Relevance for Transport-Driven Process**.

To afford you more flexibility, we provide a Business Add-In (BAI). With this BAI, you can establish your own criteria for the determination of transport-driven deliveries when saving a Delivery Management delivery.

For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BAIs) for Transportation Management > Integration > Advanced Shipping and Receiving > BAI: Change of Data Related to Transport-Driven Process**.

Executing a Transportation-Driven Process

1. Creating an Outbound Delivery

Start with an outbound delivery. Based on Customizing, the system determines whether a delivery is relevant for the transport-driven scenario.

At this stage, the **Warehouse Execution Status** of the delivery is set to **Blocked**.

2. Setting the Warehouse Processing Status to Ready for Warehouse Processing

Create a freight order in the **Create Road Freight Order** app in TM.

Up to now, the delivery status is set to **Blocked** so information about transportation planning is not transferred to the delivery.

You can change the status to **Ready for Warehouse Processing** in the freight order (FO) using the **Edit Road Freight Order** app. Alternatively, you may choose to trigger warehouse activities directly from the freight unit (FU) by setting the warehouse processing status in the FU to **Ready for Warehouse Processing**.

In a transport-driven scenario, the default settings require you to execute the trigger manually so that the blocking of warehouse processing is determined by the executor.

3. Enabling and Halting Warehouse Processing

It is possible to operate both of the Advanced Shipping and Receiving (Advanced SR) scenarios: the default Advanced SR warehouse-driven scenario and the Advanced SR transport-driven scenario. The transport-driven scenario is an extension of the warehouse-driven processes. Hence, once a block is removed, warehouse activities can be carried out as usual.

If, for any given reason, warehouse execution must be stopped, you can reverse the [Ready for Warehouse Processing](#) status as long as execution has not yet begun.

4. Updating Deliveries

When you change the status of an order to [Ready for Warehouse Processing](#), carrier, delivery date, and loading point are transferred to the delivery.

Compliance Checks

In transport-driven cases where compliance checks need to be conducted, processing is blocked until a successful compliance check is carried out. You can set the status to [Ready for Warehouse Processing](#) manually once compliance is met. Should a second compliance check be required, the [Ready for Warehouse Processing](#) status is automatically reverted to **Not Started**. Similarly, when compliance is later met, the status is automatically updated, and that information is forwarded to Delivery Management and retrieved by EWM.

Once warehouse activities begin, it is no longer possible to revert the status to [Not Started](#).

You can find the compliance status in the [Edit Freight Unit](#) app.

Customer Returns: Inbound Process

This example describes an inbound process for handling customer returns in Advanced Shipping and Receiving (Advanced SR).

Prerequisites

For more information, see [Prerequisites for Advanced Shipping and Receiving](#).

Supplier Side: Inbound Process

1. Creating a customer returns order

Create a customer returns order in Sales (SD), to define which products are being returned.

2. Creating freight units

For every schedule line, the system creates one or more **freight units** in TM.

Customer Side: Outbound Process

The return is processed at the customer.

Supplier Side: Inbound Process

1. Creating consignment orders using an advanced shipping notification (ASN)

When the supplier's system receives the ASN, it creates a **new consignment order**. The relevant freight units are determined on the basis of criteria such as returns reference (customer returns order number), supplier or material number, and assigned to the new consignment order. The ASN can also be used to update consignment orders.

2. Displaying unplanned freight units

You can display all order-based and unplanned freight units related to the customer returns order you created at the beginning of the process.

You can use the [Freight Units Worklist](#) app from the launchpad, and enter the customer returns order number under [Original Order](#).

Once the related ASN arrives and the system has assigned the freight units to the consignment order, the freight units are no longer displayed in your list.

3. Creating returns deliveries

The system can automatically create a **delivery for customer returns** in Logistics Execution (LE) and Extended Warehouse Management (EWM). The deliveries are blocked for warehouse execution in a first step.

To view the customer returns deliveries, on the launchpad open the [Change Outbound Delivery](#) app in LE or the [Maintain Inbound Deliveries](#) app in EWM.

4. For all subsequent steps, please refer to [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#).

End-to-End Process (E2E) for Advanced Shipping and Receiving (Decentralized)

Decentralized EWM and TM is an outbound delivery-based arrangement embedded in the same system. The process must be executed based on a sales order.

Decentralized EWM and TM conducts transportation planning and execution based exclusively on outbound deliveries. Outbound deliveries reach Transportation Management (TM) via Extended Warehouse Management (EWM), and you have the ability to conduct the process as either warehouse-driven or transportation-driven.

Additionally, you can carry out compliance checks and export declarations by integrating your system with SAP Global Trade Services. For more information, see [SAP GTS Integration in Decentralized EWM and TM](#).

Communication between TM and an ERP system is routed via EWM.

Prerequisites

For more information, see [Prerequisites for Advanced Shipping and Receiving](#).

Decentralized Sample Process

Decentralized EWM and TM Outbound Process and Integration

The placement of an order leads to transportation requirements. As a result, the shipper creates an outbound delivery.

1. Creating an Outbound Delivery

In Logistics Execution (LE), create an outbound delivery from a sales order.

To create an outbound delivery, open the [Create Outbound Delivery](#) app from the launchpad.

2. Creating an Outbound Delivery Order

The ERP system communicates the outbound delivery to the decentralized system automatically, where EWM creates an outbound delivery order.

EWM replicates the outbound delivery order to TM. And TM creates freight units (FU) based on the outbound delivery order.

3. Planning and Processing

With FUs, TM can then propose a transportation plan.

From this point onward, you can conduct activities that either prioritize warehouse execution or transportation planning.

Transport-Driven Process

In a transport-driven process, transportation planning takes precedence.

1. Performing Transportation Planning

EWM creates an outbound delivery order from an outbound delivery from Delivery Management. The outbound delivery order is blocked for processing until transportation planning is complete.

At this stage, create a road FO and insert the FUs in the order.

You assign FUs to a freight order (FO) by entering the outbound delivery order, delivery, or FU number.

To create an FO, open the [Create Road Freight Order](#) app or [Transportation Cockpit](#) app from the launchpad.

2. Releasing the Freight Order

Next, you change the status of the FO to [Set to Ready for Warehouse Processing](#).

With the change of status in TM from [In Process](#) to [Ready for Warehouse Processing](#), the outbound delivery order is released, and the execution of warehouse activities can proceed.

For more information about the transport-driven process, see [Transport-Driven Execution](#).

3. Performing Warehouse Activities

TM automatically communicates the change in status to EWM.

Warehouse specialist can proceed with warehouse activities as described below.

Warehouse-Driven Process

In a warehouse-driven process, warehouse activities take precedence.

1. Performing Warehouse Activities

Upon creation of an outbound delivery order, the warehouse can proceed with its activities. Warehouse workers carry out activities such as picking and packing.

With the completion of warehouse activities, EWM communicates the delivery status to TM.

For more information about statuses, see [Statuses in the Outbound Process](#) and [Statuses in Advanced Shipping and Receiving](#).

2. Performing Transportation Logistics

Once warehouse activities are complete, you can conduct transportation planning and execution in TM.

Once transportation planning is complete, you can set the status to [Ready for Shipping](#) in the EWM [Outbound Delivery Order](#).

Execution

Finally, transportation of goods can take place.

1. Reporting Truck Arrival and Departure

In your FU, you will find the warehouse status is now set to **Completed**. Also, information regarding the package is now available. On the **Item** tab, you will find information such as the package ID and packaging.

Navigate to the **Execution Status** tab to change the status to **Ready for Transportation Execution**.

With the arrival of the truck, you can set the status in TM to **Set to Arrived**, and after, **Checked In** on the **Overview** tab.

Loading activities are recorded in EWM. At the gate, the warehouse operators load the goods onto the truck.

To set delivery statuses, open the **Load or Unload Freight Orders** app from the launchpad.

To manage the check-in and -out of the truck, open the **Edit Freight Order** app from the launchpad.

Change the **Goods Movement** status to **Goods Issued** to report the truck's departure.

More Information

For more information, see the SAP Best Practices solution scenario at https://me.sap.com/processnavigator/SoIS/EARL_SoIS-055. Navigate to **Supply Chain**.

Related Information

[Prerequisites for Advanced Shipping and Receiving](#)

[Overview of Features in Advanced Shipping and Receiving](#)

[Advanced Shipping and Receiving for Decentralized EWM and TM](#)

Overview of Features in Advanced Shipping and Receiving

The table gives an overview of features in the Advanced Shipping and Receiving process.

→ Remember

In a decentralized EWM and TM deployment, only outbound processes and functions are enabled.

Feature	Good to Know	More Information
Delivering goods to several warehouses within a location	You can give detailed information to a delivery truck driver about warehouses within the company premises. You need this information when a truck driver has to deliver goods to or pick up goods from several warehouses at the same location.	Delivering Goods to Several Warehouses Within a Location
Detailed organization of large warehouses	You can organize your loading and unloading activities in detail and decide which door, or which loading or unloading point is used by which warehousing component. This allows greater control over where trucks load or unload cargo.	Assignment of STRM Doors to EWM Doors or Loading Points
Late changes to capacity documents	You can make changes to a capacity document, for example, a road freight order, even after the execution at the stop in the	Late Changes to Capacity Documents

Feature	Good to Know	More Information
	capacity document has started or been completed.	
Status management	Several statuses are used in Advanced Shipping and Receiving for the inbound and outbound processes.	Statuses in Advanced Shipping and Receiving Statuses in the Inbound Process Statuses in the Outbound Process
Consignment order	A consignment order represents a business document that you can use to group several transportation requirements based on certain criteria, such as the same source and destination location, the same delivery date/time, and the same business partner. You can create consignment orders automatically using inbound messages as well as manually.	Consignment Order Creation and Editing of a Consignment Order Automatic Consignment Building
Creating or updating of consignment orders using ASN	Your business partner can send you a consignment order or freight order as an advanced shipping notification (ASN) using EDI communication. The related freight units are determined and assigned to the new consignment order.	Creation or Update of Consignment or Freight Orders on Inbound Messages Creation and Editing of a Consignment Order
Freight unit determination	Relevant freight units are determined on the basis of certain criteria (such as supplier or material number).	Creation or Update of Consignment or Freight Orders on Inbound Messages
Using web services for external planning	You can use EDI messages to communicate to place transportation order request, receive confirmation, updates, and cancelations. Output management is based on the Post-Processing Framework (PPF) which is integrated in the Business Objects Processing Framework (BOPF).	External Planning
Creating freight orders using EDI messages	Your business partner can send you a freight order (bordero, bill of lading) with assigned consignment orders using EDI communication. The related consignment orders are determined and assigned to the new freight order.	Creation or Update of Consignment or Freight Orders on Inbound Messages
Freight unit split and mass data changes	When you receive an external delivery note number from a supplier, you can change it in business documents, such as freight orders or consignment orders, with or without an FU split.	Freight Unit Split and Mass Data Changes
Correction of errors in business documents	If the inbound consignment order contains errors, it is usually still saved as a document in TM. The errors are displayed on the Document Errors tab page and can be corrected here or in the corresponding worklist.	Correction of Errors in Business Documents

Feature	Good to Know	More Information
	With the report /SCMTMS/TOR_RETRY_ERRORS you can automatically resolve document errors in freight orders without opening and retrying each freight order manually. You can resolve reference errors to an unknown consignment order that is now available in the system.	
Synchronization of predecessor documents	For the inbound process you can enable the synchronization of predecessor documents referenced in a consignment order or freight order. You can, for example, change the data of a freight order during the check-in process of a truck at the gate and update the predecessor documents with the changes.	Synchronization of Predecessor Documents
Handling units (EWM), package ID (TM), and external package ID (TM)	In the inbound part of the Advanced Shipping and Receiving process, the package information received from the vendor (either via ASN or entered manually in the system, for example, during the check-in process) is used to create handling units. In the outbound part, the handling unit information received from the warehouse is used to create package information. Throughout the process the integration ensures that all information is synchronized between TM and the warehouse. A vendor provides a package ID and confirms that it is a standardized number, such as a global transport label (GTL) or a serial shipping container code (SSCC). The system checks the number to ensure that it is unique across systems and displays it as a package ID in a business document, such as a road freight order or a consignment order. If the vendor says that the package ID is not a standardized number, then the system displays it as an external package ID. The system also creates a unique, standardized number for the external package ID and displays it in the Package ID field. This is the number that is used for handling units in the warehouse.	
Auxiliary packaging material	The system includes auxiliary packaging materials, that have been defined in TM business documents, in the corresponding handling units when creating LE and EWM deliveries.	Use of Auxiliary Packaging Materials
Serial numbers and batch management	You can store and display information about materials, such as batch information and serial numbers, in business documents.	Use of Batches and Serial Numbers

Feature	Good to Know	More Information
	This information is sent by the supplier. You can also transfer this information to the inbound deliveries or receive them from the outbound deliveries. In the consignment and road freight order, you can manually enter or edit the batch and serial number information.	
Deliveries	You can split deliveries from TM across a number warehouse management applications. Also, you can delete deliveries created in LE from TM.	Delivery Split Delivery Deletion
Displaying detailed warehouse information (on the user interface)	In the road freight order, under the Advanced SR tab page, you can see the planned and actual warehouse details where a truck has loaded or unloaded goods. This tab page is for the product in the road freight order.	Delivering Goods to Several Warehouses Within a Location
Integration with SAP Global Trade Services (SAP GTS)	Integration between TM and SAP GTS enable you to conduct customs processes from TM.	Integration with SAP Global Trade Services SAP GTS Integration for the Outbound Process SAP GTS Integration for the Inbound Process Statuses in the Integration with SAP GTS (Inbound Process) Statuses in the Integration with SAP GTS (Outbound Process)
Performing warehouse activities in EWM app (on the user interface)	To perform warehouse activities, you can use the Load or Unload Freight Orders app.	Load or Unload Freight Orders
Subcontracting a purchase order	Integration between MM and TM enable you to subcontract purchase orders.	Subcontracting of Purchase Order
Air freight bookings for shippers	You can model an air freight booking process for shippers with message exchange between the shipper and the carrier in which consignment orders play an essential part.	Air Freight Bookings for Shippers (Outbound)
Ocean freight bookings for shippers	You can model an ocean freight process with container units and message exchange between the shipper and the carrier.	Ocean Freight Bookings for Shippers (Outbound)
Cross-delivery handling units	The integration of TM with EWM based on outbound deliveries supports cross-delivery handling units (HU) in the outbound process.	Cross-Delivery Handling Units
Ad hoc loading	The assignment of FU to freight orders (FO) on the warehouse floor after the truck has docked and checked in.	Ad Hoc Loading

Delivering Goods to Several Warehouses Within a Location

In the Advanced Shipping and Receiving process in Transportation Management (TM), you can handle mixed and multi-warehouse scenarios within one road freight order. This means that you can give detailed information to a delivery truck driver about warehouses within the company premises, which is necessary when a truck driver has to deliver goods to or pick up goods from several warehouses at the same location.

At the gate, the truck driver gets the information about where to deliver the goods, either from predecessor documents or on the spot.

Features

To provide more information about the warehouse to facilitate transportation execution in the Advanced Shipping and Receiving process, you can use the following features:

Warehouse Loading Stops and Warehouse Unloading Stops

You or the system can define the exact warehouse number, loading point or unloading point, and warehouse door that the truck needs to drive to. This information is contained in a more detailed loading or unloading stop called the warehouse loading stop (for the source location) or warehouse unloading stop (for the destination location). For intermediate stops, if the warehouse loading stop and the warehouse unloading stop are identical, that is, they have the same warehouse number, loading point or unloading point, and warehouse door, you can define a [Warehouse Unloading/Loading Stop](#). You can see the warehouse loading stop or warehouse unloading stop on the [Overview](#) tab page. As a prerequisite, you need to configure this in the Customizing for the [Overview](#) hierarchy type. For more information, see Customizing for [Transportation Management](#) under [Freight Order Management](#) > [Define Hierarchical Views for Business Documents](#).

The warehouse loading stops and warehouse unloading stops can be added to business documents, such as road freight orders, consignment orders, freight units, and transportation units. All items of a road freight order must be assigned to a warehouse loading stop or warehouse unloading stop and its assigned loading point or unloading point and warehouse door. A loading point or unloading point is defined in master data as a location and represents a group of doors where goods are delivered or picked up. This allows greater control over where trucks load or unload cargo.

You can manually assign items to different warehouse loading stops or warehouse unloading stops if the warehouse number of the item matches the warehouse number of the warehouse loading or unloading stop.

You or the system can create warehouse loading stops or warehouse unloading stops at stop level in the following ways:

- The system can derive the stops in the road freight order based on the information provided in the predecessor documents, such as a sales order or a scheduling agreement.
- You can manually create or change the stops in the capacity document, for example, a road freight order, in the [Loading Details](#) or the [Unloading Details](#) tab page in the [Overview](#) section. You might need to do so if, for example, the planned loading point and warehouse door can no longer be used because they are occupied. However, the predecessor document is not updated with this information.
- In your road schedule, you can manually create warehouse loading or warehouse unloading stops for schedule stops in road schedules. As a prerequisite, you have enabled the use of warehouse loading and warehouse unloading stops in your schedule type. This can be useful in Ad hoc loading processes. For more information, see [Ad Hoc Loading](#).

In the hierarchy, a loading stop or unloading stop may have one or more substops (warehouse loading stops or warehouse unloading stops that are defined as substops). You can assign documents such as consignment orders, freight units, and packages to the warehouse loading stops or warehouse unloading stops.

You can update the derived warehouse loading stops or warehouse unloading stops with information about the warehouse door that was actually used.

Handling Warehouses with Multiple Warehousing Components

If you have a warehouse where you use two different components to manage two different parts of the warehouse, for example, Stock Room Management (STRM) and Extended Warehouse Management (EWM), you can determine the assignment of doors and loading or unloading points for loading or unloading activities.

You can assign the STRM door to the EWM door. You can also assign STRM doors to loading points or unloading points.

For more information, see [Assignment of STRM Doors to EWM Doors or Loading Points](#).

Detailed Warehouse Information

In the road freight order, under the **Advanced SR** section, you can see the planned and actual details for warehouses where a truck has loaded or unloaded goods. This tab page is for the product in the road freight order.

When a truck arrives at the gate and is checked in, in the **Planned** section, the system displays information derived from predecessor documents about where the item is planned to be loaded or unloaded. This includes, for example, the warehouse number and the loading point or unloading point.

If the planned loading points or unloading points or doors cannot be used, you can enter new loading points or unloading points or doors in the **Overview** section. In the **Actual** section, the system displays the actual assignment to warehouse loading stops or warehouse unloading stops.

Warehouse Loading Stops and Warehouse Unloading Stops in the Transportation Cockpit

Warehouse numbers, loading points and unloading points, and warehouse doors are also displayed for road freight orders, consignment orders, freight units, and transportation units in the transportation cockpit.

If there are multiple warehouse loading stops or warehouse unloading stops beneath a loading stop or unloading stop, the value **Multiple** is displayed in the corresponding fields and a quick view lists the individual warehouse numbers, loading points or unloading points, and warehouse doors.

i Note

In the transportation cockpit, only the current warehouse loading stop or warehouse unloading stop is displayed. This means that, if you have changed the warehouse loading stop or warehouse unloading stop in TM, the original warehouse loading stop or warehouse unloading stop from the sales order, for example, is not displayed.

You can do various manual planning actions, such as:

- Inserting a warehouse loading stop or warehouse unloading stop

The new substop is then displayed as a separate line (hierarchy level) in the location hierarchy.

- Changing a warehouse loading stop or warehouse unloading stop

A dialog box is displayed in which you can change the warehouse number, the loading point or unloading point, and the warehouse door.

- Assigning a stage of a requirement document to a capacity document or reassigning a planned stage of a requirement document to a capacity document

When assigning or reassigning a stage, the system considers warehouse loading stops and warehouse unloading stops.

Package Building

If you build packages for these business documents and the items have different warehouse numbers or loading points or unloading points, then these items are packaged in different mixed packages. This is also the case if no warehouse number, loading point, or unloading point has been defined for an item.

Assignment of STRM Doors to EWM Doors or Loading Points

In the Advanced Shipping and Receiving process in Transportation Management (TM), if you have a large warehouse with multiple loading points or unloading points and multiple doors, and you use more than one application component for warehousing, such as Stock Room Management (STRM), Lean Warehouse Management (Lean WM), and Extended Warehouse Management (EWM), you can organize your loading and unloading activities in more detail by deciding which warehouse door or loading point is used by which application component.

To do so, open the [Assign STRM Doors to EWM Doors or Loading Points](#) app from the launchpad. Alternatively, on the SAP Easy Access screen choose [Logistics](#) > [Transportation Management](#) > [Master Data](#) > [Transportation Network](#) > [Location](#) > [Assign STRM Doors to EWM Doors or Loading Points](#).

You can make the following assignments using this app:

- If you use both an STRM or Lean WM warehouse and an EWM warehouse in one warehouse building, and they share a common warehouse door, you must set up the system to select the EWM warehouse door for loading or unloading activities. You can do so by assigning the STRM or Lean WM warehouse door to the EWM warehouse door. When you do this, the system uses the EWM warehouse door as the main door for loading or unloading activities and automatically replaces the STRM or Lean WM warehouse door with the assigned EWM door in the warehouse loading stop or warehouse unloading stop details. For example, one part of your warehouse uses an STRM system for bulk goods and the other uses an EWM system for liquids. They share a common door for some of their activities. Using the [Assign STRM Doors to EWM Doors or Loading Points](#) app, you assign the STRM door to the EWM door, ensuring that in the system the door used for unloading liquids is also used for bulk goods.

If you want to change the warehouse door that is assigned for loading or unloading activities, you can do so only if it is shared by the same pair of EWM and STRM or Lean WM warehouses. For example, you cannot assign a door that is only used by EWM or that is used by the same EWM warehouse but by a different STRM warehouse.

If the EWM warehouse door already has a loading point or unloading point assigned to it, the system also uses the EWM loading point or unloading point as the main one and overwrites any existing assignment.

- You can assign STRM or Lean WM warehouse doors to loading points or unloading points if they are not already assigned to one.

i Note

The following restrictions apply:

- If you use a Lean WM warehouse, you need to create a placeholder warehouse with one loading point or unloading point and one warehouse door.
- You can only assign one STRM or Lean WM warehouse door to one EWM warehouse door. The same EWM warehouse door cannot be assigned to multiple STRM or Lean WM warehouse doors.
- You can only assign one loading point or unloading point to an STRM or Lean WM warehouse door.

Late Changes to Capacity Documents

In the Advanced Shipping and Receiving process in Transportation Management (TM), you can make changes to a capacity document, for example, a road freight order, even after the execution at the stop in the capacity document has started or been completed.

i Note

The following describes a situation where you need to correct a capacity document to reflect what is actually happening, or has already happened. If you need to insert extra locations or stops in a capacity document, please see [Document Assignment When Execution Has Already Started](#).

Features

During the loading or unloading process at the warehouse, sometimes additional cargo items that are not assigned to the road freight order are loaded on the truck or unloaded at the warehouse along with the items that are already assigned. This results in a mismatch between the information in the road freight order and what actually happened at the time of loading or unloading. To correct this, you can update the road freight order by assigning the additional items to the same warehouse loading stop or warehouse unloading stop at which the other items were loaded or unloaded. For example, you can add one or more requirement documents to the warehouse unloading stop in the road freight order, even after the truck has departed from that stop. You need to do this, for example, for charge calculation or settlement with the carrier.

Assignment of a New Requirement Document

You can assign a requirement document, such as a consignment order or freight unit, to the road freight order when the truck has arrived at the gate or stop but the processing of items has not yet started. In this case, the statuses of the new items match the statuses of the other items in the road freight order.

However, if you add a requirement document after the truck has loaded or unloaded the goods at the warehouse loading stop or warehouse unloading stop, has checked out, or has departed, then the status of the items added later doesn't match the status of the stop and of the other items in the road freight order. In this case, the system creates a document error. You cannot revoke the status of the stop in the road freight order. But you can update the statuses of the items in the new requirement document to match it.

To do so, see the **Document Errors** tab page in the road freight order. A document error is created and the **Document Error** tab page is enabled. To correct these errors, you can use the actions available next to the description of the error. After you have corrected the error, the **Document Errors** tab page is disabled.

i Note

The **Document Error** tab page is enabled even if you haven't made the corresponding settings, that is, you haven't selected the **Enable Saving of Documents with Errors** value in the **Handling of Incorrect Inbound Messages** field in the Customizing for the freight order type. For more information, see [Correction of Errors in Business Documents](#).

When you add a new freight unit to a road freight order that is relevant for a customs process, you must enter the same external delivery note number for the freight unit that was used for one of the other items of the road freight order. In the inbound process, the freight unit is added after the execution at the stop is completed; therefore, the TM system then sends the customs-relevant information about the products in the freight unit to the integrated SAP Global Trade Services (SAP GTS) system.

Unassignment of a Requirement Document

You can unassign a requirement document from a capacity document after the execution at the stop has started but before goods movement is posted. You need to do this if, for example, in the outbound scenario you have a consignment order that could not be loaded at the warehouse loading stop as planned but that, according to the road freight order, is loaded.

If you want to remove a requirement document from the capacity document after loading is completed and the check-out is already reported, the following prerequisites must be met to allow the late removal process:

- The warehouse loading stop must have the handling execution status **Not Loaded**.

The status of the corresponding warehouse loading stop must be reverted to **Not Loaded**. This will not affect the logistical execution status of the document, that is, the check out and later events and statuses will not be reverted. This creates a document error indicating the temporary status mismatch between the warehouse loading stop, where loading is not completed, and the already checked out shipping point.

- The items may not have goods movement posted or requested.

Goods movement postings for the related items must be cancelled.

- The items may not have an export declaration requested or approved

Any approved or requested export declarations must be cancelled.

- The items may not have a customs invoice created.

Any existing customs invoices need to be cancelled.

Now, the loading of the items can be reverted by setting the handling execution status back to **Not Loaded** and the requirement can be unassigned from the capacity document. This can also be done by reporting the loading completion of the corresponding warehouse loading stop, as this will also remove all requirements that do not have the handling execution status **Loaded**.

The system performs the following steps when you unassign the requirement document:

- It sets the handling execution status to **Not Loaded** for all the items of the requirement document that is unassigned.
- In the unassigned requirement document, the system revokes all **Loading** and **Unloading** events related to these items.

Examples

The following examples describe a scenario where a requirement document is assigned to a capacity document after the truck has departed, and another scenario where you correct a requirement document error resulting in the late processing of the document.

Late Assignment of Requirement Documents

The following is an example for the inbound scenario:

A truck arrives at the company premises with a freight unit assigned to its road freight order. The road freight order items have a warehouse unloading stop defined. After you set the handling execution status to **Checked In**, the truck drives to the warehouse unloading stop to unload the goods. While unloading, the truck driver discovers that there is an extra pallet on the truck that is not assigned to the road freight order. The truck driver nevertheless unloads this pallet at the stop along with the other items, checks out, and departs. You update the status of the road freight order accordingly.

In the system, you now want to assign the additional pallet that was unloaded to the warehouse unloading stop in the road freight order, after the truck has departed. You can assign a new freight unit to the road freight order. But the statuses of the new item do not match the statuses of the rest of the items assigned to the stop that have already been processed. For example, the handling execution status of the new item is **Not Unloaded** while the other items have the status **Unloaded**. The system creates document errors on the **Document Errors** tab page.

If the inbound stop of the freight document has the handling execution status **Unloaded** or a status value that follows **Unloaded** in the inbound process, such as **Checked Out** or **Departed**, then on the **Document Errors** tab page, you must use the actions available next to the error description to do the following:

1. You set the warehouse processing status of the item to **Ready for Warehouse Processing**. However, to do so the item in the requirement document must be assigned to a delivery. It must be based on inbound documents like the inbound delivery. You therefore need to first create a delivery by choosing **Follow Up > Create Deliveries**.

When you set the status of the item to **Ready for Warehouse Processing**, the inbound delivery for the item to be delivered at that stop gets the status **In Yard** (for Extended Warehouse Management (EWM) deliveries) or **In Plant** (for Logistics Execution (LE) deliveries).

Once the delivery is created, you see the action to post goods movement on the **Document Errors** tab page.

2. You set the handling execution status of the item to **Unloaded**.
3. You post goods movement.

The statuses of all the items in the road freight order are now consistent.

Document Errors in Requirement Document

The following example describes how an **Execution Block** can be overruled to process the capacity document while the errors contained in one of the requirement documents in that capacity document are corrected.

You receive a road freight order with several consignment orders assigned to it via an EDI message. One of the consignment orders has document errors. It has a master data error because the product ID is wrong and a reference error that indicates that the system could not assign the freight unit to the consignment order. The system sets an **Execution Block** for the corresponding stop in the road freight order. As a result, you cannot process the road freight order at that stop and the truck is not allowed to enter the yard and start the execution at the warehouse unloading stop. However, you can overrule the block for the stop in the road freight order on the **Blocking Information** tab page.

Once you overrule the block, the consignment orders without document errors can be processed and the truck can drive into the yard and unload the cargo. The statuses of the consignment orders are set as they are processed, and the truck departs. You don't need to unassign the consignment order that has an error from the road freight order in the system. The cargo assigned to this consignment order is unloaded already and you just need to update the document in the system.

You do the following to process the consignment order:

1. You correct the errors in the consignment order. On the **Document Errors** tab page, you correct the master data error by entering the correct material. This also automatically resolves the reference data error and a freight unit is assigned to the consignment order. The consignment order is automatically assigned to the road freight order.
2. You now have a consignment order assigned, but its statuses do not match the statuses of the road freight order. You can create a delivery by choosing **Follow Up > Create Deliveries**.

Using the actions available on the **Document Errors** tab page in the road freight order, you set the warehouse processing status to **Ready for Warehouse Processing**, set the handling execution status of the item to **Unloaded**, and post goods movement.

The statuses of all the items in the road freight order are now consistent.

Statuses in Advanced Shipping and Receiving

The table below lists the statuses used in the Advanced Shipping and Receiving process in Transportation Management (TM) and links them to their descriptions in the **Explanation of Statuses** section below the table.

Statuses

Status	Status Values
Handling Execution Status	Cargo Ready for Loading Arrived Checked In Loaded Unloaded Checked Out Departed
Warehouse Door Status	Arrived at Door Departed from Door
Warehouse Processing Status	Not Determined Not Started Ready for Warehouse Processing In Process Completed In Transit Partially Ready for Warehouse Processing Partially In Process Partially Completed Partially In Transit
Goods Movement Status (see below, under Statuses in the Goods Movement Process)	Not Relevant Not Processed Yet Partially Processed Completely Processed
Goods Movement Processing (see below, under Statuses in the Goods Movement Process)	No Processing Requested Cancellation Requested Posting Requested Posting/Cancellation Failed Posting Partially Requested Cancellation Partially Requested Posting/Cancellation Requested Posting/Cancellation Partially Requested

Explanation of Statuses

The statuses used in the Advanced Shipping and Receiving process are explained below:

Handling Execution Status

For more information about the handling execution statuses, see [Explanation of Statuses of Business Documents](#).

For more information about checks for handling execution status-relevant events, see [Checks for Handling Execution Status-Relevant Events](#).

Warehouse Door Status

This status indicates whether a warehouse door is currently physically occupied by a truck or is available to load or unload goods. You can get an overview of warehouse doors that are presently available and can be used.

You can set the warehouse door status manually or the system sets it automatically at the level of the warehouse loading stop or warehouse unloading stop. If you have set the status to the wrong value by mistake, you can reverse the changes that you have made.

The status has the following values:

- **Arrived at Door**

You can set this status value to indicate that the truck has reached the warehouse door to load or unload goods and that the door is now occupied. The door cannot be used by other trucks until the loading or unloading is completed and the door is made available again.

You can only set the status value if the door is not currently occupied. You can only set it for one warehouse loading stop or warehouse unloading stop at a time and only if the handling execution status of the stop is set to **Checked In**. However, if the same door is being used for both loading and unloading activities, then the door of the warehouse loading stop and the warehouse unloading stop is set to **Arrived at Door** at the same time. You cannot set the status to **Arrived at Door** for the same door at the same time in another business document.

However, if you select the **Allow Multiple Door Assignments** checkbox for a loading point in the **Define Location** app, the system does not check if a warehouse door has the **Arrived at Door** status in other business documents. The warehouse door for that location is always available, even if it is occupied and has the **Arrived at Door** status in another business document. This can be used if, for example, you have created a placeholder warehouse door.

After you set the **Arrived at Door** status, the system reports the event **Arrival at Door**.

- **Departed from Door**

You can set this status value to indicate that the truck has finished loading or unloading the goods and has driven away from the warehouse door. The door is available again and can be used by another truck.

The system automatically sets the **Departed from Door** status if you directly set the handling execution status to **Checked Out** or **Departed**.

After you set this status, the system reports the event **Departure from Door**.

Warehouse Processing Status

You can use the warehouse processing status to track the progress of an item or business document through the warehouse and to determine the next steps in the transportation process based on the status. This status is from the perspective of TM. For information on the warehouse processing status from the perspective of Extended Warehouse Management (EWM), see [Transport Plan](#).

In the inbound process, the warehouse processing status helps control the handover of responsibility for the cargo items from TM to the warehouse. In the outbound process, this status can be used to track if the cargo items are ready to be shipped from the perspective of the warehouse. Once the items are ready to be shipped, TM can start the loading process or trigger follow-on processes, such as subcontracting and requesting customs declarations.

The following warehouse processing status values are available on the item level. They apply to single items or single documents. Note that the status values have a slightly different connotation in the inbound and outbound processes.

- **Not Determined**

This status value is the initial value that the system displays in both the inbound and outbound process.

- **Not Started**

This status value indicates that the warehouse has not started processing the cargo items. Depending on the loading direction, one of the following scenarios may apply:

- **Inbound:** In the inbound scenario, this status value is displayed in TM when the business documents of the truck have not been checked against the documents in the system. In the warehouse, the inbound delivery is blocked and cannot be processed until the documents have been checked.
- **Outbound:** In the outbound scenario, this status value is optional and is not relevant for processing cargo items in TM. Transportation planning can start irrespective of the warehouse processing status, but it is recommended to wait until the warehouse processing status is **Completed** and the warehouse has prepared the handling units that are to be shipped.

- **Ready for Warehouse Processing**

This status value indicates that the cargo items can be processed by the warehouse, but the processing hasn't started yet. Depending on the loading direction, one of the following scenarios may apply:

- **Inbound:** In the inbound scenario, the business documents of the truck are checked at the gate, and if they match the system documents, the truck arrives at the warehouse door. You then set this status value to transfer the responsibility for the cargo items from TM to the warehouse in the system. After this status value is set, the delivery is unblocked in the warehouse and the possibility of making changes to items is limited in TM.
- **Outbound:** In the outbound scenario, this status value is optional and is not relevant for the processing of cargo items in TM. Transportation planning can start irrespective of the warehouse processing status, but it is recommended to wait until the warehouse processing status is **Completed** and the warehouse has prepared the handling units that are to be shipped.

- **In Process**

This status value indicates that the warehouse has created warehouse tasks and is actively working on the cargo items. Depending on the loading direction, one of the following scenarios may apply:

- **Inbound:** In the inbound scenario, this status value is optional and is not relevant for processing cargo items in TM because the responsibility for the cargo items was handed over to the warehouse after the **Ready for Warehouse Processing** status was set.
- **Outbound:** In the outbound scenario, this status value indicates that the warehouse is working with the cargo items. You can start cargo planning but cannot change the items.

- **Completed**

This status value indicates that the warehouse has completed the activities related to the processing of the cargo items. Depending on the loading direction, one of the following scenarios may apply:

- **Inbound:** In the inbound scenario, this status value is optional and is not relevant for processing cargo items in TM because the responsibility for the cargo items was handed over to the warehouse after the **Ready for Warehouse Processing** status was set. Warehouse activities, such as re-packaging received cargo items, are not relevant for transportation.
- **Outbound:** In the outbound scenario, this status value indicates that from the perspective of the warehouse, the cargo items are ready to be shipped. The responsibility for the cargo is transferred from the warehouse to TM in the system.

- **In Transit**

This status value is only relevant for the stock transport order (STO) process. Handling is performed at the outbound warehouse and the document progresses through the different warehouse processing statuses as normal. However, in the case of an STO, once outbound handling is completed, the freight unit is reassigned from an outbound delivery to an inbound delivery. The freight unit has the inbound delivery as its direct predecessor document, though the outbound delivery still appears in the list of predecessor documents. At this moment the warehouse processing status is set to **In Transit**. The inbound transport will arrive in the inbound warehouse.

After this switch, you are able to set the warehouse processing status for the inbound warehouse and the document progresses through the different warehouse processing statuses as any other inbound delivery.

For more information about STO, please see [Integration of Stock Transport Orders](#).

The warehouse processing status also has aggregated status values. The status is aggregated from the item level to the next higher level in the hierarchy. The status also appears on the header level of the business document. For example, if all the items in the road freight order have the status **Ready for Warehouse Processing**, the status on the header level will also be **Ready for Warehouse Processing**.

However, if all items do not have the same status, then the system displays the aggregated status of these items on the header level, indicating partial processing. For example, if one item has the status **Completed**, one has the status **In Process**, and one has the status **Not Started**, then the system displays the highest aggregated status for these items, that is, **Partially Completed**.

The following are the aggregated warehouse processing statuses:

- **Partially Ready for Warehouse Processing**

The system displays this status value when one or more of the items or documents have the status **Ready for Warehouse Processing** and at least one other item or document has a lower status indicating that it is at an earlier stage in the warehouse processing lifecycle. For example, the other items may have the status **Not Started**.

The **Partially Ready for Warehouse Processing** status can change to **Partially In Process** or **Partially Completed** if the status of one of the items or documents in the road freight order changes to **In Process** or **Completed**.

- **Partially in Process**

The system displays this status value when one or more of the items or documents have the status **In Process** and at least one other item or document is at an earlier stage in the warehouse processing lifecycle. For example, the other items may have the status **Ready for Warehouse Processing**.

The aggregated status can change to **Partially Completed** when the status of one or more items changes to **Completed**.

- **Partially Completed**

The system displays this status value when one or more of the items or documents have the status **Completed** and at least one other item or document is at an earlier stage in the warehouse processing lifecycle. For example, the other items may have the status **In Process**.

- **Partially In Transit**

This status value is only relevant for the stock transport order (STO) process. The system displays this status value when one or more of the items or documents have the status **In Transit** and at least one other item or document is at an earlier stage in the warehouse processing lifecycle. For example, the other items may have the status **In Process**.

Statutes in the Goods Movement Process

In TM, you can trigger the posting or cancellation of goods movement for a delivery assigned to a warehouse loading stop or warehouse unloading stop. You can do so using the **Post** or **Cancel** options in the **Goods Movement** drop-down menu. The request

is sent from TM to the warehouse and the warehouse posts or cancels goods movement. TM waits for the goods movement posting to be confirmed or cancelled by the warehouse.

In the inbound process, you can trigger the posting or cancellation of goods movement only after the warehouse processing status of the items or business documents assigned to the warehouse unloading stop is set to **Ready for Warehouse Processing**.

In the outbound process, you can trigger the posting or cancellation of goods movement only after the warehouse processing status of the items or business documents assigned to the warehouse loading stop is **Completed** and the handling execution status of the warehouse loading stop is **Loaded**.

The posting request or cancellation request is sent to the warehouse in the system after you save the changes in the business document. Until you save, the posting or cancellation request is saved locally in the business document and you can cancel the changes using the **Reverse Changes** option in the **Goods Movement** drop-down menu.

Goods movement for a handling unit or a delivery can also be triggered by the warehouse itself and this information is communicated to the related item, such as a pallet or container.

The goods movement information is displayed in the **Goods Movement Status** field and the **Goods Movement Processing** field in the business document. The **Goods Movement Status** field displays the overall status of goods movement processing in the warehouse. The **Goods Movement Processing** field gives you more information about the intermediate stages during the processing. For example, if you have not posted goods movement, the goods movement status is **Not Relevant**, and the goods movement processing status is **No Processing Requested**. After you post goods movement, the goods movement status is **Not Processed Yet** because only a request has been received, and the goods movement processing status is **Posting Requested**. Eventually, once the warehouse has processed the request, it updates the goods movement status to **Completely Processed**. The status values are explained in detail below.

- **Goods Movement Status**

In TM, the system displays the information communicated by the warehouse about the goods movement status of an item. The goods movement status is displayed on the item and header level of a road freight order, consignment order, and freight unit (FU).

The status has the following values:

- **Not Relevant**

This status value indicates that the item is not yet considered for goods movement.

- **Not Processed Yet**

This status value indicates that TM has sent the request for goods movement processing of an item to the warehouse, but the warehouse has not processed that item yet. The warehouse has also not communicated the current status to TM yet because, for example, the request is in the queue.

- **Partially Processed**

This status value indicates that the warehouse has performed goods movement for some of the items, but not for all.

- **Completely Processed**

This status value indicates that the warehouse has sent a goods movement confirmation for all the subitems.

- **Goods Movement Processing**

This status is from the perspective of TM. The status values indicate the intermediate stages in the goods movement process. The goods movement processing status is displayed on the item level of a road freight order.

The status has the following values:

- **No Processing Requested**

This status value indicates that goods movement processing has not been requested yet.

- **Posting Requested**

This status value indicates that TM has triggered goods movement posting, and the warehouse has received the request and is working on it. But it has not posted goods movement yet.

- **Cancellation Requested**

This status value indicates that TM has triggered goods movement cancellation, and the warehouse has received the request and is working on it. But it has not cancelled goods movement yet.

- **Posting/Cancellation Failed**

The system displays this status when the posting or cancellation of goods movement by the warehouse is not successful.

The goods movement processing status is also aggregated from the item level to the next higher level in the hierarchy. It has the following aggregated status values:

- **Posting Partially Requested**

The system displays this status value for an item or business document. This status value indicates that goods movement posting has been requested for one or more items, but not for all. For example, in the outbound process, you have created a consignment order with two FUs for a warehouse loading stop. If one of the FUs has the handling execution status **Loaded**, warehouse processing status **Completed**, and the goods movement status **Posting Requested** and the other has the statuses **Not Loaded**, **In Process**, and **No Processing Requested**, the aggregated goods movement processing status of the consignment order will be **Posting Partially Requested**.

- **Cancellation Partially Requested**

The system displays this status value for an item or business document. This status value indicates that goods movement cancellation has been requested for one or more items, but not for all. For example, in the outbound process, you have created a consignment order with two FUs for a warehouse loading stop. If one of the FUs has the handling execution status **Loaded**, warehouse processing status **Completed**, and the goods movement status **Cancellation Requested** and the other FU has the statuses **Not Loaded**, **In Process**, and **No Processing Requested**, the aggregated goods movement status of the consignment order will be **Cancellation Partially Requested**.

- **Posting/Cancellation Partially Requested**

The system displays this status value for an item or business document, such as a consignment order, when, for example, one FU belonging to the consignment order has the goods movement processing status **Posting Requested**, the second FU has the status **Cancellation Requested**, and the third FU has the status **No Processing Requested**.

The system can also display this status value as an aggregation of the statuses of the business documents. For example, if one consignment order of the warehouse loading stop has the goods movement status **Posting Partially Requested** and the other consignment order has the status **Cancellation Partially Requested**, then the aggregated status of the warehouse loading stop will be **Posting/Cancellation Partially Requested**.

- **Posting/Cancellation Requested**

The system displays this status value if all the items or business documents have the goods movement processing status **Posting Requested** or **Cancellation Requested** and there are no items with the status **No Processing Requested**. For example, two of the three FUs belonging to a consignment order have the goods movement processing status **Posting Requested** and one has the status **Cancellation Requested**. In this case, the aggregated status of the consignment order will be **Posting/Cancellation Requested**.

Statuses in the Inbound Process

The inbound part of the Advanced Shipping and Receiving process in Transportation Management (TM) covers the activities that take place between the time that a loaded truck arrives at the warehouse gate, unloads the goods at the warehouse door, and departs from the company premises. The responsibility for the goods is handed over from TM to the warehouse. Below is a sample sequence of activities in terms of the statuses used in the inbound process.

You can see the statuses on the **Statuses** tab page on the product level or the header level of the business document or in the fields in the **Overview** section. For more information about statuses in general, see [Statuses in Advanced Shipping and Receiving](#). For more details about statuses in a road freight order that is relevant for a customs process, see [Statuses in the Integration with SAP GTS \(Inbound Process\)](#).

The basic sequence of statuses and activities in the inbound process is as follows:

1. When the truck arrives at the gate, you set the handling execution status for the stop to **Arrived**. The warehouse processing status of the items is **Not Started**.
2. If the business documents of the goods on the truck match the documents in the system, you set the handling execution status to **Checked In**. All the inbound deliveries for the goods to be delivered at that stop get the status **In Yard** (for Extended Warehouse Management (EWM) deliveries) or **In Plant** (for Logistics Execution (LE) deliveries), thus communicating to the warehouse that the ordered stock has reached the company premises. If you set the **Checked In** status without first setting the **Arrived** status, then the system sets the **Arrived** status automatically.

If the documents of the goods don't match the documents in the system, you can make changes manually in the business document. The referenced predecessor documents are also updated if you have enabled this in the Customizing of the document type. You can make changes that affect the warehouse process until the warehouse processing status is set to **Ready for Warehouse Processing**.

3. The truck drives to the assigned warehouse unloading stop to unload the goods. When it reaches the door, you set the warehouse door status to **Arrived at Door**. This status value indicates that until this truck departs, the door cannot be used by other trucks. You can allow a door to be used in several business documents at the same time using the **Define Location** app. For more information, see "Warehouse Door Status" under [Statuses in Advanced Shipping and Receiving](#).

After you set this status, the system reports the event **Arrival at Door**.

4. You then set the warehouse processing status to **Ready for Warehouse Processing** at the warehouse unloading stop level to signal to the warehouse that they can begin their activities. With this status value, you transfer the responsibility for the goods from TM to the warehouse in the system. After this status value is set, the delivery is unblocked in the warehouse and the possibility of making changes to items is limited in TM. In an EWM integration scenario, after you set the **Ready for Warehouse Processing** status for the freight units or for the inbound delivery from LE, the freight units that were initially assigned to the inbound delivery from LE are assigned to the warehouse inbound delivery from EWM. You can track the progress of this assignment on the **Unloading Details** tab of a warehouse unloading stop in the **Overview** section of the road freight order.

If you don't have an explicit check-in process at your company gate, you can directly set the status to **Ready for Warehouse Processing** at the warehouse unloading stop level. The system will automatically set the **Checked In** status at the stop level and **In Yard** (for EWM deliveries) or **In Plant** (for LE deliveries).

When you set the **Ready for Warehouse Processing** status for LE inbound deliveries, the system checks if it's defined for the product that it must be packed or unpacked, before you can set the status.

5. If information about the unloading activities cannot be electronically communicated, you can use the handling execution status option **Set Items to Unloaded** to manually set the status of the items that are unloaded from the truck to **Unloaded**.
6. When unloading is completed, you can use the handling execution status option **Set Stop to Unloading Completed** to set the status of the warehouse unloading stop to **Unloaded**. If you set this status without setting the status of the items at this stop to **Unloaded**, the system creates document errors. On the **Document Errors** tab, you can use the actions available

next to the error description to set the handling execution status of the items to **Unloaded**. For information about making changes to a capacity document even after the execution at the stop has started or been completed, see [Late Changes to Capacity Documents](#).

7. You can post goods movement at the warehouse unloading stop level only after the warehouse processing status is set to **Ready for Warehouse Processing**. The system displays the status value **Posting Requested** and triggers goods receipt by the warehouse. After the warehouse has successfully processed the request, the system updates the goods movement status to **Completely Processed**. If the posting of goods movement by the warehouse is not successful, the system displays the goods movement processing status **Posting/Cancellation Failed**. In the freight order, you can view the details of the errors related to posting of goods on the **Document Errors** tab page.
8. Irrespective of the result of the goods movement posting, you can manually set the warehouse door status to **Departed from Door**. The door is available again for other trucks to use. After you set this status, the system reports the event **Departure from Door**.
9. At the stop level, you set the handling execution status to **Checked Out**. If you set the **Checked Out** status without first setting the warehouse door status to **Departed from Door**, the system automatically sets the **Departed from Door** status.
10. You set the handling execution status to **Departed**.

Statuses in the Outbound Process

The outbound part of the Advanced Shipping and Receiving process in Transportation Management (TM) includes those activities that occur from the start of planning to the time the truck departs from the company premises after loading. The responsibility for the goods is handed over from the warehouse to TM. Below is a sample sequence of activities in terms of the statuses used in the outbound process.

You can see the statuses on the **Statuses** tab on the product level or the header level of the business document or in the fields in the **Overview** section. For more information about statuses in general, see [Statuses in Advanced Shipping and Receiving](#). For more details about statuses in a freight order that is relevant for a customs process, see [Statuses in the Integration with SAP GTS \(Outbound Process\)](#).

The sequence of statuses and activities in the outbound process depends on which scenario for transportation planning and warehouse processing is used. The following scenarios are supported:

- **Warehouse-Driven Scenario** - In this scenario, warehouse processing takes places before transportation planning.

Transportation planning can start irrespective of the warehouse processing status, but it is recommended to wait until the warehouse processing status is **Completed** and the warehouse has prepared the handling units that are to be transported.

When the warehouse creates outbound deliveries and freight units but doesn't start its warehouse activities, such as picking and packing, the system displays the warehouse processing status **Ready for Warehouse Processing** in TM.
- **Transport-Driven Scenario** - In this scenario, transportation planning takes place, in whole or in part, before warehouse processing.

In a transport-driven scenario, you have to manually set the new freight units or the road freight order to **Ready for Warehouse Processing**. For more information, see [Transport-Driven Execution](#).

The basic sequence of statuses and activities in the outbound process is as follows:

1. Warehouse processing and transportation planning take place.
2. When the warehouse starts packing handling units, the system displays the warehouse processing status **In Process** in TM. You can start planning the transport of the cargo items but cannot make changes to the items.

3. When the warehouse tasks have been completed and the handling units are ready, the warehouse sets the shipping readiness status of the delivery to **Ready for Shipping** in Extended Warehouse Management (EWM) or sets the warehouse execution status to **Delivery is Ready for Shipping** in Delivery Management (LE-SHP). This updates the freight units with the packing details of the handling units and triggers the warehouse processing status **Completed** in TM. This means that you can begin your activities, such as creating a consignment order or a road freight order from the freight units.

If you use a logistics service provider (LSP) to ship your goods, then information about the updated freight units is sent to the LSP. The LSP creates a road freight order from the freight units.

4. You can set the handling execution status to **Cargo Ready for Loading** in the freight order. With this, you can initiate customs processes and have the good cleared before the truck's arrival and check in. Any items whose warehouse processing status is not set to **Completed** at this point are removed from the freight order.

Status **Cargo Ready for Loading** as a prerequisite and automatic trigger for customs processes requires customizing. For more information, see [SAP GTS Integration for the Outbound Process](#) and [Integration with SAP Global Trade Services in Advanced Shipping and Receiving](#).

5. You can then assign the freight unit to a freight order and specify on which truck the products contained in the delivery are to be transported. Additionally, the system can automatically create a consignment order, which bundles all deliveries that have the same ship-to party. The truck then drives to the warehouse to load the goods. When it arrives at the gate, you set the handling execution status for the stop to **Arrived**.

For more information about automatic consignment creation, see [Creation and Editing of a Consignment Order](#) and [Automatic Consignment Building](#).

6. After the documents of the truck are verified, you set the handling execution status for the stop to **Checked In**. The system automatically defines the loading point and warehouse door from which the truck has to pick up the goods, if this information is provided in the predecessor documents. You can also manually define a loading point and warehouse door.

7. When the truck reaches the warehouse door, you set the warehouse door status to **Arrived at Door** and the warehouse can start loading the truck. After you set this status, the system reports the event **Arrival at Door**. You can allow a door to be used in several business documents at the same time using the **Define Location** app. For more information, see "Warehouse Door Status" under [Statuses in Advanced Shipping and Receiving](#).

8. You can use the handling execution status option **Set Items to Loaded** to manually set the status of the items at the warehouse loading stop to **Loaded**.

9. After you have finished loading the truck, you can use the handling execution status option **Set Stop to Loading Completed** to set the status of the warehouse loading stop to **Loaded**. After this status has been set, no further items can be loaded onto the truck, and the system automatically removes all the items that do not have the status **Loaded** from the road freight order.

10. You can post goods movement at the warehouse loading stop level only after the warehouse processing status is set to **Ready for Warehouse Processing**. The system displays the status **Posting Requested** and triggers goods issue by the warehouse. After the warehouse has successfully processed the request, the system updates the goods movement status to **Completely Processed**. If the posting of goods movement by the warehouse is not successful, the system displays the goods movement processing status **Posting/Cancellation Failed**. In the freight order, you can view the details of the errors related to posting of goods on the **Document Errors** tab.

11. Irrespective of the result of the goods movement posting, you can manually set the warehouse door status to **Departed from Door**. After you set this status, the system reports the event **Departure from Door**.

12. At the stop level, you set the handling execution status to **Checked Out**. If you set the **Checked Out** status without first setting the warehouse door status to **Departed from Door**, the system automatically sets the **Departed from Door** status.

13. You set the handling execution status to **Departed**.

Consignment Order

In Transportation Management (TM), a consignment order represents a business document that you can use to group several transportation requirements based on certain criteria, such as the same source and destination location, the same delivery date/time, and the same business partner.

Use

A consignment order serves as an important basis for the communication between the partners that are involved in a business process. For example, as a supplier you can communicate to your ordering party which products you will deliver to a certain destination on a specific delivery date. The consignment order receives an ID that serves as a reference for the business partners.

Based on a consignment order, you can also perform subcontracting, charge calculation, cost distribution, settlement with the carrier, and shipment tracking. It is not, however, possible to perform tendering based on a consignment order.

You can then assign the consignment order to a road freight order and in doing so stipulate on which truck the products contained in the consignment order are to be transported. A consignment order can be assigned to only one freight order at a time and must be completely assigned. A freight order can, however, contain several consignment orders. The transportation is then planned and executed based on the freight order.

You can choose whether to perform subcontracting, charge calculation, settlement, and so on, for the freight order or the consignment order. You can do this in Customizing for the consignment order type or freight order type by enabling charge calculation, cost distribution, and settlement or by indicating that the documents of this document type are relevant for subcontracting. For more information, see Customizing for **Transportation Management** under ► [Freight Order Management](#) ► [Freight Order](#) ► [Define Freight Order Types](#) and ► [Freight Order Management](#) ► [Consignment Order](#) ► [Define Consignment Order Types](#).

You can also assign a consignment order to a transportation unit, such as a container or trailer unit, and then assign this to the freight order. However, the container or trailer unit must then contain a transportation stage that corresponds to that of the consignment order.

Creating consignment orders for capacity documents that refer to requirement documents containing multiple main items is not supported. Examples for requirement documents with multiple main items are:

- Container unit with several container items
- Package unit with several package items

Consignment Order Creation Based on Top-Level Packages

You can use consignment order creation based on the **top-level package** (handling unit). When you use this function, each consignment contains only one package item or unpacked product. This means, the system creates one consignment order per handling unit (pallet).

For this function, the grouping rule [TOP_CARGO Topmost Cargo Item \(Package or Unpacked Product\)](#) is available in the [Consignment Building Profile](#) app.

The consignment order creation based on top-level packages is, for example, useful in air freight processes for shippers. For more information, see [Ocean Freight Bookings for Shippers \(Outbound\)](#) and [Air Freight Bookings for Shippers \(Outbound\)](#).

BAdI to Determine Consignment Order Type

In the [Consignment Building Profile](#) app, you can configure how the system determines the consignment order type. You can either choose [Document Type Defined in Profile](#) or [Document Type Defined by BAdI](#). This is useful especially if you need two consignment order types that might have a different custom profile to be used in one freight order, for example, to distinguish consignment order for "normal" deliveries from consignment orders for deliveries with empties.

You can use [BAdI: Consignment Order Type Determination](#) to determine the consignment order type in the [Consignment Building Profile app](#). For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Freight Order Management](#) > [Consignment Building](#) > [BAdI: Consignment Order Type Determination](#).

Structure

You include requirements in a consignment order, that is, freight units or package units that are based on predecessor documents, such as sales orders, scheduling agreements, or deliveries. These requirements must have the same source and destination location as well as the same delivery date/time. Moreover, the business partners involved, such as shipper, consignee, and carrier, must be the same.

A consignment order can contain only one transportation stage between the source and destination location, that is, no intermediate stages. This stage must be strictly linear. The consignment order can also have only one shipper and one consignee.

The main item of the consignment order has the category "Consignment Order". Freight units and package units, packages, and products appear below the consignment order in the item hierarchy. You can also perform package building for the items of the consignment order.

Creating a Consignment Order

You can create a consignment order either manually or automatically using an inbound message (an advanced shipping notification via EDI). To create the consignment order manually, open the [Create Consignment Order](#) app on the launchpad.

In the consignment building profile, you can also specify that consignment orders are automatically built locally in a freight order when a freight order is created and saved, and that the corresponding consignment order documents are created later.

For more information, see [Creation and Editing of a Consignment Order](#) and [Automatic Consignment Building](#).

Worklist and Search Options for Consignment Orders

You can display your consignment orders in your personal worklist for consignment orders and call them from there. Open the [Consignments - Worklist](#) app on the launchpad. You can also search for consignment orders by freight order, for example.

Related Information

[Creation and Editing of a Consignment Order](#)

[Automatic Consignment Building](#)

[Ocean Freight Bookings for Shippers \(Outbound\)](#)

[Air Freight Bookings for Shippers \(Outbound\)](#)

Creation and Editing of a Consignment Order

In Transportation Management (TM), you can create consignment orders automatically using inbound messages as well as manually. The options for creating consignment orders and an example of how to create and edit consignment orders are described below.

Prerequisites

You have defined consignment order types in Customizing. Within the consignment order type, you configure the most important settings for the consignment order. For more information, see Customizing for Transportation Management under [Freight Order](#)

Creation Options

You have the following options for creating a consignment order:

- Manual creation

To create the consignment order manually, open the [Create Consignment Order](#) app on the launchpad. For more information, see the process description below.

- Creation from a freight order

On the [Items](#) tab page in a freight order, choose [Create Consignment Order](#). The system uses the consignment order type that was defined in the freight order type. If a consignment order type was not defined in the freight order type, the system uses the consignment order type that you have defined in the consignment building profile or planning profile or it uses the default consignment order type predefined in the system (in this sequence). For more information, see Customizing for Transportation Management under [Freight Order Management > Freight Order > Define Freight Order Types >](#).

- Creation using an inbound message

Your business partner can send you the consignment order as an advanced shipping notification using EDI communication. You then receive the message using the service interface `TransportationOrderGenericRequest_In`. Based on this message, the system automatically creates a consignment order.

For more information, see [Creation or Update of Consignment or Freight Orders on Inbound Messages](#).

If the inbound consignment order contains errors, it is usually still saved as a document in TM. If the freight units (FUs) in the consignment order contain errors, the system first creates deliveries for the FU without errors. After you have corrected the errors, the system either assigns the remaining FUs to existing deliveries or creates new deliveries for FUs with different external delivery numbers. The errors are displayed on the [Document Errors](#) tab page and can be corrected here. For more information, see [Correction of Errors in Business Documents](#).

- Creation during manual planning using the transportation cockpit

In the transportation cockpit, you have different options for creating a consignment order and assigning it to a freight order or truck, for example. For more information, see [Use of Consignment Orders in Planning](#).

- Creation by copying

You can create a consignment order by copying an existing one (reference document). The system copies the header data and the logistical data. However, it does not copy references to freight units.

You can also create more than one copy of an existing consignment order at the same time. In addition to the number of copies, you can specify whether the numbers of the created documents are to be displayed in the dialog box.

If you copy an existing consignment order, you can enter a different consignment order type for the new consignment order. This is useful if you want to copy certain attributes of an existing document into a new one, such as the location or carrier. For example, you can copy an existing consignment order with the type [Consignment Order for Load Consolidation](#) and create a new consignment order with a different document type, such as [Consignment Order for Load Planning](#).

- Automatic building for freight units of a freight order

You can let the system group freight units in a freight order into consignment orders based on grouping rules. For more information, see [Automatic Consignment Building](#).

The [Origin of Consignment Order](#) field in the general data of the consignment order shows you how the consignment order was created.

Process

The following is a description of an example process for manually creating a consignment order. It is assumed that you already know the most important transportation data, such as source and destination location and business partners.

1. Entering the general data

You first enter the general data for your consignment order, for example:

- Transportation data, such as shipper, consignee, carrier
- Locations and dates/times
- Notes

2. Defining the items

You have the following options:

- You add freight units or package units to your consignment order. You do so on the **Overview** tab page by choosing the **Insert** pushbutton and then **Freight Unit** or **Transportation Unit**.

When you add freight units, note that a consignment order can contain only one transportation stage, in other words, only one source and destination location. If you add freight units that have the same goods supplier, the system copies this data to the consignment order. If the freight units have different goods suppliers, you can manually enter the required goods supplier for this consignment order as a business partner. The goods suppliers and the corresponding source locations of the freight units are automatically adjusted. The predecessor documents, such as purchase orders, remain unchanged.

- You directly enter items to be transported. You do so on the **Items** tab page by choosing the **Insert** pushbutton.
- You can have the system build packages. For more information, see [Unified Package Building in TM](#).

3. Calculating transportation charges

Optionally, you can have the system calculate the expected transportation charges for your consignment order. They are displayed on the **Charges** tab page. For more information, see [Charge Calculation](#).

4. Checking and saving

Once you have entered all the important data, you can check the consistency of the data. You then save your consignment order.

5. Sending the consignment order

You send the consignment order to your carrier. You do so by choosing **Subcontracting** and then **Send to Carrier**. You can also assign the consignment order to a freight order first and perform subcontracting there.

When you subcontract a consignment order, the system recalculates the charges. If these charges differ from the previously calculated charges, the system cancels subcontracting. You must then check the charges manually and adjust them, if necessary. Subsequently, you can resend the consignment order to the carrier. For more information, see [Charge Calculation](#).

6. Confirming or rejecting the consignment order

You receive a confirmation or rejection of the consignment order from your carrier in the form of electronic messages. If you have not set up message communication with your carrier, you can also enter this information manually in the system.

7. Printing

Optionally, you can print the corresponding print documents. For more information, see [Printing](#).

8. Fixing the consignment order

You can fix your consignment order to protect it from undesirable changes arising from background processing (planning runs or carrier selection).

9. Creating a freight settlement document

After execution, you can create a freight settlement document (see [Freight Settlement Document](#) and [Enablement of Freight Settlement Documents for Creation](#)). This serves as the basis for checking the invoice or issuing a voucher.

You can also perform freight settlement for the freight order after you have assigned the consignment order to it.

10. You assign the consignment order to a freight order. To do this, open the **Edit Road Freight Order** app on the launchpad. Then choose the **Insert** option on the **Items** tab page. You can insert consignment orders based on the ID, the external ID, or your selection.

The consignment order is displayed in the document flow of the freight order, from where you can call it up. It is also displayed in the item hierarchy on the **Items** tab page of the freight order and on the **Overview** tab page, where you can perform further planning steps.

For more information, see [Freight Order](#).

i Note

A possible variant of this process is that you simply enter the general data. You then call the transportation cockpit and assign freight units to your consignment order. You subsequently perform all additional planning steps in the transportation cockpit (for example, you have the system calculate the transportation charges). For more information, see [Manual Planning](#) and [Use of Consignment Orders in Planning](#).

Related Information

[Consignment Order](#)

[Additional Business Document Attributes in FOM](#)

[Automatic Consignment Building](#)

Automatic Consignment Building

In Transportation Management (TM), you can let the system build consignment orders for freight orders automatically. Based on rules that you have specified in a profile as well as default rules, freight units of the freight order are grouped into consignment orders.

Features

If you want to build consignment orders for an existing freight order that you need for follow-up processes, such as a customs declaration, you can let the system execute this operation automatically. You can specify that the system first creates the consignment orders locally in the freight order and creates the corresponding consignment order documents later. The freight units can be grouped into consignment orders based on a profile in which you have specified the grouping rules for the freight units. Furthermore, the system applies predefined default rules that belong to the prerequisites of a consignment order in TM (for example, the rule that the freight units must have the same source and target location).

Consignment Building Profile

In the consignment building profile, you can select grouping rules for freight units that are already predefined in the system, such as grouping by loading point or Incoterm. You can also combine several rules. The system then groups only those freight units into

a consignment order that correspond to these rules (for example, freight units that have the same loading point and the same Incoterm). You can also specify the following in the profile:

- You can select your own rules that you have defined in Customizing. For more information, see Customizing for Transportation Management under [Freight Order Management > Consignment Order > Define Grouping Rules for Automatic Consignment Building](#).
- You can specify whether incompatibilities are to be considered as a grouping rule. To do so, enter the name of the settings that you have defined in the [Create Incompatibility Settings](#) app in the [Incompatibility Settings](#) field. Note, however, that you must have selected the [Identical Values Only](#) checkbox in your incompatibilities for them to be considered.
- You can specify the consignment order type of the consignment orders that are created automatically. The consignment order type also determines the number range from which the system takes the consignment order number. This number is then retained consecutively in the system, even if consignment order documents are created later for local consignment orders.
- You can specify whether the system automatically builds consignment orders locally in the freight order or updates only existing consignment orders. For more information, see [Creating Local Consignment Orders](#) below.
- You can specify whether the system automatically creates consignment order documents. For more information, see [Creating Consignment Order Documents](#) below.

To create the profile, choose the [Create Consignment Building Profile](#) app on the launchpad. The system takes the profile into account if you enter it in Customizing for the freight order type. For more information, see Customizing for Transportation Management under [Freight Order Management > Freight Order Type > Define Freight Order Types](#).

Creating Local Consignment Orders and Consignment Order Documents

You have the following options for automatically building consignment orders locally and creating consignment order documents in the freight order:

- **Creating local consignment orders**

- Manual triggering from the freight order UI

Select the menu option [Follow-Up > Build Local Consignment Orders](#) in the [Create Freight Order](#) or [Edit Freight Order](#) app. The system then groups the freight units of the freight order into consignment orders based on the default rules and according to any rules defined in the consignment building profile and creates local consignment orders with the item type SHP and the item category [Consignment Order](#). The consignment orders are also marked with the qualifier [local](#) in the item hierarchy.

If you do not enter a profile in the freight order type and trigger the automatic building of consignment orders manually from the UI of the freight order, the system applies only the default rules and uses the consignment order type.

- Automatic building when the freight order is created and saved

In the [Automatic Building](#) field of the consignment building profile, you can specify that the system automatically builds consignment orders locally when a freight order is created and saved manually or based on an inbound message.

- Manual triggering with subsequent automatic update

In the [Automatic Building](#) field of the profile, you can also specify that although you have to trigger the initial building of local consignment orders manually on the freight order UI, the update takes place automatically. Depending on the setting, the system then only updates the local consignment orders for consistency reasons according to the defined grouping rules, or it updates the consignment orders completely, taking into account newly added freight units. For more information, see the field help.

The advantage of local consignment orders is that changes can still be made to the freight order after consignment building that are taken into account when the consignment order documents are generated. The consignment order documents are then no longer updated with freight order changes. You can, for example, add freight units to the freight order and have the consignment building executed again. The system then assigns new freight units to the existing consignment orders or, if this is not possible according to the grouping rules, it groups them into new consignment orders.

Note that the system does not take local consignment orders into account during further planning.

- **Creating consignment order documents**

- Manual triggering of document creation for all consignment orders

If you do not expect any further changes to the freight order and you need the consignment order documents for follow-up processes, save your freight order data again and choose ► **Follow-Up** ► **Create Consignment Order** ► to create the relevant documents for all local consignment orders of the freight order.

- Manual triggering of document creation for individual consignment orders

You can select individual local consignment orders from the item hierarchy of the freight order and create documents only for these consignment orders.

- Automatic document creation based on the profile settings

In the **Create Consignment Documents** field of the consignment building profile, you can specify that consignment order documents are created automatically by the system following specific actions, status changes, or events. You can, for example, specify that the consignment order documents are created automatically when you save the freight order, when the execution status is set to **Ready for Transportation Execution**, or when the truck has departed from the company premises. When the freight order is saved and the execution status is set to **Ready for Transportation Execution**, the documents are created for all consignment orders of the freight order. In the case of events, only the consignment order documents are created for the location for which the event is executed.

However, the prerequisite for this automatic document creation is that the consignment orders have already been built locally in the freight order. For more information, see the field help.

Related Information

[Consignment Order](#)

[Creation and Editing of a Consignment Order](#)

Creation or Update of Consignment or Freight Orders on Inbound Messages

You can create or update a consignment order or freight order either manually or by an inbound message using EDI.

Your business partner can send you a consignment order as an advanced shipping notification (ASN) or a freight order using EDI communication. You then receive the message using the service interface **TransportationOrderGenericRequest_In**. Based on this message, the system automatically creates or updates a consignment order or freight order.

For more information about **manual** creation of consignment orders, see [Creation and Editing of a Consignment Order](#).

Prerequisites

In Customizing for the freight order type or consignment order type, you can configure your settings for the handling of inbound messages. For more information, see the following Customizing activities in Customizing for Transportation Management:

- [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) >
- [Freight Order Management](#) > [Freight Order Management](#) > [Consignment Order](#) > [Define Consignment Order Types](#) >

Separate or Combined Inbound Messages

You have the following options to create or update freight orders or consignment orders:

- **Separate Inbound Messages (Consignment-Based Freight Order)**

Information for freight orders and consignment orders can be sent in **separate** inbound messages (ASN). The freight order contains references to the related consignment orders. Ideally, the consignment order is sent first.

If a consignment order is error-free, the system can automatically create an inbound delivery in Logistics Execution (LE) if the Customizing is set accordingly. If a consignment order or freight order that was sent in separate inbound messages contains errors, it is usually still saved as a document in TM as long as the consignment order contains at least a source and destination location. The errors are displayed on the [Document Errors](#) tab and can be corrected there. Whether the system creates errors, depends on the setting that you have made in the Customizing for the freight order type or consignment order type. If the freight order message is sent before the consignment order was created, and the consignment order is created later, the assignment of the consignment order is done automatically by choosing the respective external consignment ID.

TM creates deliveries for error-free FUs when you create or save a consignment order. Upon correction of errors in FUs, the system will either assign these FUs to an existing delivery or create new deliveries for FUs with different external delivery numbers. Once FUs are assigned to a delivery, the system cannot make changes to these deliveries during an MRP run and the subsequent freight unit building step. As with consignment order errors, you can find and correct errors on the [Document Errors](#) tab.

For more information, see [Example Checks to Find Errors in Business Documents](#) and [Correction of Errors in Business Documents](#).

- **Combined Inbound Messages (Transport-Based Freight Order)**

Information for freight orders and consignment orders and their item structure can be combined and sent in a **single** inbound message. With combined inbound messages you can create freight orders and consignment orders and all the related items underneath.

A combined inbound message contains both freight order header data and information about the full item structure (such as, information about the active vehicle resource, shipment items, and references to the requirement documents). With this information the system creates consignment orders with assigned freight units and a freight order with references to the consignment orders. The consignment order type is then the default consignment order type defined in Customizing.

The system checks to make sure the information of the inbound message is error-free. If a combined message contains serious errors, the system rejects the inbound message and no freight order is created.

It is not possible to share items of a consignment order between an active and a passive vehicle resource. Such item structures can currently not be sent with EDI-messages.

Creating or Updating Consignment Orders (Information in Separate Inbound Messages)

You can create or update consignment orders using inbound messages.

Creating Consignment Orders

You receive, for example, an ASN from your supplier. The inbound message informs you about the expected cargo and contains information about the external consignment. When your system receives the ASN, it creates a new consignment order including a

reference to the external consignment number that the supplier's system uses to identify the consignment. The relevant freight units are determined on the basis of search criteria and assigned to the new consignment order.

Freight Unit Determination (Inbound)

How does the system find existing freight units? First, the system checks if the ASN contains references to a purchase order or scheduling agreement.

If no purchase order or scheduling agreement reference is available for searching, the system searches for the freight units based on the following search criteria:

- A combination of material number and vendor
- Unloading point (optional additional search criteria, used if the result is not clear)

No Matching Freight Units Found

If the system does not find any matching freight units, the new consignment order is marked as erroneous but can still be saved in the freight order. The errors are displayed on the **Document Errors** tab and can be corrected by assigning a freight unit manually later. For more information, see [Correction of Errors in Business Documents](#).

Updating Consignment Orders

The ASN can also be used to update consignment orders. You can, for example:

- Update to correct item data like material master errors or reference errors
- Update with changes in deliveries, for example, new product items were added or removed
- Update product or package data

Creating or Updating Freight Orders (Information in Separate Inbound Messages)

You can create or update freight orders using inbound messages.

You receive, for example, an EDI message from your carrier with information about a freight order (bordero, bill of lading). When your system receives the EDI message, it creates a new freight order and assigns the corresponding consignment orders to the new freight order. The relevant consignment orders are determined on the basis of external references (such as external consignment number and supplier).

Freight Unit Determination (External Transportation Planning)

For external transportation planning, the system searches the freight units with the following search criteria: First, the system checks if the EDI message contains references to freight units, if not, the outbound delivery is taken as search criteria.

Fixing of Packages

For consignment orders that are created or updated via EDI message (ASN), it is necessary, during the inbound process, to prevent unintended changes to package information that might happen either manually during the check-in process at the gate or by package building.

Therefore, the system automatically sets the load plan status of the packages and the load plan item status of the product items to **Finalized**.

After the status has been set, the packages are fixed for further planning. This means that once a consignment is planned on a freight order, you can only change the packages if you manually reset the status of the respective items within the freight order or consignment order from **Finalized** to **Planned**. After changing (for example, updating data or adding new package items in the item hierarchy) you set the load plan status back to **Finalized**. The status of the predecessor documents (either consignment order or freight order) is also updated.

If an ASN message contains a package-related error (for example, a packaging material typo or a wrong packaging material) the system sets the load plan status of the package to **Invalidated**. You can correct the error on the **Document Errors** tab. Once the error is corrected the status of the packages and the package items is set to **Finalized**.

For more information about the check for package-related errors, see [Example Checks to Find Errors in Business Documents](#).

Handling of Updates by Inbound Messages After Manual Changes

When customizing a freight order or consignment order type, you can specify whether further updates of a business document by inbound messages are allowed after you have already made manual changes in that document.

For more information, see Customizing for Transportation Management under ► **Freight Order Management** ► **Freight Order** ► **Define Freight Order Types** ► or ► **Freight Order Management** ► **Consignment Order** ► **Define Consignment Order Types** ►.

Displaying External ID Numbers

You can display external ID numbers your business partner sent, such as:

- Number of external consignment (on tab **Subcontracting**)
- External delivery note number (Select a product item and navigate to **Document Reference** (field **External**)).
- External package ID

Related Information

[Automatic Consignment Building](#)

Creation of Packages Across Freight Unit Items

In consignment orders or freight orders you can create packages across freight unit items. The system splits and merges the freight units as needed, and updates the existing item hierarchy and the successor documents, that is the corresponding capacity documents.

Use

In the Advanced Shipping and Receiving inbound process you can assign product items from different freight units to the same package, for example, by packing them on a single pallet. In consignment orders this can be done manually on the UI or by inbound messages. For freight orders this can be done manually on the UI or by combined inbound messages (transport-based freight order). If possible, the system performs a freight unit merge for these packages that contain items across several freight units. Merging means, that all items of a freight unit are moved to another freight unit. The empty freight unit is then cancelled. In all other cases only single items are moved to another freight unit.

Example

Repacking might be necessary when a truck arrives at the gate and the load of the truck differs from the data in the system. The load has, for example, two product items (from different freight units) packed on one pallet. In the consignment order or freight order you can directly reassign product items to other packages to reflect the truck load. When you save, the system performs a freight unit merge: the two freight units are merged into one freight unit. Now empty freight units are canceled. Based on predefined criteria, such as a delivery being referenced to a freight unit, the system then determines which freight unit is to be kept after the merge. The existing item hierarchy in the business documents is updated accordingly.

The system does not support the building of packages across freight unit items with different deliveries (cross-delivery).

Prerequisites

You have activated the synchronization of predecessor documents. For more information, see [Synchronization of Predecessor Documents](#).

Example

Freight Unit Split and Mass Data Changes

In the inbound process of Advanced Shipping and Receiving in Transportation Management (TM), your supplier can send you external delivery note numbers with a consignment order using an advanced shipping notification (ASN). You can change the number in business documents, such as freight orders or consignment orders.

Use

The external delivery note number is the number that identifies the supplier's delivery note and groups all the items of this delivery note.

In the business document, the external delivery note number is displayed in the **Document References** tab page of a product or package. You can also see it in the details section of a freight unit.

All freight unit items in an inbound delivery must have the **same** external delivery note number. If you assign a freight unit to a capacity document, for example, a road freight order, after the execution at the stop in the capacity document has started or been completed, you must enter the same external delivery note number for the freight unit that was used for one of the other items of the road freight order. This is only relevant for road freight orders that have customs-relevant goods. For more information, see [Late Changes to Capacity Documents](#).

- If a consignment order is sent as ASN containing one or several freight units with different external delivery note numbers, the system automatically splits the freight unit. No further manual split is needed.
- If the external delivery note number of one or all freight unit items sent as ASN does not match the number provided in the documents of the carrier, you can change the numbers in the business document. A mass change of data is also possible.

The predecessor documents (such as a purchase order or scheduling agreement) are updated accordingly.

You can change the external delivery note numbers in the following ways:

- **Change Number Without FU Split**

In a business document, you can change the external delivery note number for a freight unit and its underlying items all at once. When you enter the new external delivery note number in the details section of the freight unit, the freight unit and its underlying items are updated with the new number.

- **Change Number With FU Split**

If one or several of the external delivery note numbers differ, you can change them for those freight unit items by splitting the freight unit. The system splits the original freight unit and creates a new one with the items that have the new external delivery note number. It also creates a new inbound delivery in the hierarchy.

Example

In a freight unit, only 50 product items of 100 have the same external delivery note number as provided in the documents of the carrier. On the **Items** tab page of the business document, you can select the items with a different number and initiate a manual freight unit split for them. After entering the new external delivery number, you then have two different freight units and inbound deliveries in the hierarchy.

Correction of Errors in Business Documents

You can use this function to display errors in freight orders and consignment orders that were created based on incorrect EDI messages in Transportation Management (TM). You can then correct the errors manually.

Prerequisites

In Customizing for the freight order type or consignment order type, you have selected the **Enable Saving of Documents with Errors** value in the **Handling of Incorrect Inbound Messages** field. For more information, see the following Customizing activities in Customizing for **Transportation Management**:

- [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) >
- [Freight Order Management](#) > [Consignment Order](#) > [Define Consignment Order Types](#) >
- [Freight Order Management](#) > [Freight Booking](#) > [Define Freight Booking Types](#) >

i Note

The system determines whether it creates a document with errors depending on the error. For some errors, documents are still not created despite your Customizing setting.

Features

In Transportation Management (TM), you can create freight orders or consignment orders automatically based on inbound messages (such as advanced shipping notifications or transfers from an external TM system). However, these messages can contain errors. As a result of some of the possible errors, the documents cannot be created in TM.

If you have activated the Customizing setting described above and the errors belong to the following error categories, the system creates the documents despite the errors contained in the messages:

- Master data error
- Reference error (references to predecessor documents)

You can then display and correct the errors on the **Document Errors** tab page.

You can use the Business Add-In (BAI) **BAI: Addition of Categories for Error Handling** to add your own error categories to the Transportation Order (TO) business object in addition to the standard error categories. For more information, see Customizing for **Transportation Management** under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Freight Order Management](#) > [Freight Order Management](#) > [BAI: Addition of Categories for Error Handling](#) >.

Correcting Master Data Errors

Errors can occur for the following master data:

- Products
- Resources
- Locations
- Business partners

The errors are often related to IDs of products or other master data that are not known in the TM system. This could be because the product has not yet been created, for example. The ID may also have been spelled incorrectly. You can correct both types of error on the **Document Errors** tab page. You can do this as follows:

- A product does not (yet) exist:

You choose the pushbutton for an action next to the description of the error. If the product has been created in the meantime, the error is corrected automatically.

- The product ID is incorrect:

- You can enter the correct ID directly in the **Correct Value** column. A search help is also available.
- You can first choose **Error Details**. The system selects the incorrect products on the **Items** tab page. You can also enter the correct values here.

This same behavior also applies to locations, business partners, and so on. In this case, you can enter the correct values in the relevant places in the document, such as on the **Business Partner** tab page.

i Note

A help text for the error message is also displayed when you choose **Error Details**. It explains how you can correct the master data error.

If you have activated the function of intelligent error resolution in Customizing for Transportation Management, the system provides the user with suggestions on how to solve errors based on previous solutions. For more information, see [Intelligent Resolution of Document Errors](#).

Correcting Reference Errors

Document errors can also relate to references to predecessor documents, such as freight units, that are not known in the TM system. To correct these errors, you can choose the pushbutton for an action next to the description of the error. A dialog box appears in which you can, for example, select existing freight units.

Automatic Correction of Document Errors

You can use the report /SCMTMS/TOR_RETRY_ERRORS to automatically resolve document errors in freight orders so that you do not have to resolve them manually in each freight order individually. For example, you can resolve reference errors to an unknown consignment order that is now available in the system.

We recommend that you run the report as a background job using a variant. However, you can also use the report UI to run it.

For more information, see the system documentation for the report.

Automatic Correction of Reference Errors in Related Documents

If you correct errors, for example, on the **Document Errors** tab page of a consignment order or in the product item itself, the system tries again to resolve reference errors in the related freight order. It checks if reference errors for an unknown consignment

order can be resolved. If the consignment order is now available in the system, the missing reference is automatically added in the freight order.

Blocking the Document

If the document contains errors, the system automatically sets planning, execution, and invoicing blocks for the document. These blocks are displayed on the **Block Information** tab page with the corresponding block reasons.

As soon as you have corrected the errors as described above, the system automatically removes the blocks.

For more information, see [Blocking Information of Freight Units and Capacity Documents](#).

Related Information

[Freight Order](#)

[Consignment Order](#)

[Example Checks to Find Errors in Business Documents](#)

Example Checks to Find Errors in Business Documents

The system runs various checks when freight orders or consignment orders are created or changed either manually or by an inbound message using EDI (for freight orders) or advanced shipping notification (ASN) (for consignment orders). Some checks are also run for freight orders when events are updated by transportation event bulk notifications during execution.

Overdelivery of Quantity

In the inbound process the system checks for an overdelivery of quantity when the system creates or updates a consignment order on the basis of an inbound message or when you manually change quantities in a consignment order or freight order.

Prerequisites

In the Advanced Shipping and Receiving inbound process, you must enable the setting for **Enable Synchronization of Predecessor Documents** in the Customizing for freight order type and consignment order type to be able to update the item quantity in the document. For more information, see Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types** or **Freight Order Management** > **Consignment Order** > **Define Consignment Order Types**. For more information about the Advanced Shipping and Receiving process, see [Advanced Shipping and Receiving](#).

Features

For purchasing documents (such as purchasing orders or scheduling agreements) you can define a quantity tolerance for a product item, a percentage by which the quantity can be exceeded. You can also specify that there is no limit on overdelivery quantities (**Unlimited**).

You maintain these settings in the purchase order or scheduling agreement on the **Delivery** tab page.

For the quantity calculation, the system considers the product item quantity of the inbound message (or a manually entered value), the open quantity, the tolerance, and the unlimited flag:

- If the open quantity of an item plus the sent or manually entered quantity is below or within the tolerance, the system creates the consignment order without an error message. "Open quantity" means quantity not yet assigned by a freight unit to a consignment order or quantity already executed within a freight order.

- If the open quantity of an item plus the sent or entered quantity exceeds the tolerance, the system creates the consignment order despite the errors and saves the respective item locally. You can display and resolve the errors on the Document Errors tab page. When the error is resolved (by changing the purchasing document or the quantity of the item in the freight order or consignment order) the system automatically assigns the corresponding freight units to the consignment order.
- If **unlimited overdelivery** is defined in the related purchase order or scheduling agreement, the system creates the consignment order without an error message. The system adjusts the quantity in the latest freight unit and assigns it to the consignment order.

If no further freight unit exists, the system creates an error message and you need to adapt the purchasing document. For scheduling agreement you need to create more call-offs. The system then generates more related freight units due to the automatic freight unit building.

How to Resolve Errors?

- Send a new inbound message (advanced shipping notification, ASN) to update the quantity with a quantity within the tolerance.
- Enter a new item quantity within the overdelivery tolerance limit on the **Document Errors** tab page in the business document.
- Increase the quantity of the ordered items or increase the overdelivery tolerance limit in the purchase order or scheduling agreement.

Vendor, Goods Supplier

Features

When an inbound message (ASN) arrives to update a consignment order, the system checks the sent vendor. It checks if the vendor of the ASN matches the vendor of the referenced purchasing document (purchase order or scheduling agreement).

How to Resolve Errors?

If the vendor does not match, the consignment order is created with a document error.

- You can then correct the vendor information in the purchase order or consignment order.
- Alternatively, you can copy the vendor information from the predecessor document, or enter a new purchasing document that contains the correct vendor information.
- Or, you decide to save the new **goods supplier** sent with ASN, for example, if the original supplier cannot deliver. The stage in the related consignment order and freight units is then adjusted accordingly.

Package-Related Errors

When a consignment order is created or updated either manually or with an inbound message (ASN), the system checks for package-related errors.

Prerequisites

You have enabled the **packaging material check** for item types. For more information, see Customizing for Transportation Management under ► **Freight Order Management** ► **Define Item Types for Freight Order Management** ►.

Features

If errors are found, the consignment order is saved with a document error. Packages with errors are saved locally in the consignment order. If one of the packages in a nested package has an error, the system also saves the top-level package locally, though the top-level package doesn't contain errors.

If, for example, one of the packages in a nested package contains the wrong packaging material, the system saves the top-level package locally though the top-level package doesn't contain errors. You can display and resolve the error on the **Document Errors** tab page.

Examples for package-related errors:

- The package information contains a "real" product instead of a packaging material.
- It contains an auxiliary packaging material instead of a packaging material.
- It contains a packaging material instead of an auxiliary packaging material.
- The package information is empty or has a typo.

How to Resolve Errors?

You can display and correct the errors on the Document Errors tab page of the consignment order.

If an ASN message contains a package-related error (for example, a packaging material typo or a wrong packaging material) the system sets the load plan status of the package to **Invalidated**. Once the error is corrected, the status of the packages and the package items is set to **Finalized**.

For more information about the load plan status and the fixation of packages against changes, see [Creation or Update of Consignment or Freight Orders on Inbound Messages](#).

Customs Reference Number

Features

In the inbound process, the system checks the customs reference number that you receive from the carrier as part of the freight order with assigned consignment orders, or that you enter in the business documents. This number is displayed in the **Previous Document** field. If the format of the check digit in the customs reference number is not in compliance with the ISO standard 6346, the system creates a document error on the **Document Errors** tab page in the freight document.

How to Resolve the Errors?

You can resolve the error by providing the correct customs reference number of the previous document. You can do so in the following ways:

- In the **Items** details for the consignment order, on the **Customs** tab page, you can enter the correct customs reference number in the **Previous Document** field.
- You can enter the correct number on the **Document Errors** tab page in the business document.
- If the customs reference number was sent using an inbound message, you can request a new version of the inbound message with the correct customs reference number.

For more information about the customs processes, see [Global Trade](#).

Delivery-Related Errors

Features

Certain process steps, such as an update from a TM freight order to an LE delivery, or the posting of goods movement to an LE or EWM delivery are asynchronous. For example, users in different areas may attempt to update the same delivery at the same time or post goods movements before warehouse processing is possible. This can cause errors, for example locking of business documents, inconsistent data in business documents, or failed updates to business documents.

You can use the **Document Errors** tab page to view details of errors. The following are some examples of possible errors when working with the freight order:

- Goods Movement Posting

If a TM user attempts to post goods movement on the freight order before the warehouse processing status of the EWM or LE delivery is set to **Ready for Warehouse Processing** the system cannot post the goods movement.

- Freight Unit Building

If a TM user changes a freight order to update, for example, date information, while at the same time a user in LE makes a change to the same delivery the system cannot make both of the updates. One of the changes has to be prioritized. The system determines which update should be prioritized based on the warehouse processing status. During planning, the system gives priority to updates from TM. Once transportation planning is completed and the warehouse processing status is set to **Ready for Warehouse Processing**, warehouse tasks can begin, so an update from LE takes priority.

- Delivery update from TM to LE

For example, a user in TM makes a change to the freight order and saves it but the system is not able to update the information in the LE delivery due to a technical fault.

- Delivery Update from EWM to TM

For example, a user in EWM updates a delivery but the system is not able to update the information in the TM system due to a technical fault.

In all cases the system records the error on the **Document Errors** tab page with additional details from the application log.

Estimated Time of Arrival or Departure

Features

If an inbound transportation event bulk notification updates the planned arrival or departure dates in a freight order or freight booking, the system creates a corresponding event **Estimated Time of Arrival** or **Estimated Time of Departure**.

In a freight booking, the system takes over the updated times because there is no scheduling.

In a freight order, the system performs scheduling to check the updated dates and times within the document and then saves the freight order with the updated times. If the time conflict can't be resolved using scheduling, the system creates document errors. For example, the event **Estimated Time of Departure** is reported for the source location, but the departure time from the source location of the freight order is later than the arrival time at the destination location of that freight order. The system performs scheduling but the time conflict is not resolved. The system then creates a document error.

The planned arrival or departure dates can be updated several times with transportation event bulk notifications. You can also see the original planned dates and time.

How to Resolve the Errors?

You can resolve the document error in the following ways:

- You can correct the planned departure or arrival times manually to remove the time conflict.

- You can send another transportation event bulk notification with the correct dates and times.

Related Information

[Correction of Errors in Business Documents](#)

Synchronization of Predecessor Documents

In the Advanced Shipping and Receiving **inbound** process you can enable the synchronization of predecessor documents referenced in a consignment order or freight order. You enable this setting, for example, if you want to compare the transportation documents of the truck with the system entries at the gate.

Prerequisites

For more information, see Customizing for [Transportation Management](#) under [Integration > Settings for Advanced Shipping and Receiving](#).

Check-In Process at the Gate

When a truck arrives at the gate you can compare the transportation documents of the truck with the freight order in the system. If the load of the truck differs from the freight order data in the system, you can directly change discrepancies in the freight order. As an example, changing the data is necessary, if the truck contains more, less, or other cargo than expected.

You can change the item structures in the freight orders or consignment orders until the [Ready for Warehouse Processing](#) status is set. The status is relevant for changes that affect the warehouse process. After the [Ready for Warehouse Processing](#) status is set, you can still change data that does not affect the warehouse process.

The system can automatically update the referenced predecessor documents of the consignment order or freight order and keep them up-to-date.

For a freight order or consignment order you can change data such as:

- Add unplanned freight units or remove freight units. The systems unassigns the removed freight units.
- Change goods supplier in consignment order. The goods supplier and its source location are then changed accordingly in the corresponding freight unit.
- Change fields related to unit of measure
- Change product item quantity (decrease or increase quantity)

If you decrease the quantity, the freight unit is split and a new freight unit with the leftover-quantity is created.

- Add or delete product items
- Add or delete package items
- Change or delete package data
- Change nested packages (package hierarchy)
- Change packaging material

This might be necessary if a different packaging material was used. After the change, the amount of products stays the same and is not recalculated automatically. The package hierarchy above or below remains unchanged.

- Change auxiliary packaging material
- Add or delete a delivery note item, change delivery note item fields (which results in add or delete freight units or product items)
- Change fields related to Materials Management (MM), such as:
 - Number of handling unit (package ID)
 - Serial number
 - Batch-related data

Related Information

[Correction of Errors in Business Documents](#)

[Example Checks to Find Errors in Business Documents](#)

Use of Auxiliary Packaging Materials

You can also use auxiliary packaging materials in the business documents of Transportation Management (TM), such as freight orders and consignment orders. Auxiliary packaging materials include lining in cartons, foil, protective paper, and pallets used to secure the load, for example.

Prerequisites

The packaging material type **Auxiliary Packaging Material** has been defined in the material master data. You have made the following settings in Customizing for Transportation Management to be able to use this packaging material type in TM:

- You have assigned a packaging material category to the packaging material type in the product master data. Only the packaging material categories **Packaging Materials** (C) and **Auxiliary Packaging Material** (D) can be assigned to auxiliary packaging materials.

For more information, see Customizing for Transportation Management under ► **Master Data** ► **Product** ► **Define Packaging Material Types** ►.

- You have defined an item type and assigned the item category **Auxiliary Packaging Material** to it. In the item type, you can also specify that for items in the business document the system checks whether you have entered a valid packaging material or auxiliary packaging material. You can then define this item type as an allowed item type for your freight order type or consignment order type.

For more information, see Customizing for Transportation Management under ► **Freight Order Management** ► **Define Item Types for Freight Order Management** ►.

- If you have defined your own item hierarchy in Customizing, you have added the auxiliary packaging material to this hierarchy under **Hierarchy Levels**. The auxiliary packaging material is already contained in the standard item hierarchy **All Items** (ITEM).

For more information, see Customizing for Transportation Management under ► **Freight Order Management** ► **Define Hierarchical Views for Business Documents** ►.

Features

You can add auxiliary packaging materials to the item hierarchy of a business document manually or transfer these automatically using EDI messages, such as an advanced shipping notification. In addition, items for auxiliary packaging materials can be

automatically created under package items during package building. Note, however, that auxiliary packaging materials must always be created as the lowest item in an item hierarchy, meaning that no further items may be assigned below the auxiliary packaging material. Since auxiliary packaging materials do not have their own capacity, they cannot contain any further items.

If the auxiliary packaging material is a main item, it is considered during load planning.

The packaging material type and packaging material category are displayed in the detailed information for the auxiliary packaging material items in the business documents.

Auxiliary packaging materials are also taken into account when integrating TM with Extended Warehouse Management (EWM). When EWM deliveries are created by the system, the related handling units also contain the auxiliary packaging materials. For more information, see [Advanced Shipping and Receiving](#).

Related Information

[Items](#)

[Defining Auxiliary Packaging Materials](#)

Delivery Split

In Advanced Shipping and Receiving, you can split deliveries during the planning, warehouse execution, and loading stages of an outbound process. Delivery splits influence the related freight units (FU).

From Transportation Management (TM), you can trigger a split while assigning FUs to a freight order (FO) and during the loading process, while in Extended Warehouse Management (EWM), a split can be triggered during warehouse processing.

Split Types

When you split a delivery, the system creates a new delivery and assigns the relevant FUs to it. If you split a delivery that contains just one FU, the system also splits the FU.

The activities in the process are as follows:

- Planning Activities

During planning you can change assignments of FUs, for example assign a FU from an existing delivery to a different FO. If you make such a change, a new delivery is required for the new FO.

- Warehouse Activities

You can split delivery items or handling units (HUs) from an outbound delivery order into a new order during warehouse execution. The corresponding FUs in TM are also split or reassigned to the new outbound delivery based on the requirement. For example, if the items in a split outbound delivery order correspond to the items in the FU, the FU would be reassigned to the new outbound delivery order. In other cases, where the FU contains other items which are not in the new ODO, the corresponding FU must be split and a new FU is assigned to the new outbound delivery order. Newly created or reassigned FUs remain unplanned even when the initial FU had been planned before the split occurs.

- Loading Activities

While loading the truck, it may also be necessary to split deliveries, for example, if the truck can't accommodate all of the HUs assigned to the delivery on the FO. To report the loading of a FO, you can select the HUs that have been loaded and change the **Execution Status** to **Set Items to Loaded**.

Once you set the **Execution Status** of the stop to **Loading Completed**, the system removes from the FO any HU that has not been loaded, then creates the following:

- a new FU to which the HU is assigned
- a new delivery

If you have automated the status reporting for items or packages, you must trigger the split from EWM before item loading is reported.

Delivery Deletion

You can delete LE inbound and outbound deliveries from Transportation Management (TM) in SAP S/4HANA.

You can delete deliveries from Delivery Management (LE-SHP) in embedded scenarios. The corresponding freight units (FU) must have been created out of a delivery and must not have been assigned to a freight order.

In addition to standard orders, you can delete deliveries linked to customer and supplier returns and 2-step stock transport orders (STO).

i Note

- Side-by-side setups are not supported
- In scenarios involving EWM outbound delivery orders and EWM TU-based deliveries, deleting LE deliveries is not supported
- In scenarios involving EWM inbound deliveries, you can delete deliveries whose status is not yet set to **Ready for Warehouse Processing**

Example

You create inbound deliveries based on an external delivery note number. Shortly after, the truck arrives at the gate, and the team unloads the goods. During their administrative cargo processing, the team discovers a missing external delivery note, which has also been removed from the consignment order in the system.

Supposing the supplier cannot deliver the note within a certain time horizon, or the new delivery date is undetermined, you open your **Freight Units – Worklist** app from the Launchpad and delete the inbound deliveries created based on the note.

i Note

If there are two or more FUs in a delivery, you must select all FUs in a delivery.

Related Information

[Advanced Shipping and Receiving](#)

[End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#)

[Overview of Features in Advanced Shipping and Receiving](#)

[Prerequisites for Advanced Shipping and Receiving](#)

Delivery Updates

You can update specific header and item level fields in Logistics Execution (LE) outbound deliveries that are relevant for Transportation Management (TM). The fields can also be updated in integration scenarios involving Extended Warehouse

Management (EWM) after the deliveries have been transferred to EWM and freight units (FU) have been created in TM.

The updates are possible up to certain points in the process and depend on the specific scenario.

When updating a delivery, you must consider at what stage of a warehouse- or transport-driven process you find yourself in. Secondly, whether you use an LE-managed warehouse or an EWM-managed warehouse, you must consider the warehouse execution (LE) or processing (EWM) status.

Integration with LE-Managed Warehouses

Transport- and Warehouse-Driven Processes

The following fields can be updated and changed. The updates can take place before the process reaches the status **Ready for Warehouse Execution** in transport-driven scenarios; or while in status **In Warehouse Execution**, but no warehouse task has been created and the FUs have not yet been assigned to a freight document in warehouse-driven scenarios.

Header Level

- Delivery Date
- Shipping Type
- Incoterm Version
- Incoterm 1
- Incoterm 2
- Incoterm Location 1
- Incoterm Location 2

Item Level

- Delivery Quantity
- Additional items

You can also add new items to your outbound delivery.

All Processes and Statuses

The following fields can be updated at any point in time regardless of the execution status or scenario:

Header Level

- Texts
- Delivery Priority

Item Level

- Quantity (only increase or decrease; no freight unit building (FUB) applied)
- Batch Number
- Texts
- Material Freight Group

- Customer Material Number
- Material Description

Integration with EWM-Managed Warehouses

Transport- and Warehouse-Driven Processes

The following fields can be updated and changes made after distribution. The updates can take place before the process reaches the status **Ready for Warehouse Execution** in transport-driven scenarios; or while in status **In Warehouse Execution**, but no warehouse task has been created and the FUs have not yet been assigned to a freight document in warehouse-driven scenarios.

Header Level

- Delivery Date
- Shipping Type
- Delivery Priority
- Incoterm 1
- Incoterm 2

Item Level

- Quantity
- Additional Items

All Processes and Statuses

The following fields can be updated at any point in time regardless of the execution status or scenario:

Header Level

- Texts

Item Level

- Texts
- Material Freight Group
- Customer Material Number
- Material Description

Related Information

[Delivery Updates in TM Triggered by Logistics Execution](#)

Subcontracting of Purchase Order

In the subcontracting process, a company outsources certain stages in the manufacturing process to subcontractors.

The component parts of the final product are transported to the subcontractor, where the next step of the manufacturing process takes place. After, the subcontractor returns the finished products to the original manufacturer. In a subcontracting process where the subcontractor is in another country, compliance check and customs clearance must be considered during the transportation process.

The subcontracting purchase order contains the final product and its components, which are listed in a bill of material (BOM) with item category **L** for subcontracting. The BOM is used to define the relationship between one component and several subcomponents and contains information such as the ratio of the component to the subcomponents.

Sample Process

The following is an example of how a subcontracting process might develop from the perspective of the manufacturer:

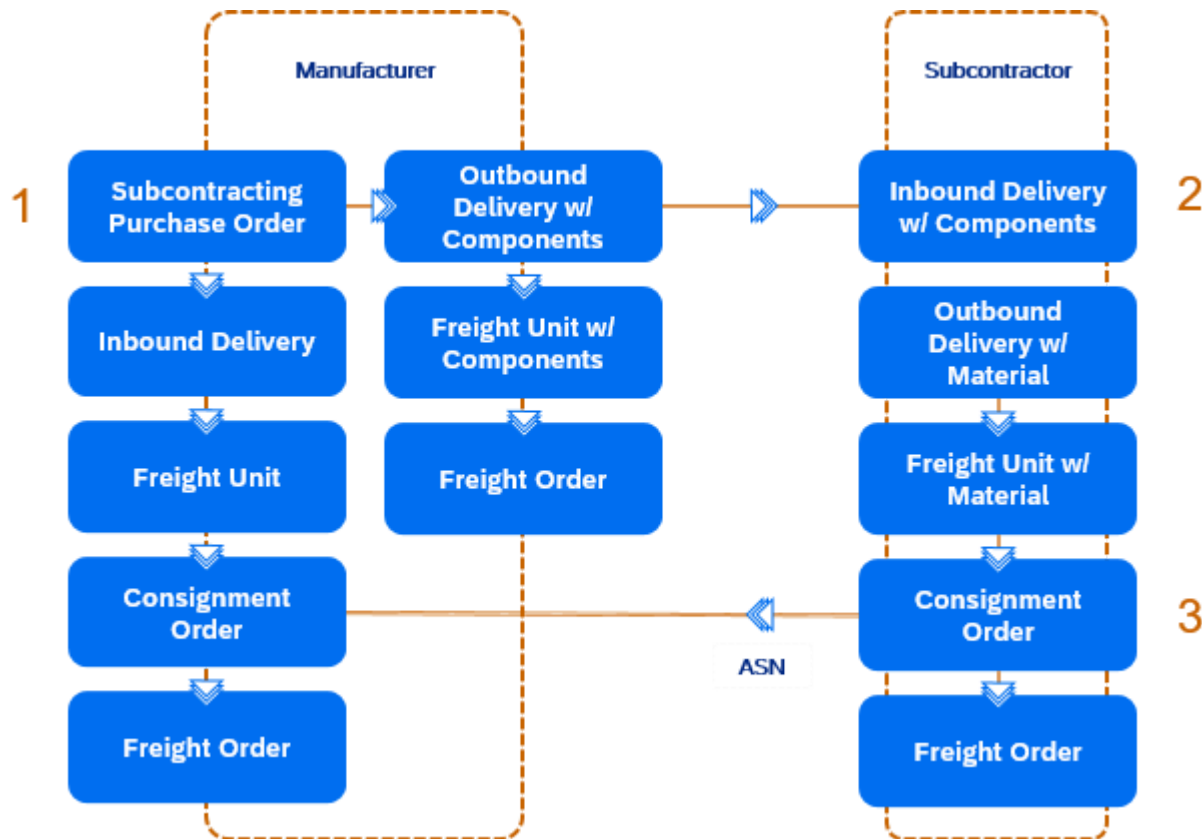


Figure 1: The Subcontracting of a Purchase Order

1. Creating a purchase order

The manufacturer creates a subcontracting purchase order or scheduling agreement with item type **L**.

→ Remember

Run transaction ME21N to create a subcontracting purchase order.

2. Creating an outbound delivery

They run transaction ME20N to open the **Subcontracting Cockpit**. They enter the supplier and plant to find the subcontracting purchase order and create the outbound delivery.

Integration between Transportation Management (TM) and Materials Management (MM) triggers the creation of a freight unit for the expected finished material in TM. This freight unit, which also lists the subcomponent, is relevant for the later inbound process from the subcontractor.

The manufacturer assigns the freight unit to a freight order.

In a subcontracting process where the subcontractor is located across borders, compliance and customs must first be checked during the transportation process.

3. Creating an inbound delivery

Finally, the subcontractor uses the consignment order to send an ASN to the manufacturer.

The manufacturer relies on the ASN from the subcontractor to create a consignment order for the inbound delivery. The consignment order contains the freight unit created in step 1, which is now used to conduct the inbound process and the subsequent unloading activities.

Related Information

[Integration of Orders](#)

[Configuring Integration of Orders and Deliveries](#)

Ocean Freight Bookings for Shippers (Outbound)

With Transportation Management (TM) you can manage multimodal transportation scenarios with multiple transportation stages and different transportation modes. This topic provides details for shippers on outbound transportation with ocean freight. The scenario is designed for Advanced Shipping and Receiving.

You can model end-to-end scenarios with container units and the message exchange between you, as the shipper, and the carrier.

For a high-level sample, see [Sample Ocean Freight Process for Shippers \(Outbound\)](#).

Prerequisites

- If you want to save freight bookings that contain errors, you have activated this function for the ocean freight booking types you want to use. You do so in Customizing for Transportation Management under **Freight Order Management > Freight Booking > Define Freight Booking Types**. In section **Integration Settings**, choose **Enable saving of documents with errors** for dropdown box **Handling of incorrect inbound messages**.

The activation is only valid for freight bookings you create **after** the activation.

- The usage of **container units** is mandatory for the ocean freight process for shippers and doesn't support local container items.
- The process supports full container load (FCL) cargo. Less than container load (LCL) cargo is not supported.
- When using Advanced Shipping and Receiving with Extended Warehouse Management for outbound processes, you must create a pick-up freight order for **each** container unit. The system requires these freight orders for the loading process.
- The system can handle **cargo that was left behind** at the port. To use this function, you have activated the **Itm Load. Stat. Port** checkbox for the event profile and assigned it to the freight booking type you use. You do so in Customizing for Transportation Management under **Freight Order Management > Execution Settings > Define Event Profiles**.

For more general information, see [Cargo Left Behind at Port/Airport](#).

- When creating container units, the system can automatically check if the container registration number for the container is correct according to ISO 6346. To use the check, configuration is necessary. For more information, see section "container units".
- To communicate via EDI with your carrier, you need to configure the for sending of container update messages to the (pre-carriage) carrier. For more information, see the corresponding section below.

Ocean Freight Bookings

The ocean freight scenario for shippers supports door-to-door (D2D) and door-to-port (D2P) shipping. As usual, you use an ocean freight booking to reserve freight space on a vessel.

You can receive the confirmation for the oceanfreight booking from the carrier using EDI communication. Alternatively, you enter the confirmation manually in the ocean freight booking by setting the status to [Set to Confirmed by Carrier](#).

Container Units

You create container units and assign freight units. The freight units can originate from sales orders or outbound deliveries. After creating the container units, you assign them to an ocean freight booking. Finally, you create pick-up freight orders (pick-up FOs). In case you need to change the assignment of container units to pick-up or delivery freight orders, you can also assign or unassign the container units in the freight booking.

When creating container units, the system can **automatically check** if the container registration number for the container is correct according to ISO 6346. The system runs the check when you enter the container registration number **together** with an equipment type. To use the check, you need to specify the equipment type in Customizing for Transportation Management under [►Master Data ► Resources ► General Settings ► Define Equipment Groups and Equipment Types ►](#). Set the resource class for the container group to [10 Container](#).

You plan the container units on the ocean freight booking. To assign container units, you can search for the containers using their registration numbers.

On the [Booking Data](#) tab in the container unit, you can display the booking information.

Pick-Up Freight Orders and Delivery Freight Orders

When you create a pick-up FO for the pre-carriage or a delivery FO for the on-carriage, you can assign **one** or **several** container units to it.

Define Transportation Mode

On the [Stages](#) tab of the freight order, you can display the transportation mode column.

When a pick-up FO or delivery FO is created out of an ocean freight booking, the system considers a couple of settings to find out the transportation mode that is to be used. You configure if the system creates a **road** freight order or a **rail** freight order for the pre-carriage or on-carriage. You can define freight order default types in the **freight order type** or **freight booking type** in Customizing.

You have several setting options. Here are two examples:

- The transportation mode you need varies from time to time. Sometimes you need a road freight order, sometimes a rail freight order. You want to be flexible and easily adjust the transportation mode to your needs.

To achieve this, you configure a freight order default type for pick-up FOs or delivery FOs in section [Default Types](#) in Customizing for Transportation Management under [►Freight Order Management ► Define Freight Booking Types ►](#).

In the pre- or on-carriage **stage** of the ocean freight booking, you have then the option to **adjust** the default transportation mode.

In addition, you can define freight order default types in the **freight order type** Customizing or in the **freight booking type** Customizing. The system uses these FO default types when creating a pick-up FO or delivery FO for a freight booking.

- The transportation mode you need is always the same.

To achieve this, you define a freight order type for road (and one for rail) that is defined as **Default FO Type for FB**. You do so in section **Default Types** in Customizing for Transportation Management under **Freight Order Management** **Define Freight Order Types**.

Loading Cargo

You have several options to set the loading status in the system:

- In the **Load or Unload Freight Orders** app
- In the **Edit Road Freight Order** app or **Edit Rail Freight Order** App
- Via Radio Frequency (RF)

As **storage location**, you can use an inventory-managed (IM) storage location for Lean Warehouse Management or a storage location for Extended Warehouse Management (EWM).

Consignment Orders

You can create consignment orders for pick-up FOs associated with an ocean freight booking. The consignment orders include booking information from the assigned ocean booking, for example, vessel, voyage, IMO Ship ID, UN/LOCODE, expected arrival date and time at the port of destination or at the customer (depending on the scenario in use, that is door-to-door or door-to-port).

Automatic or Manual Creation of Consignments

For the creation, you can choose to manually create consignment orders, or you can use the automatic consignment building. For more information, see [Automatic Consignment Building](#).

You manually create consignment orders in the pick-up FO under **Follow up** **Build Local Consignment Orders** or **Follow up** **Create Consignment Order**. For more information about consignment orders in general, see [Consignment Order](#).

You can display the booking data on the **Booking Data** tab in the consignment order.

i Note

When consignment orders are created in the ocean freight process, the container units on the pick-up FO must contain only a single container item.

Additional Infos About Ocean Freight Bookings

Port of Destination and Incoterms

When the agreed Incoterm location with the customer is “port”, use a door-to-port freight booking. The door-to-port scenario has some impacts:

- Depending on the settings in the logistics integration profile the Incoterm is considered during freight unit building. You can, for example, configure the freight units to have a planning-relevant stage from the shipper’s location to the port of destination and a statistical stage (not in the shipper’s responsibility) from port of destination to the consignee.
- The consignment order will still have one stage from the shipper’s location to the port of destination. The planned arrival date refers to the date at the port of destination.

You can set the port of destination to **Unloaded**. The system prevents setting a container unit to **Unloaded** in the ocean freight booking if the container unit has a document error. Once the error is resolved by setting the container unit to **Loaded**, the system

automatically updates the unloading status for the container unit.

Handle Containers Left Behind at the Port

The ocean freight process can handle containers that were left behind at the port of loading. For more information, see [Cargo Left Behind at Port/Airport](#).

Invoicing

The invoicing process starts only after the ocean vessel's departure. It also requires that there are no pending decisions for any containers left behind. For more information, see [Carrier Invoice](#).

Message Exchange Between Shipper and Carrier

The ocean freight scenario offers an end-to-end process with container units and message exchange between you and the carrier.

Sending Container Units to (Pre-Carriage) Carrier

The container message is sent to the carrier when **all** the following conditions are met:

- When the container unit (CU) is assigned to an ocean freight booking and important information such as steamship line booking number, vessel ID, and container registration number is available.
- When goods movement is posted and the status changes to **Completely Processed**.
- When loading is completed in the pick-up FO.

i Note

If multiple container units are assigned to a freight order, the system can only send the respective container message to the carrier after **all** containers have been loaded.

Settings for Sending CU-Update Messages to (Pre-Carriage) Carrier

The following settings are necessary to automatically send EDI messages to the (pre-carriage) carrier:

- You have configured the Post Processing Framework (PPF) in Customizing for [Cross-Application Components](#) under [Processes and Tools for Enterprise Applications](#) > [Reusable Objects and Functions for BOPF Environment](#) > [PPF Adapter for Output Management](#) > [Maintain PPF Settings](#).

In application [/SCMTMS/TRANSPORTATION](#) you activated the action definition [TOR_TPNOR_NTF \(Send Notification\)](#) of action profile [/SCMTMS/TOR](#).

- In addition, you have assigned the output profile [/SCMTMS/TOR](#) to the container unit type you use. You assigned the output profile in Customizing for Transportation Management under [Freight Order Management](#) > [Transportation Unit](#) > [Define Transportation Unit Types](#) in Section [Output Options](#).

With the following status changes the systems sends a notification to the carrier:

- Execution status changes to [Ready for Transportation Execution](#)
- Execution status changes to [In execution](#)
- Departure event is posted
- Checkout event is posted

If the status change takes place in the **freight booking**, a `TransportationOrderGenericRequest_Out` message (with business scope "Notification", categoryCode "BO" (booking)) is sent to the **main-carriage carrier**.

If the status change takes place in the **pick-up FO**, a `TransportationOrderGenericRequest_Out` message (with business scope "Notification", categoryCode "TO" (freight order)) is sent to the **pre-carriage carrier**.

For companies that store containers in a local container depot, it might be useful to send an EDI message when the container is loaded to inform the **pre-carriage carrier** about the necessary pick-up of the container.

Receiving Freight Booking Updates from the Carrier

i Note

The EDI messages you receive from the carrier must contain certain information. For some XML samples, see SAP note [3514066](#) .

The carrier can send you an freight booking update using EDI communication. You can, for example, receive the house bill of lading as well as updates for the departure dates from the port of loading, the containers that were loaded, the expected time of arrival, or change the port of destination in a door-to-door scenario. The system supports updates for freight bookings and for individual container units within a freight booking.

Based on this inbound message, the system updates the freight booking with the received information. The message can include information, such as vessel, voyage number, house bill of lading, expected time of arrival, general booking information like freight booking reference (that is freight booking ID in the shipper's system), freight booking type and document category (that is the business object for freight booking), item information like item category and container registration number. To identify the location, the message can contain a UN/LOCODE.

i Note

A **separate** message for **each** container can be sent (single container update messages).

Alternatively, you can manually enter updates in the freight booking.

Identifying Locations by UN/LOCODES

The system finds locations using the UN/LOCODES that were sent with an inbound message. If the handling of errors for business documents is activated for the freight booking type, the system displays document errors if the UN/LOCODE couldn't be found, isn't unique or is wrong.

To enable this check, you need to activate the handling of errors for business documents in Customizing for the ocean freight booking type.

Update Port of Destination

The port of destination can be updated by inbound messages.

When you change the port of an ocean freight booking manually in a **door-to-door scenario**, the system checks if a delivery freight order already exists. You need to confirm that the corresponding location in the delivery freight order can be changed too.

In some situations, such as weather conditions, breakdowns, or repairs, it might be necessary to change the port of discharge for an ocean freight booking. This allows the shipper to make necessary adjustments in the system and to inform involved parties.

Changing the port of destination is allowed in the following cases:

- The delivery freight order is not in execution.

- The delivery freight order is not assigned to multiple freight bookings. This might be the case if a truck picks up deliveries of two different vessels.

Sending ASN to Customer

In the following cases, the system sends a **separate** advanced shipping notification (ASN) message for **each** consignment order to the customer when the vessel departed. At this point in time, additional booking information is available and can be included in the ASN to the customer.

- When goods movement status on consignment order is set to **Completely Processed**.
- When the ocean freight booking is set to **Departed From Port of Loading**.
- Alternatively, when manually requested in the consignment order under **Follow Up > Send Advanced Shipping Notification**.

Especially in a door-to-port scenario, consignments, and their updates are useful for customers. Depending on the Incoterms, they might need to know the voyage number, and the expected date and time of arrival to organize the pickup and transport of the goods for the last stage.

Related Information

[Sample Ocean Freight Process for Shippers \(Outbound\)](#)

[Consignment Order](#)

[Cargo Left Behind at Port/Airport](#)

[Air Freight Bookings for Shippers \(Outbound\)](#)

[Sample Air Freight Process for Shippers \(Outbound\)](#)

[Advanced Shipping and Receiving](#)

Air Freight Bookings for Shippers (Outbound)

With Transportation Management (TM) you can manage **multi-modal** transportation scenarios with multiple transportation stages and different transportation modes. This topic provides details for shippers on outbound transportation with air freight. The scenario is designed for Advanced Shipping and Receiving.

You can model an end-to-end process with the message exchange between you, as the shipper, and the carrier in which consignment orders play an essential part.

For a high-level sample process description, see [Sample Air Freight Process for Shippers \(Outbound\)](#).

Prerequisites

- If you want to save freight bookings that contain errors, you must activate this function for the air freight booking types you want to use. You do so in Customizing for Transportation Management under **Freight Order Management > Freight Booking > Define Freight Booking Types**. In section **Integration Settings**, choose **Enable saving of documents with errors** for dropdown box **Handling of incorrect inbound messages**.

The activation is only valid for freight bookings you create **after** the activation.

- When you decide to work with local consignment building based on top-level packages, some configuration is necessary. For more information, see section "Settings for Consignment Building on Top-Level Packages".
- To communicate via EDI with your carrier, you need to configure the sending of consignment orders to the carrier. For more information, see section "Settings for Sending Consignments to the Carrier".

Pick-Up Freight Orders and Delivery Freight Orders

The air freight scenario for shippers supports door-to-door (D2D) and door-to-port (D2P) shipping. You can model the transport of goods for a customer by airplane and order the transport service (D2D or D2P) from a carrier. The carrier organizes the air freight bookings on its own and informs you once it is decided which flight the freight will be on.

The air scenario starts with creating a road pick-up freight order (FO). The pick-up FO includes all goods (freight units) that need to be transported via the same airport and can be created before the air freight booking exists. The freight units can originate from sales orders or outbound deliveries.

i Note

Only freight units destined for an airport are allowed. This means you can't combine air freight cargo with other truck deliveries, such as requirements for direct transport to customers.

You create pick-up FO orders without reference to an air freight booking in the [Freight Bookings Work List](#) or the [Freight Orders Work List](#). Choose query [Pick-Up/Delivery](#), and then [New](#) [Road Freight order for Pick-Up/Delivery](#).

Consignment Orders for Carriers and Customers

Out of the pick-up FO to the airport, you create consignment orders to be able to communicate with the carrier. Based on these consignment orders, you inform the carrier about the goods that need to be transported.

Carrier Plans Consignments on Flights

The consignment orders are sent to the carrier. The carrier plans the consignment orders on appropriate flights, creates the corresponding air freight bookings, and informs you which consignment orders are planned on which flight. The carrier may send this information via EDI communication or email. The consignment orders of a pick-up FO can be distributed over several air freight bookings.

i Note

The system doesn't support splitting consignment orders. In air freight operations, such a split becomes necessary when a carrier allocates cargo items from a single consignment order across multiple air freight bookings. If the carrier communicates such an allocation situation, the system can't process the air freight booking message.

Due to this restriction, it is important that **all** cargo items of a consignment order are assigned to the **same** flight.

To ensure that no consignment order split is needed, you can either define an appropriate process in the system or configure the system so that consignment orders contain only one cargo item when they're created. To do this, use the consignment order creation based on the top-level package.

Consignments Based on Top-Level Package

When you configure the consignment order creation based on the top-level package, each consignment order contains only one package item or unpacked product. This means, the system creates one consignment order per handling unit (pallet). By that, the package-based consignment orders (pallets) of one pick-up FO can be transported on different flights.

Settings for Consignment Building on Top-Level Packages

To enable local consignment building based on the top-level package, you need to configure your consignment building profile in the [Create Consignment Building Profile](#) app. As grouping rule, you define [TOP_CARGO Topmost Cargo Item \(Package or Unpacked Product\)](#).

Automatic or Manual Creation of Consignments

For the creation, you can choose to manually create consignment orders, or you can use the automatic consignment building. For more information, see [Automatic Consignment Building](#).

You manually create consignment orders in the pick-up FO under ► **Follow up** ► **Build Local Consignment Orders** ►. You then create consignment order documents for the local consignment orders under ► **Follow up** ► **Create Consignment Order** ►.

For more information, see [Consignment Order](#).

The consignment orders include booking information from the assigned air freight booking, for example, airline code, flight number and airport of destination, expected arrival date and time at the port of destination or at the customer (depending on scenario in use, that is door-to-door or door-to-port).

You can display the booking data on the **Booking Data** tab in the consignment order.

Air Freight Booking

If you want to change the assignment of a consignment order to an air freight booking, you first need to unassign it from the original freight booking. After that, you're able to assign the consignment order to a different freight booking.

You can receive messages regarding freight bookings from the carriers using EDI communication. Alternatively, you manually enter the data you received.

Document Errors

As described in the prerequisites, you can save freight bookings that contain errors. If activated, the system reports document errors, in the following situations, for example:

- For the pick-up FO or consignment order if consignment orders aren't yet assigned to an air freight booking. This shows that there are still consignments orders or parts of deliveries that are not transported yet.

This happens, for instance, in a door-to-door scenario where the delivery's destination location is the customer's side, but the freight order is destined for the airport. The document errors are resolved automatically when the consignment order is assigned to a freight booking.

- For a freight unit if the locations don't match and the freight unit and its successor document have a different destination location. The error is automatically resolved once the consignment order is created.

Invoicing

The invoicing process starts only after the departure of the airplane. For more information, see [Carrier Invoice](#).

Message Exchange Between Shipper and Carrier

The air freight process offers a message exchange between you, as the shipper, and the carrier.

Sending Consignments to the Carrier

Consignments are automatically sent to the carrier when the **goods issue** is reported and the goods movement status for the pick-up FO and the consignment order is set to **Completely Processed**.

Settings for Sending Consignments to the Carrier

When goods issue is posted and the system sets the goods movement status of the freight order and the consignment order to **Completely Processed**, the system sends a `TransportationOrderGenericRequest_Out` message (with Business Scope "EXTERNAL_PLANNING") to the main-carriage carrier. The status in the consignment order is the status that is relevant for sending this message.

The following settings are necessary to send the consignment orders to the carrier for external planning:

- You have defined a consignment order type with transportation mode "air", to automatically send ASN messages to the carrier.

You define the transportation mode in section **Service Definition** in Customizing for Transportation Management under **Freight Order Management** > **Consignment Orders** > **Define Consignment Order Types**.
- You have configured the Post Processing Framework (PPF) in Customizing for **Cross-Application Components** under **Processes and Tools for Enterprise Applications** > **Reusable Objects and Functions for BOPF Environment** > **PPF Adapter for Output Management** > **Maintain PPF Settings**. In application `/SCMTMS/TRANSPORTATION`, you activated the action definition **TOR_TPNOR_CO (Replication of Consignment Order)** of action profile `/SCMTMS/TOR`.
- In addition, you have assigned the output profile `/SCMTMS/TOR` to the consignment order type you use. You assign the output profile in Customizing for Transportation Management under **Freight Order Management** > **Consignment Order** > **Define Consignment Order Types** in section **Output Options**.

Receiving Air Freight Bookings from the Carrier

i Note

The EDI messages you receive from the carrier must contain certain information. For some XML samples, see SAP note [3514066](#).

The carrier can send you EDI messages with information about the air freight bookings. The message can be used for creating and updating the air freight booking in your system. It can be used for **creating** the air freight booking when the planning of the freight booking by the carrier is done. Alternatively, it can be used for **updating** the air freight booking.

First Message from the Carrier (Creating)

With the first message, the carrier informs you which consignment orders are planned on which flight. The message for creating the air freight booking must contain information, such as:

- Carrier number
- Stage information, such as stage between airport of departure and airport of destination, source and destination location specified by IATA code, stage category, flight code, expected departure time, expected arrival time
- Item information, such as item category, referenced consignment orders, house air waybill (HAWB) per customer
- Event information (optional), event type, stop identification specified by IATA code

When the carrier sends the message before flight departure, the event information is not necessary.

With this information, the air freight booking with the referenced consignment orders can be created in your system.

Update Messages

When the flight departed, the carrier can send you an update message. Maybe the planning was changed, or some cargo was not transported with the flight.

The messages can include information such as the referenced consignment orders, uplift confirmed status when the airplane departed or additional event updates when the airplane arrived at the destination airport.

Alternatively, you can manually enter updates in the freight booking.

Identifying Locations by IATA Codes

The system finds locations using the IATA codes that were sent with an inbound message. If the handling of errors for business documents is activated for the freight booking type, the system displays document errors if the IATA code couldn't be found, isn't unique or is wrong.

In exceptional cases, creating a freight booking on an inbound message fails, for example, when the message contains a non-unique IATA code and at the same time, the departure is already set for the air port of loading. Such a message will fail, and the air freight booking will not be created as the system can't identify the location and therefore can't set the departure status for the location.

To enable this check, you need to activate the handling of errors for business documents in Customizing for the air freight booking type.

Update Port of Destination

Changing the port of destination on inbound messages works the same in the ocean and in the air process. For more information, see [Ocean Freight Bookings for Shippers \(Outbound\)](#).

Sending ASN to Customer

The consignment orders and their updates are also sent **to the customer** to inform them about the transport.

In the following cases, the system sends a separate advanced shipping notification (ASN) message for each consignment order to the customer when the airplane departed. At this point in time additional booking information is available and can be included in the ASN.

- When the goods movement status on consignment order is set to **Completely Processed**.
- When the air freight booking is set to **Departed** from port of loading.
- Alternatively, when manually requested in the consignment order under **Follow Up > Send Advanced Shipping Notification**.

Especially in a door-to-port scenario, consignment orders and their updates are useful for customers. Depending on the Incoterms, they might need to know the voyage number and the expected date and time of arrival to organize the pickup and transport of the goods for the last stage.

Related Information

[Sample Air Freight Process for Shippers \(Outbound\)](#)

[Consignment Order](#)

[Cargo Left Behind at Port/Airport](#)

[Sample Ocean Freight Process for Shippers \(Outbound\)](#)

[Ocean Freight Bookings for Shippers \(Outbound\)](#)

[Advanced Shipping and Receiving](#)

Cross-Delivery Handling Units

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on outbound deliveries supports cross-delivery handling units (HU) for the outbound process.

Use

You can create handling units like cartons or pallets for items from two or more outbound deliveries. This information is passed on to TM and reflected in the freight unit (FU).

Transportation-driven scenario: Outbound delivery orders are blocked until transportation planning is complete and the status is set to **Ready for Warehouse Processing** in the FU, which releases the outbound delivery order. Once you create cross-delivery HUs and set the status to **Ready for Shipping**, the system updates TM.

Warehouse-driven scenario: Outbound deliveries are released upon creation so you can proceed with warehouse tasks. Once you set the execution status to **Ready for Shipping**, EWM updates TM with the cross-delivery HUs' packaging information. In a warehouse-driven scenario, it is also possible to do transportation planning in parallel. If freight orders (FO) already exist, the system also updates them with the packaging information.

The system handles the merging of FUs whose items have been packed in cross-delivery HUs in the same way as HUs that contain items from the same delivery. In TM, a cross-delivery HU is represented as package item in the FU. Therefore, once warehouse activities are completed, the system merges FUs with identical package items.

Splits can occur at two points in the process. First, an FU would be split after planning if only some items in the order are planned on the FO and **Ready for Warehouse Processing** is set. Second, any items not loaded before handling execution status **Loading Completed** is set are removed, resulting in an FU split. The FU split subsequently leads to the split of the corresponding outbound delivery.

Additionally, deliveries packed into cross-delivery HUs can have different sold-to parties but must have the same ship-to party.

Ad Hoc Loading

You can assign freight units (FU) to road freight orders (FO) directly from the staging area in your warehouse.

The ad hoc loading process consists of the early creation of FOs and the assignment of FUs to said FOs during the loading of the corresponding handling units (HU) onto the truck.

Prerequisites

- **Optional:** You have activated the use of warehouse loading stops and warehouse unloading stops for road schedules in Customizing for **Transportation Management** under ► **Master Data** ► **Transportation Network** ► **Schedule** ► **Define Schedule Types** ►

Warehouse loading and unloading stops can then be defined in the road schedule and are taken over automatically when you create a road freight order.

Features

Ad hoc loading enables the warehouse worker to assign FUs to an existing FO while loading.

When the HUs are scanned in the staging area, the system checks for matching criteria and decide whether the corresponding FUs can be assigned to a given FO.

For FU and FO to match, three criteria must be met: the FO must be docked to a warehouse door, and the carrier and destination locations in both documents are identical. If the criteria are met, all FUs linked to the outbound delivery order that make up the HUs are assigned to the FO.

There are two approaches to working with ad hoc loading. To begin loading, you can either scan the door or handling units (HU) in the staging area. For more information about the warehouse processes, see [Ad Hoc Loading in Extended Warehouse Management](#).

Activities

You can implement [BAdI: Definition of FU-FO Matching Criteria for Ad-Hoc Loading](#) to adapt the matching criteria. You can remove certain requirements and enhance the program logic with requirements of your own. For more information, see Customizing for **Transportation Management** under [Business Add-Ins \(BAdIs\) for Transportation Management](#) [Freight Order Management](#) [Freight Order Management](#).

Integration with SAP Global Trade Services in Advanced Shipping and Receiving

Advanced Shipping and Receiving (Advanced SR) supports integration between Transportation Management (TM) and SAP Global Trade Services (SAP GTS).

Prerequisites

- For both inbound and outbound processes, you can activate the compliance check in the logistics integration profile. For more information, see Customizing for **Transportation Management** under [Integration](#) [Logistics Integration](#) [Internal TM Component Integration](#) [Define Logistics Integration Profile for SD, MM, and LE](#).

Once you activate the compliance check, the system determines compliance for each freight unit. If the criteria for compliance aren't met, the system prevents any subsequent warehouse steps. For additional information on this status, see [Statuses in Advanced Shipping and Receiving](#).

- You've defined customs activities and customs profiles, and assigned one or more customs activities to each customs profile. For more information, see Customizing for **Transportation Management** under [Basic Functions](#) [Global Trade](#) [Define Customs Activities and Profiles](#).
- Optionally, you've defined an event profile with [Cargo Ready for Loading](#) and defined its settings. For more information, see Customizing for **Transportation Management** under [Freight Order Management](#) [Execution Settings](#) [Define Event Profiles](#).

GTS Integration in Advanced Shipping and Receiving

Communication between TM and GTS includes compliance-relevant checks, for example, for embargoes or sanction party lists, and for obtaining clearances such as unloading permits and customs clearance after approval of the export declaration. SAP GTS performs compliance checks based on freight units and freight orders and export declarations based on consignment or freight order.

In the outbound process, customs activity is based on either consignment or freight order. In the inbound process, customs activity is based on the freight order.

The following sections detail the steps in the Advanced SR process that are relevant for GTS integration. For full details of the process, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#). For additional information on the statuses mentioned, see [Statuses in Advanced Shipping and Receiving](#).

Final Delivery Determination

To correctly run the compliance checks, the system needs to check whether the final delivery document is available. In a decentralized scenario it does this by directly checking whether the delivery exists. In an embedded scenario, when checking for the existence of a delivery, the system checks indirectly by looking at the distribution status of the delivery. It uses this status to determine whether the final delivery is available. By default, in an embedded scenario, the system checks indirectly but you can activate the direct check in the Customizing.

To enable the direct check, you must ensure that the status type **DSC - LE Distribution Confirmation** is active in all relevant outbound delivery status profiles, in Customizing for Extended Warehouse Management under **Cross-Process Settings > Delivery - Warehouse Request > Status Management > Define Status Profiles**.

i Note

The direct check is more accurate. If you are experiencing issues with the final delivery not being correctly identified using the default settings, SAP recommends activating the direct check.

Related Information

[Integration with SAP Global Trade Services](#)

[SAP GTS Integration for the Outbound Process](#)

[SAP GTS Integration for the Inbound Process](#)

[Statuses in the Integration with SAP GTS \(Inbound Process\)](#)

[Statuses in the Integration with SAP GTS \(Outbound Process\)](#)

[Global Trade](#)

[Event Profiles](#)

SAP GTS Integration for the Outbound Process

Several steps in the outbound part of the Advanced Shipping and Receiving process are relevant for the integration of a Transportation Management (TM) system with a SAP Global Trade Services (SAP GTS) system. These steps are detailed below as a sample process.

For full details of the process, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#). For additional information on the statuses mentioned, see [Statuses in Advanced Shipping and Receiving](#).

1. When you save or update a sales order that you have created, the SAP GTS system creates the customs documents and runs a trade compliance check. SAP GTS returns an overall status for the order with details on the embargo check, watch list screening status, and legal control statuses. These three statuses are visible on item level, and the system aggregates these statuses in the header status in the sales order. You can see additional information on a blocked sales order on the **Status** tab in the sales order. You can configure the system to assign compliant items of a sales order to an outbound delivery order.

Each sales document can include a legal split indicator. When you create a freight unit from a sales document, the system considers the legal split indicator and groups items in freight units accordingly. All items in a freight unit have the same split indicator. When creating or splitting deliveries, the system considers the legal split indicator on header level of the sales order and groups the items accordingly.

2. In the delivery-based scenario, when you create the delivery, TM creates the freight units. The SAP GTS system creates the customs documents and runs a compliance check for all freight unit items. If the trade compliance check hasn't been completed or if the freight unit is not compliant, the delivery is blocked for warehouse execution. From the delivery, you can see the document flow on the **TM Status** tab. The compliance status is listed with a status for a planning block, execution block, delivery block, and invoicing block.

i Note

In a decentralized EWM and TM setup, the outbound delivery leads to the creation of an outbound delivery order in EWM. Therefore, TM relies on the outbound delivery order as an indicator of delivery completeness to trigger the creation of freight units. From there, the process is identical to the one described above.

For freight units that meet the requirements for compliance, the **Warehouse Execution** status is unblocked and the delivery is released for warehouse execution.

In the transport-driven process, you must manually set the warehouse processing status of the freight unit to **Ready for Warehouse Processing** once compliance is met. For more information, see [Transport-Driven Execution](#).

3. You then create a freight order and can create a consignment order from the freight order. The system runs a compliance check for the freight order. If you've activated the compliance check in Customizing and, for example, you create a freight order from a delivery, the system considers the compliance status of the freight units of that delivery. Any blocks affecting the freight units are displayed on the **Blocking Information** tab of the freight order.

For more information on the creation of freight orders and consignment orders, see [Creation and Editing of Freight Orders](#) and [Creation and Editing of a Consignment Order](#).

4. After you've set the handling execution status of the stop to **Checked In**, you can start the export declaration process (on customs group level) for customs groups to be built and assigned to the stop. If you've configured the system to require loading completion, the export declaration process can be triggered automatically once **Set Stop to Loading Completed** is set. TM considers the legal split indicator when it builds customs groups. Once the export declaration processing status is set to **Ready for Export Declaration**, the system can send a request to SAP GTS to create a customs declaration. You can configure the system to include customs invoice data in the request. Alternatively, if you have not configured the automatic request of customs declarations, choose **Request Customs Invoice** in the consignment order. If there is a missing customs clearance, you have the option of preventing the **Checked Out** status for the stop in the freight order.

Alternatively, if you use handling execution status **Cargo Ready for Loading**, the system can trigger customs invoice and export declaration requests before the truck checks in or has even arrived. Depending on your customizing settings in your **Customs Profile**, the requests are triggered automatically. You can set **Cargo Ready for Loading** after the warehouse processing status is set to **Completed**. After the truck arrives and items have been loaded, you set the handling execution status to **Loading Completed**.

5. Once SAP GTS has approved the customs declaration, the system updates the TM document with a customs reference number (CRN). With the CRN in place, TM can post goods issue and the freight order can be checked out.

Related Information

[SAP GTS Integration for the Inbound Process](#)

[Statutes in the Integration with SAP GTS \(Outbound Process\)](#)

[Integration with SAP Global Trade Services](#)

[Explanation of Statutes of Business Documents](#)

SAP GTS Integration for the Inbound Process

Several steps in the inbound part of the Advanced Shipping and Receiving process are relevant for the integration of a Transportation Management (TM) system with a SAP Global Trade Services (SAP GTS) system. These steps are detailed below as a sample process.

For full details of the process, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#). For additional information on the statutes mentioned, see [Statutes in Advanced Shipping and Receiving](#).

1. When you create a purchase order, scheduling agreements, or a scheduling agreement delivery schedule, the system creates freight units. The system checks these freight units for trade compliance and, if the freight unit is not compliant, the system places a planning block or execution block on them.
2. You create a freight order, and you can create a consignment order from the freight order. For more information on the creation of freight orders and consignment orders, see [Creation and Editing of Freight Orders](#) and [Creation and Editing of a Consignment Order](#).

If you have configured the system in such a way, it runs a compliance check for the freight order. You can only set the handling execution status for the stop to **Checked In** if the freight order is compliant.

3. You must assign a customs reference number (CRN) for each consignment order that is assigned to the freight order before you can set the warehouse processing status to **Ready for Warehouse Processing**.
4. To set the warehouse processing status of the warehouse unloading stop to **Ready for Warehouse Processing**, you need an unloading permission. The TM system creates a request for unloading permission for each consignment order of the freight order and sends it to SAP GTS. In addition, the freight unit needs an assignment to the external delivery note number (document reference with type code 1137) and to the item number.

For more information about the receipt of the unloading permission and the related setting of the **Ready for Warehouse Processing** status, see [Statuses in the Integration with SAP GTS \(Inbound Process\)](#).

5. After the unloading permission is received, the goods are unloaded. For a basic customs process, you set the status of the customs groups from the freight order to **Ready for Import Declaration**. A prerequisite for this step is that the goods receipt has been performed for all the items of the customs group.
6. You set the handling execution status of the stop to **Checked Out**.

Related Information

[SAP GTS Integration for the Outbound Process](#)

[Integration with SAP Global Trade Services](#)

SAP GTS Integration in Decentralized EWM and TM

Integration between decentralized EWM and TM and SAP GTS supports the execution of the outbound, or export, process. It enables you to conduct compliance checks for freight unit and freight order and complete export declaration for consignment and freight order.

Supported Systems

The central ERP system must be an S/4HANA system.

Prerequisites

- You have selected the **Enable Compliance Check** checkbox in your logistics integration profile for decentralized EWM and TM to activate compliance check at a freight unit level. For more information, see Customizing for **Transportation Management** under [►Integration►Logistics Integration►Decentralized EWM-TM Integration►Define Logistics Integration Profile for EWM](#) .
- You have indicated the transportation relevance for sales orders and assigned a logistics integration profile. For more information, see Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Transportation Relevance of Sales Documents](#) .

- You have indicated your output profile under **Output Options**. Optionally, you can define a customs profile so as to enable customs activity at the freight unit level. For more information, see Customizing for **Transportation Management** under **►► Planning ► Define Freight Unit Types ►**.
- You have enabled compliance check for freight orders and, optionally, entered the customs profile to make your customs profile relevant for certain customs activities. For more information, see Customizing for **Transportation Management** under **►► Freight Order Management ► Freight Order ► Define Freight Order Types ►**.
- You have indicated your output profile so the PPF actions that send the customs and compliance messages to GTS are triggered automatically. For more information, see Customizing for **Transportation Management** under **►► Freight Order Management ► Freight Order ► Define Freight Order Types ►**.
- You have indicated your customs and output profiles for consignment orders. For more information, see Customizing for **Transportation Management** under **►► Freight Order Management ► Consignment Order ► Define Consignment Order Types ►**.
- You have defined your customs activities and profiles. For more information, see Customizing for **Transportation Management** under **►► Basic Functions ► Global Trade ► Define Customs Activities and Profiles ►**.

i Note

In a decentralized setup, the **Customs Invoice Required** checkbox is mandatory and must remain activated.

⚠ Caution

In a decentralized setup, GTS is integrated with TM. To prevent processing errors, you must disable the following functions in Logistics Execution (LE):

- Legal Control Check
- Customs Invoice Creation
- Export Declaration Request

Context

Compliance Check

To check for compliance, GTS carries out the following three checks:

1. **Sanctioned Party List Screening:** Checks if a business partner is listed on a Sanctioned Party List. If a business partner is listed, GTS returns a negative compliance confirmation.
2. **Embargo Check:** Checks if there is an existing embargo between source and destination countries. If an embargo prohibits trade between those two countries, GTS returns a negative compliance confirmation.
3. **Legal Control Check:** Some goods require legal control licenses. In case the goods sent to GTS do not comply with the licensing requirements, GTS returns a negative compliance confirmation.

The legal control check is based on material data including quantities, weights or exchange rates.

Export Declaration

The export declaration request in TM triggers the creation of the customs documents needed for exporting the goods in the SAP GTS system.

The customs authorities need data like parties, items, and predecessor documents to confirm the export of the items in a freight or consignment order. You can resort to your customs invoice, also known as a proforma invoice, to export your goods, but it is not mandatory. In its absence, the export declaration is based on the goods value of the items in question.

You can cancel an existing export declaration either directly from GTS or request a cancellation from TM.

Export Declaration Statuses

You will encounter the following status while working with export declarations:

Status	Description
Loading Not Finalized	The loading of goods is yet to be reported
Pending Customs Invoice Preparation	The loading has been finalized, but the deliveries are not yet up-to-date
Ready for Customs Invoice Creation	All requirements are met, the loading has been finalized, and deliveries are up-to-date
Customs Invoice Requested	The request for the creation of a customs invoice has been sent to the ERP system
Ready for Export Declaration	Customs invoice has been created, and export declaration request can be sent to the GTS system

i Note

The export declaration statuses are linked to your customs profile. To set requirements and automate creation, you must first customize your system. In Customizing, you have the option to activate the following requirements:

- Loading Completed Required for Export Declaration
- Customs Invoice Required for Export Declaration
- Enable Automatic Customs Invoice Creation
- Automatic Export Declaration Request

To enable **Customs Invoice Required for Export Declaration**, you must first enable **Loading Completed Required for Export Declaration**. Furthermore, for **Enable Automatic Customs Invoice Creation** to function, you must first enable **Customs Invoice Required for Export Declaration**.

For more information, see Customizing for **Transportation Management** under **Basic Functions > Global Trade > Define Customs Activities and Profiles** .

Customer Invoice

In a decentralized EWM and TM setup, you create a customer invoice from your ERP system.

Sample Process

You can find a sample outbound process description and information concerning statuses under **Related Information**.

Related Information

Statuses in the SAP Global Trade Services Integration

When Transportation Management is integrated with SAP Global Trade Services as part of the Advanced Shipping and Receiving process, a specific set of customs processing statuses is used, depending on whether the inbound or outbound process is active.

Related Information

[Statuses in the Integration with SAP GTS \(Inbound Process\)](#)

[Statuses in the Integration with SAP GTS \(Outbound Process\)](#)

[Export Declaration Processing Statuses and Customs Profile Customizing](#)

Statuses in the Integration with SAP GTS (Outbound Process)

If a road freight order, consignment order, or freight unit is relevant for a customs or compliance process, the following sample sequence of activities takes place in Transportation Management (TM) during integration with an SAP Global Trade Services (SAP GTS) system.

The activities are described in terms of the statuses used in the outbound process. For full details about statuses in the outbound process of Advanced Shipping and Receiving, see [Statuses in the Outbound Process](#).

Prerequisites

You have made the following settings:

- You have assigned a customs profile in the Customizing for a freight order type and consignment order type. A customs profile contains one or more customs activities that control customs handling for a specific process (for example, export or transit). For more information, see Customizing for [Transportation Management](#) under [Freight Order Management](#) [Freight Order](#) [Define Freight Order Types](#) or [Freight Order Management](#) [Consignment Order](#) [Define Consignment Order Types](#).
- You have enabled a compliance check in the road freight order and freight unit.

For more information on setting up customs activities, for example, to display customs statuses in a document, see [Configuration of Customs Processing](#).

To determine whether a business document is relevant for a customs process, see [Customs Relevance Check](#).

Outbound Process

Compliance Check

In the outbound process, when you create or modify freight units, a compliance check is started by the SAP GTS system for all freight units. If SAP GTS can't complete the compliance check or if the freight units are not compliant, the outbound delivery order and the freight units are blocked for execution. The system doesn't set the warehouse processing status of the freight unit to [Ready for Warehouse Processing](#).

A freight unit that was previously compliant may also receive an update that makes the freight unit non-compliant. For example, a business partner is placed on the sanctioned party list after a screening has taken place. In this case, if the warehouse processing status is **Ready for Warehouse Processing**, the status will be reversed and set to **Not Started**. If the warehouse processing status was already set to **In Process**, it cannot be reversed.

After the compliance-related block is resolved, the system automatically sets the status of the freight unit to **Ready for Warehouse Processing** and you can proceed with the warehouse execution.

In the transport-driven process, you must manually set the warehouse processing status of the freight unit to **Ready for Warehouse Processing**. You can also manually reverse the warehouse processing status to **Not Started**, as long as warehouse activities have not begun. For more information about the transport-driven process, see [Transport-Driven Execution](#).

If you use the compliance process and the transport-driven process together, you must manually set the status to **Ready for Warehouse Processing**, after compliance has been met. However, if there is a need for a second compliance check, the system will automatically set the **Ready for Warehouse Processing** status after compliance is met again, and only if you had manually set it before.

Customs Process

For the customs processes, if a road freight order has several shipping points, then the consignment order can be used as the main document.

The basic sequence of statuses and activities in the outbound process is as follows:

1. When you set the execution status of the road freight order to **Ready for Transportation Execution**, the system checks all the execution blocks that are set for the road freight order and the consignment order. If the execution block is only relevant for customs processes, you can proceed with execution of the road freight order and allow the truck to load. You can see the blocks and the block reasons on the **Blocking Information** tab in the document. The reason for the block for customs processes can be, for example, that there's a missing export declaration for the road freight order.

→ Remember

You have the option of using handling execution status **Cargo Ready for Loading**, which allows you to trigger customs invoice and export declaration requests before the truck checks in or has even arrived. With **Cargo Ready for Loading**, the requests are triggered automatically after the warehouse processing status is set to **Completed** instead of after **Loading Completed**.

2. After the truck arrives at the gate, you set the handling execution status for the stop to **Arrived**.
3. If the documents of the goods on the truck match the documents in the system, you set the handling execution status for the stop to **Checked In**.
4. After the goods are loaded on the truck, you can set the handling execution status of the items and the warehouse loading stop to **Loaded**, even if there's a block that's relevant for customs processes. If you've configured the system to require loading completion, when the status is set to **Loaded**, the export declaration status changes to **Pending Customs Invoice Preparation**. Once the deliveries are up-to-date, the export declaration status changes to **Ready for Customs Invoice Creation**, and TM requests a customs invoice from the ERP.

After receiving the customs invoice, the export declaration processing status changes to **Ready for Export Declaration**. TM sends a request to SAP GTS to create a customs declaration and includes a customs invoice in the request.

You can see the export declaration processing status and the customs status on the **Customs** tab of the road freight order or consignment order.

i Note

In a decentralized EWM and TM setup, you will find one additional status: **Customs Invoice Requested**, which indicates that a request for a customs invoice has been sent to the ERP system.

5. If the export declaration is missing or incomplete, you can't set the handling execution status for the stop to **Checked Out** because the block related to the customs process still exists. After the customs authorities approve the customs declaration, SAP GTS updates the road freight order and the assigned consignment order. You can then set the status to **Checked Out**.

For more information about SAP GTS integration in the outbound process of Advanced Shipping and Receiving, see [SAP GTS Integration for the Outbound Process](#).

6. After you post goods movement for the complete road freight order, you can create a customer invoice for the Logistics Execution (LE) or Extended Warehouse Management (EWM) outbound delivery order assigned to the road freight order. You can also cancel the customer invoice.

For more information about SAP GTS, see SAP Library for SAP Global Trade Services on SAP Help Portal at <http://help.sap.com/grc> under SAP Global Trade Services.

Related Information

[Integration with SAP Global Trade Services](#)

[Statuses in Advanced Shipping and Receiving](#)

[Statuses in the Integration with SAP GTS \(Inbound Process\)](#)

[SAP GTS Integration in Decentralized EWM and TM](#)

Statuses in the Integration with SAP GTS (Inbound Process)

If a road freight order or consignment order is relevant for a customs process, the following sample sequence of activities takes place in Transportation Management (TM) during integration with an SAP Global Trade Services (SAP GTS) system.

The activities are described in terms of the statuses used in the inbound process. For full details about statuses in the inbound process of Advanced Shipping and Receiving, see [Statuses in the Inbound Process](#).

Prerequisites

You have assigned a customs profile in the Customizing for a freight order type and consignment order type. A customs profile contains one or more customs activities that control customs handling for a specific process (for example, export or transit). For more information, see Customizing for [Transportation Management](#) under [Freight Order Management > Freight Order > Define Freight Order Types](#) or [Freight Order Management > Consignment Order > Define Consignment Order Types](#).

For more information on setting up customs activities, for example, to display customs statuses in a business document, see [Configuration of Customs Processing](#).

To determine whether a business document is relevant for a customs process, see [Customs Relevance Check](#).

Inbound Process

The basic sequence of statuses and activities in the inbound process is as follows:

1. If the documents of the goods on the truck match the documents in the system, you set the handling execution status for the stop to **Checked In**. This triggers the building of customs groups. The system sends a request for unloading permission to the SAP GTS system and sets an execution block for the road freight order because the approval for the unloading

permission is pending. The system automatically sets the status of the customs activity to **Unloading Requested** on the **Customs** tab in the road freight order.

If you reverse the **Checked In** status after a block caused by a missing unloading permission is set for the road freight order, the block is not reversed.

For more information on the process to secure approval from SAP GTS to unload a road freight order, see [Customs Transit Discharge](#).

2. You set the warehouse processing status to **Ready for Warehouse Processing** at the warehouse unloading stop level to signal to the warehouse that they can begin their activities. However, because the unloading permission from the customs authorities hasn't been received yet, the system creates the **Ready for Warehouse Processing** event but also blocks it from being processed. You can see the events under the **Execution** tab of the road freight order. The warehouse processing status is **Not Started**.

If you don't have an explicit check-in process at your company gate, you can directly set the warehouse processing status to **Ready for Warehouse Processing** at the warehouse unloading stop level. The system automatically sets the **Checked In** status at the stop level, creates the **Ready for Warehouse Processing** event, and blocks it from being processed.

3. After SAP GTS sends an approval for the unloading permission, the system sets the status of the customs activity to **Unloading Approved**. The system then processes the **Ready for Warehouse Processing** event and sets the warehouse processing status of the warehouse unloading stop to **Ready for Warehouse Processing**. Alternatively, you can manually overrule the block under the **Blocking Information** tab to trigger the processing of the **Ready for Warehouse Processing** event.

On the item level, you can set the **Ready for Warehouse Processing** status for freight units only if you have entered an external delivery note number.

4. When unloading is completed, you can set the handling execution status for the items and the warehouse unloading stop to **Unloaded**. You can then create the customs import declaration for the stop.

You can see the import declaration status and the unloading permission status on the **Overview** tab.

For more information about SAP GTS integration in the inbound process of Advanced Shipping and Receiving, see [SAP GTS Integration for the Inbound Process](#).

For more information about SAP GTS, see SAP Library for SAP Global Trade Services on SAP Help Portal at <http://help.sap.com/grc> under SAP Global Trade Services.

Related Information

[Integration with SAP Global Trade Services](#)

[Statuses in Advanced Shipping and Receiving](#)

[Statuses in the Integration with SAP GTS \(Outbound Process\)](#)

Export Declaration Processing Statuses and Customs Profile Customizing

The export declaration processing statuses indicate what stage of the Advanced Shipping and Receiving export process you find yourself in and what steps you need to take to progress the process.

The statuses appear based on your Customizing settings. For example, if you set a customs invoice as a prerequisite to the export declaration. Additionally, you can choose to automate both customs invoice and export declaration requests. For more information, see Customizing for **Transportation Management** under **Basic Functions** > **Global Trade** > **Define Customs Activities and Profiles** .

Export Declaration Processing Statuses

Status	Description
Loading Not Finalized	The handling execution status of the relevant stop has not been set to loading completed.
Pending Customs Invoice Preparation	The LE and EWM deliveries have not been updated so the warehouse distribution status (LIKP - VLSTK) is not yet C.
Ready for Customs Invoice Creation	The customs invoice can be requested. When EWM is not involved, this stage is reached after loading has been completed. When EWM is involved, the status shows after you have created the final outbound delivery order and the LE delivery has been updated. EWM updates LE after you create the outbound delivery order in EWM, LE confirms, and EWM updates TM. In TM, you can find the final outbound delivery number under item document reference.
Customs Invoice Requested	The request for a customs invoice has been sent via web service. This status is only visible in decentralized EWM and TM.
Ready for Export Declaration	The final status indicates that all prerequisites have been met, and the export declaration can be requested.

Related Information

[Integration with SAP Global Trade Services](#)

Integration with Extended Warehouse Management

You can integrate Transportation Management (TM) with Extended Warehouse Management (EWM) in various ways:

- **Direct integration**

Within the direct integration, you have the following options:

- Integration **based on freight orders** (this type of integration is often used by shippers)
- Integration **based on deliveries and EWM transportation units** (this type of integration is often used by shippers)
- Integration based on execution documents resulting from **forwarding orders** (transit warehousing; this type of integration is often used by logistics service providers)

- **Warehouse billing**

Systems

Integration Based on Freight Orders (Advanced Shipping and Receiving)

For an integration based on freight orders, you can integrate TM in SAP S/4HANA with EWM in SAP S/4HANA, both located in the same system.

For more information, see [Advanced Shipping and Receiving](#).

Integration Based on EWM Transportation Units

For an integration based on EWM transportation units, you can integrate TM in SAP S/4HANA with the following EWM systems:

- EWM embedded in SAP S/4HANA
- Decentralized EWM based on SAP S/4HANA (as an external system)
- SAP EWM (decentral SAP EWM as an external system)

For more information about EWM integrated as an external system, see SAP Note [2952651](#) .

The integration of TM and EWM both embedded in SAP S/4HANA and the integration of TM with EWM as an external system support the same processes.

For more information about the processes in EWM in SAP S/4HANA, see <https://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management** .

For more information about the processes in SAP EWM, see <http://help.sap.com/ewm>  under ► **Application Help** ► **SAP Extended Warehouse Management (SAP EWM)** .

Features

Integration Based on Freight Order (Advanced Shipping and Receiving)

Advanced Shipping and Receiving is a streamlined and integrated, end-to-end process that allows users to send, receive, and transport products. It supports mixed and multi-warehouse scenarios within one freight order. It helps you, to coordinate the process steps from arrival of trucks at the gate to the final goods receipt in the warehouse.

For more information, see [Advanced Shipping and Receiving](#).

Integration Based on EWM Transportation Units

You can integrate TM and EWM directly. Communication is based on the deliveries that are sent from Logistics Execution (LE) to EWM and TM. The TM planning results are transferred directly to EWM. Transportation units are automatically created in the warehouse. Together with the delivery information, these form the basis for warehouse planning and execution. Confirmations of the execution results (such as loading and quantity discrepancies) are sent directly from EWM to TM.

For more information, see [Integration Based on EWM Transportation Units](#).

Transit Warehousing

You can integrate TM and EWM directly. Communication is based on execution documents resulting from forwarding orders. The TM planning results are transferred directly to EWM. Transportation units and deliveries are automatically created in the warehouse and form the basis for warehouse planning and execution. Confirmations of the execution results (such as loading and quantity discrepancies) are sent directly from EWM to TM.

For more information, see [Transit Warehousing](#).

Warehouse Billing

You can sell and purchase warehouse services using the direct integration of TM with EWM. Integration is based on agreement documents, service orders, and forwarding orders.

For more information, see [Warehouse Billing](#).

Integration Based on EWM Transportation Units

You can integrate Transportation Management (TM) with Extended Warehouse Management (EWM) based on the EWM transportation unit (TU).

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Supported Processes

The integration is **based on the EWM-TU**. For more information about the integration **based on the freight order**, see [Advanced Shipping and Receiving](#).

For the integration based on the EWM TU, the following processes are supported for **road freight orders**:

- **Delivery-Based and Order-Based Planning (Outbound)**

In TM, you can plan your transportation activities **based on an outbound delivery** or **based on an order** and send your freight orders to EWM. In EWM, the warehouse workers carry out the steps relating to the warehouse, for example, loading, picking, packing and staging. Because the two systems are integrated with one another, EWM sends a message to update TM when the truck departs. TM updates the freight order and the associated freight units.

In addition, you can enable updates from EWM to TM with status **Loading Finished**. For more information, see [Early Updates from EWM](#).

A detailed standard **sample process** is available. For more information, see [Example Process: Delivery- and Order-Based Planning \(Outbound\)](#).

For more information about delivery-based and order-based integration in Logistics in general, see [Integration of Orders and Deliveries in Transportation Planning](#).

- **Delivery-Based and Order-Based Transportation Planning (Inbound)**

In TM, you can plan your transportation activities **based on an inbound delivery** or **based on an order** and send your freight orders to EWM. In EWM, the warehouse workers carry out the steps relating to the warehouse, for example, unloading.

For more information, see [Example Process: Delivery- and Order-Based Planning \(Inbound\)](#).

i Note

In the inbound scenario, EWM does not send information on cargo back to the freight order in TM when the truck departure is posted in EWM. Therefore the item hierarchy is not changed in the freight order, even if the items were not all received in EWM.

- Further processes:

- [Transportation Planning with Trailers](#)
- [Stock Transport Order Process](#)
- [Early Picking](#)
- [Multi-Pickup, Multi-Drop](#)
- [Cross-Delivery Handling Units](#)
- [Execution-Driven Planning \(Outbound\)](#)

i Note

When you define the **freight unit building rule**, make sure that freight units are **not** split for order items or delivery items.

❖ Example

You have defined a freight unit building rule with a split quantity of 1,500 kg. If you have an item that weighs 2,000 kg, TM creates two freight units: one freight unit with 1,500 kg and another with the remaining quantity of 500 kg. If a **LoadingAppointment** message is then sent to EWM as part of the integration, the request **cannot** be executed correctly.

i Note

Deliveries to a transportation stop that originate from warehouses with different warehouse numbers are not supported. It is not relevant whether the warehouse numbers are from the same EWM system or from different EWM systems.

For information about the supported functions and the restrictions, see SAP Note [3018355](#) .

DTR and OTR

Some features, like delivery-based transportation requirements (DTR) or order-based requirements (OTR), are only relevant if TM is integrated with external logistics systems. This means, the logistic systems (such as [Logistics Execution](#), [Sales and Distribution](#), or [Materials Management](#)) are not located in the same system as TM. For more information, see [External TM System Integration](#).

Features

When is the System Updated?

- TM
 - When you have finished planning your transportation activities, you set the load plan status or unload plan status on stop level to **Finalized**. The system automatically sends a loading or unloading instruction to EWM with the message **LoadingAppointmentRequest**.
 - When you choose **Follow Up > Send Loading Instruction or Send Unloading Instruction** . The system sends a **LoadingAppointmentRequest** message to EWM.
- EWM
 - If you enable "early updates", the system sends an additional **LoadingAppointmentNotification** message to TM when the warehouse workers set the status **Loading Finished**.
 - When the warehouse workers set the status **Arrival at Checkpoint** and **Departure from Checkpoint**. The system sends a **LoadingAppointmentNotification** message to TM.
 - When the warehouse workers post the **goods issue** or **goods receipt**. The system updates the outbound or inbound delivery in Logistics Execution (LE).

Only the messages sent with **Finish Loading** and **Departure from Checkpoint** are content-related and contain a full image of the item hierarchy. All other messages and reversals only contain status and event information.

Displaying Data

The warehouse numbers from EWM are displayed in the following locations:

- Delivery-based transportation requirement: [Locations and Dates/Times](#) tab page

- Freight order: [Overview](#) tab page for the stop

Changing Data

Data changes in EWM are replicated to TM, for example:

- Item hierarchy
- Carrier
- Relevant dates and times, such as arrival time at the yard
- Gross weight of the vehicle
- Packaging material

Updating Data

- The **item hierarchy** of the freight order is updated with the information about the EWM transportation unit. **Associated freight units** are updated accordingly.
- **Updates on Freight Units**
 - If the warehouse workers have packed items in **different handling units** in EWM, the items for these handling units are created automatically in TM.
 - **Reassignment of freight units:** If the warehouse workers load **deliveries from a different freight order**, the system automatically changes the assignment of the corresponding freight units. This means that these freight units are removed from the other freight order and reassigned to the new freight order.
 - If you **assign a freight unit** to a transportation unit in EWM but the freight unit is not assigned to a freight order in TM, this information is also transferred to TM. The system then automatically assigns the freight unit to the corresponding freight order.
- If you **reassign a complete delivery** in EWM from one transportation unit to another, EWM reports the reassignment to TM. In TM, the delivery is automatically reassigned from the original freight order to the other freight order (corresponding to the assigned transportation unit in EWM).
- **Actual quantities** from EWM are automatically transferred to TM. For more information, see [Quantity Deviation](#).
- **Actual dates** are also reported automatically to TM.

Status Updates

- **LoadingAppointmentRequest**
 - If you set the load plan status on stop level to **Load Plan Finalized** for your freight order, the system automatically triggers the event **Ready for Loading**. The system transfers the event to the predecessor document and changes its status to **Load Plan Finalized**. If so, the freight unit is no longer updated from LE but from EWM.
 - The same applies for the load plan status **Unload Plan Finalized** and the event **Ready for Unloading**.
- **LoadingAppointmentNotification**
 - If TM receives the **LoadingAppointmentNotification** message during outbound-delivery-based transportation planning, the **cargo receipt status** for the item is automatically set to **Shipped**. If the item in question is to be unloaded at the next transportation stop, the handling execution status is automatically set to **Not Unloaded**. In the associated freight unit, the cargo receipt status is also set automatically to **Shipped**. As a result, the freight unit is no longer updated from the freight order or the delivery-based transportation requirement.

- The **communication history** for inbound documents is updated regarding arrival at checkpoint and departure of checkpoint.
- The **execution status** is set to **In Execution**.

Reversal of Departure From Checkpoint

Reversal of the departure and the sending of a second departure message from EWM to TM is enabled for inbound and outbound processes. For more information, see [Reversal of Departure](#).

Canceling Documents

For inbound processes you can still cancel freight orders up until the goods have arrived in the warehouse and have been reported by EWM as **Arrived**. If the status **Unload Plan (Stop) Finalized** has already been set and you cancel the freight document, the system sends a cancellation request to EWM. This request is checked and processed by the EWM system as follows:

- If the goods have not yet arrived in the warehouse, EWM confirms the cancellation, and the life cycle status of the freight document is changed to **Canceled**.
- If the handling execution status at stop level has already been changed to **Arrived**, the system sets the **Transmission-to-EWM Status** to **Partially Received** or **Received**. Execution of the transportation continues and the freight document cannot be canceled. In some cases, you can reverse the arrival and then cancel the freight document.

The freight document in TM can be canceled only if cancellation has been confirmed by EWM and reported to TM.

For more information about canceling freight orders and stops in multi-pickup, multidrop scenarios, see [Multi-Pickup, Multi-Drop](#).

Messaging

Transmission-to-EWM Status provides information about the status of the exchange of messages. For more information, see [Explanation of Statuses of Business Documents](#).

More Information

[Logistics Integration](#)

[Transportation Stops](#)

Prerequisites for Integration Based on EWM TUs

This topic describes prerequisites for the integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on EWM transportation units (TU).

Prerequisites for Special Functions

Prerequisites for special functions that need to be set in the system are described under these functions. For example:

- [Stock Transport Order Process](#)
- [Early Updates from EWM](#)
- [Delivery Split](#)
- [Transportation Planning with Trailers](#)

- [Early Picking](#)
- [Multi-Pickup, Multi-Drop](#)
- [Execution-Driven Planning \(Outbound\)](#)

General Prerequisites

The use of deliveries is a prerequisite for the TM-EWM integration.

Settings for Freight Order Type

- You have configured the following settings in your **freight order type**:
 - You have selected the **Transportation Activity Creation in EWM** option in the **Document Creation Relevance** field.
 - You have assigned an existing **EWM integration profile** or have created your own profile.
 - You have specified an **output profile** that contains actions for sending loading/appointment messages, for example, /SCMTMS/TOR.
 - In section **Basic Settings**, you have defined the **stage sequence type** by selecting one of the following options:
 - **Defined and Linear**
 - **Star-Shaped Based on FU Stages**

For more information, see Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types**.

Settings for Codes and Event Reason Codes

- **Predefined Codes**

SAP provides the following, predefined event codes for cancelations as well as the corresponding event reason codes. These event codes are set as default codes:

Event Code	Name	Event Reason Code	Name
CANCEL_EXEC_DOC	Cancel Execution Document	CANCELLED_BY_CARRIER	Canceled by Carrier
CANCEL_STATUS_RESET	Cancel and Reset Cargo Execution Status	TM_PLAN_NOT_READY	TM Planning Not Yet Ready
CANCEL_UNASSIGN_FUS	Cancel and Unassign FUs of Current Stop	STOP_CANCELLED	Loading Appointment Canceled (loading cannot be executed in the warehouse as planned)

- **Definition of Own Codes and Event Reason Codes**

You can define your own event codes and event reason codes. For each event type, you must specify at least one of your event codes as the default code.

For more information, see Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **Define Event Reason Codes**.

Prerequisites for Logistics Integration with External Systems

The following settings are only relevant if TM is integrated with external logistics systems.

In Customizing you have activated the direct integration with EWM for your delivery-based transportation requirements. This means that you have set the option **Direct EWM Integration** in the **Dir. EWM Int. Active** field. For more information, see Customizing for **Transportation Management** under ► **Integration** ► **Logistics Integration** ► **External TM System Integration** ► **Delivery-Based Transportation Requirement** ► **Define Delivery-Based Transportation Requirement Types** ►.

For more information, see [External TM System Integration](#).

Early Updates from EWM

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on EWM transportation units (TUs) supports sending early updates from EWM to TM. “Early update” means updating the item hierarchy in TM with status change **Finish Loading**, which is earlier than with status **Departure from Checkpoint**.

Prerequisites

Make the following settings in **EWM Customizing** to enable early updates for the integration based on EWM TUs:

- In the settings for TM message processing define that messages are sent multiple times for your logical system.

For more information, see Customizing for **SCM Extended Warehouse Management** under ► **Extended Warehouse Management** ► **Interfaces** ► **TM Integration** ► **Make Settings for TM Message Processing** ►.

- For the used process profile mark **HU Update**.

For more information, see Customizing for **SCM Extended Warehouse Management** under ► **Extended Warehouse Management** ► **Cross-Process Settings** ► **Delivery – Warehouse Request** ► **Process Management and Control** ► **Define Process Profile for Document Header** ►.

Use

With “early updates” in TM-EWM integration customers can print the documents earlier in the process.

By default, the data in TM is updated with the departure of the EWM transportation unit from the checkpoint.

In EWM Customizing you can define that this **additional**, earlier **LoadingAppointmentNotification** (LDAP) message is sent from EWM to TM with loading completion. The item hierarchy of the freight order is updated with the information about the EWM transportation unit. Associated freight units are updated accordingly. The LDAP message signalling departure from checkpoint is still sent, even though an earlier LDAP message was sent. The deliveries in Logistics Execution (LE) are also updated before **Goods Issue**.

It is possible to withdraw the status **Finish Loading**. This gives a chance to perform last minute changes to the load.

Update Check for Delivery or DTR

Check to verify that content-related messages are not processed before the delivery has been updated. For more information, see [Reprocessing of Failed Loading Appointment \(LDAP\) Messages](#).

Reprocessing of Failed Loading Appointment (LDAP) Messages

In the integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on EWM transportation units (TUs), you can run into issues with the LDAP message processing. Temporary issues cause errors when an LDAP message is processed. The temporary issue resolves by itself, but the LDAP message remains in a failed state.

❖ Example

- The freight order is locked by another user and is not updated by the LDAP message.
- **Update check for delivery or DTR:** There is a check to verify that content-related messages are not processed before the delivery has been updated.

For the LDAP messages sent with **Loading Finished** or **Departure from Checkpoint**, the system checks if all relevant deliveries in Logistics Execution are already updated by EWM. In distributed system scenarios, the system checks if all delivery-based transportation requirements (DTRs) are already updated. If the up-to-date check fails, an error message with the reason for the failure is displayed, and the system will not update the TM data.

Reprocess Failed LDAP Messages

If an LDAP message fails, we recommend using the Forward Error Handling service (FEH). FEH is a concept that enables errors that are detected during the execution of asynchronous services to be processed on the provider side. This concept is implemented using the Error and Conflict Handler (ECH).

Define a regular **automatic retry** in a resolution strategy to process the LDAP message as soon as the issue is resolved (for example, the delivery or DTR has been updated). For more information, see Customizing for **Cross-Application Components** under **Processes and Tools for Enterprise Applications > Enterprise Services > Error and Conflict Handler > Define Resolution Strategy**.

To define the resolution strategy use component CA-SOA-ESM-SCM-SCE and define the retry conditions for error category PRE . TEE . LRE.

For more information, see SAP Note [3372289](#).

Related Information

[Early Updates from EWM](#)

[Multi-Pickup, Multi-Drop](#)

Example Process: Delivery- and Order-Based Planning (Outbound)

In Transportation Management (TM), you can plan your transportation activities based on an order or a delivery and send your freight orders to Extended Warehouse Management (EWM). In EWM, you can then carry out the steps relating to the warehouse, for example, loading, picking, packing and staging.

Because the two systems are integrated with one another, EWM sends a message to update TM when the truck departs. The **example process** below describes a standard outbound process for TM-EWM integration.

If you enabled "early updates" the system sends an additional **LoadingAppointmentNotification** message to TM when the warehouse workers set the status **Loading Finished**.

Prerequisites

Preparation in SD

You create a sales order in Sales and Distribution (SD).

Order-Based Planning	Delivery-Based Planning
Outbound deliveries are created later on.	You create outbound deliveries for the sales order and distribute them to TM and EWM.

Transportation Planning in TM

1. Freight units are created automatically.

Order-Based Planning	Delivery-Based Planning
Once the sales order is saved, TM creates freight units.	With the delivery in Logistics Execution (LE), freight units are automatically created in TM, and an outbound delivery order is created in EWM.

As an optional step the system can create packages when freight units are created. For more information, see [Unified Package Building in Integration Based on EWM TUs](#).

Only relevant for integration with external logistics systems: In TM, a delivery-based transportation requirement is automatically created along with the associated freight units. For more information, see [External TM System Integration](#).

2. You create a road freight order.

You plan your transportation activities by creating a road freight order based on the freight units created in the previous step.

3. You create an outbound delivery.

Order-Based Planning	Delivery-Based Planning
You create outbound deliveries for the freight order. The deliveries are then transferred back from LE to TM and the delivery reference is added automatically to the freight order. The delivery reference is displayed in the document references for the item. In EWM, an outbound delivery order is created. i Note The deliveries are only created if enough stock is available in the plant or storage location defined in the sales order.	Deliveries have already been created.

4. You finalize the load plan.

When you have finished planning your transportation activities, you set the **load plan status** of the freight order on stop level to **Load Plan Finalized**. The system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

i Note

In this step, the system checks whether outbound deliveries have already been created. The system can change the status and send the message only if outbound deliveries have already been created.

You can plan multiple warehouses on one road freight order. For more information, see [Multi-Pickup, Multi-Drop](#).

Execution in EWM

When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a **transportation unit**.

1. Picking (and possibly packing) is carried out by the warehouse workers.
2. When the truck arrives at the yard, the warehouse workers assign a door to it and report its arrival.

i Note

By posting [Arrival at Door](#), [Arrival at Checkpoint](#) is posted automatically (if not already done manually).

3. The warehouse workers load the truck.

The following functions are supported:

- [Cross-Delivery Handling Units](#)
- [Quantity Deviation](#)
- [Delivery Split](#)
- [Batch Split](#)
- [Seals](#)

Optional, if configured: Warehouse workers post when they finish loading the truck. EWM sends an additional [LoadingAppointmentNotification](#) message to TM. For more information, see [Early Updates from EWM](#).

For more information about loading, see [Loading and Unloading](#).

4. The warehouse workers post the **goods issue**. The outbound delivery is automatically updated in LE.
5. The warehouse workers post the **departure** of the truck. EWM sends the `LoadingAppointmentNotification` message to TM.

Update and Further Steps in TM

Freight Order

1. When the `LoadingAppointmentNotification` message is received, TM updates the **item hierarchy** of the freight order with the EWM transportation unit data and updates the **associated freight units**.

Order-Based Planning	Delivery-Based Planning
Only relevant for integration with external logistics systems: When the data is updated in LE, the order-based transportation requirement (OTR) is automatically updated in TM.	Only relevant for integration with external logistics systems: When the data is updated in LE, the delivery-based transportation requirement (DTR) is automatically updated in TM.

For more information about the data and status updates, see [Integration Based on EWM Transportation Units](#).

2. You complete freight order execution.

More Information

- For more information about the processes in EWM, see <https://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management** ►.
- For more information about the processes in SAP EWM, see <http://help.sap.com/ewm> under ► **Application Help** ► **SAP Extended Warehouse Management (SAP EWM)** ►.

Example Process: Delivery- and Order-Based Planning (Inbound)

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports delivery-based planning and execution of products.

In TM, you can plan your transportation activities based on an **inbound delivery** and send your freight orders to EWM. In EWM, you can carry out the steps relating to the warehouse, for example, unloading. The example below describes a process for delivery-based and order-based transportation planning (inbound).

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Process

Preparation in MM

You create a **purchase order** in Materials Management (MM).

Order-Based Planning	Delivery-Based Planning
You can plan in MM and distribute it to TM.	You create inbound deliveries for the purchase order and distribute them to TM and EWM. With the delivery in Logistics Execution (LE), freight units are automatically created in TM, an inbound delivery order is created in EWM.
Only relevant for integration with external logistics systems: In TM, an order-based transportation requirement (OTR) is automatically created along with the associated freight units.	Only relevant for integration with external logistics systems: In TM, a delivery-based transportation requirement (DTR) is automatically created along with the associated freight units.

Transportation Planning in TM

- You plan your transportation activities by creating a freight order based on the freight units created in the previous step.
- When you have finished planning your transportation activities, you set the **load plan status** of the freight order on stop level to **Unload Plan Finalized**. The system automatically sends an unloading instruction to EWM with the message `LoadingAppointmentRequest`.

i Note

In this step, the system checks whether inbound deliveries have already been created. The system can change the status and send the message only if inbound deliveries have already been created.

You can plan multiple warehouses on one road freight order. For more information, see [Multi-Pickup, Multi-Drop](#).

Execution in EWM

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit.
2. When the truck arrives at the yard, the warehouse workers post its arrival.
3. The warehouse workers unload the truck. For more information, see [Loading and Unloading](#).
4. The warehouse workers post the **goods receipt**. The inbound delivery is automatically updated in LE.
5. The warehouse workers carry out any additional steps in EWM, such as putaway.
6. With posting truck departure, EWM sends an `LoadingAppointmentNotification` message to TM.

Update in TM

When the `LoadingAppointmentNotification` message is received, TM updates freight order.

i Note

In the inbound scenario, EWM does not send information on cargo back to the freight order in TM when the truck departure is posted in EWM. Therefore the item hierarchy is not changed in the freight order, even if the items were not all received in EWM.

Order-Based Planning	Delivery-Based Planning
Only relevant for integration with external logistics systems: When the data is updated in LE, the order-based transportation requirement (OTR) is automatically updated in TM.	Only relevant for integration with external logistics systems: When the data is updated in LE, the delivery-based transportation requirement (DTR) is automatically updated in TM.

More Information

[Quantity Deviation](#)

[Delivery Split](#)

[Batch Split](#)

[Seals](#)

Stock Transport Order Process

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on EWM transportation units (TUs) supports processes with stock transport orders (STOs). If a relocation of stock is necessary, you can use STOs to move stock between two plants within one company (code).

Use

The stock transport order process can be used, for example, by shippers that own more than one EWM warehouse and use the TU-based integration with EWM installed on a separate system.

For an **order-based** STO process, you start with a stock transport order (a purchasing document). You plan the resulting freight units on a road freight order and create an outbound delivery. The reason the first delivery in the process is an outbound delivery is because the **outbound process** from the supplying EWM warehouse is processed first. Posting goods issue then initializes the creation of an inbound delivery that is needed for the follow-on **inbound process** in the receiving EWM warehouse.

For a **delivery-based** STO process, the outbound delivery is created directly from the STO. You then plan the resulting freight units on a road freight order. The rest of the process works like the standard order-based STO process described below.

In the end, there are two EWM transportation units (TUs) created in the STO process: one outbound TU in the sending, issuing warehouse, and one inbound TU in the receiving warehouse.

For information about restrictions on the STO process, see [3158252](#) .

Prerequisites

General Prerequisites

Delivery Management and Transportation Management must run on the same system.

Other Prerequisites

- **Materials Management**

- You have defined a stock transport order type with an item category that allows stock transfer.

For more information, see Customizing for **Materials Management** under **Purchasing > Purchase Order > Define Document Types for Purchase Order**.

- You have configured the delivery type for the combination of the STO type and plant.

For more information, see Customizing for **Materials Management** under **Purchasing > Purchase Order > Set up Stock Transport Order > Configure Delivery Type & Availability Check Procedure by Plant**.

- **Delivery Management**

For the stock transport order process, you have defined the inbound delivery creation in the receiving plant. For the delivery creation, you need to enter a **confirmation control key** that allows the creation of inbound deliveries in the STO tab **Confirmation**.

You can automate the determination of the confirmation control key in the STO. For more information, see Customizing for **Logistics Execution** under **Shipping > Deliveries > Define Order Confirmations for Inbound Deliveries**.

- **Transportation Management**

- In the **EWM integration profile**, you have defined the specific settings for STO processes.

You have defined whether the system automatically sends **additional** unloading instructions in stock transport order (STO) processes, for example, on truck departure. This means that the shipping warehouse does not have to send the unloading instructions manually.

Furthermore, you have defined whether you want to send partial unloading instructions for multi-pickup, multi-drop scenarios. The sending of partial unloading instructions is relevant for multi-pickup scenarios when STOs are delivered from different warehouses to a single receiving warehouse. The unloading instruction is then sent to the receiving warehouse, although the pickup from all sending warehouses is not complete yet.

For more information, see Customizing for [Transportation Management](#) under [Integration > Integration with Extended Warehouse Management > Define EWM Integration Profile](#).

- o In the logistics integration profile that you use for the stock transport order process, you have configured the improved location determination. To do this, select the [Improved Location Determination](#) checkbox in the [Order-Related Settings](#) section of the logistics integration profile.

For more information, see Customizing for [Transportation Management](#) under [Integration > Logistics Integration > Internal TM Component Integration > Define Logistics Integration Profile for SD, MM, and LE](#).

- o If you want to use **early updates** from EWM, make sure you have configured the system as described in the prerequisite section in [Early Updates from EWM](#).

- **Extended Warehouse Management: Handling Unit (HU) Number Ranges**

For STO processes, the HU number ranges of the issuing and the receiving EWM warehouses must not have any overlaps. Make sure that the HU number ranges are disjunct.

For more information, see Customizing for [SCM Extended Warehouse Management](#) under [Extended Warehouse Management > Cross-Process Settings > Handling Units > External Identification > Define Number Range for HU Identification](#).

Outbound Process (Sending Warehouse)

This sample process for the integration based on EWM transportation units describes a high-level STO process with its combined inbound and outbound processes.

For more information about the outbound process, see [Example Process: Delivery- and Order-Based Planning \(Outbound\)](#).

Preparation in Materials Management

For an order-based STO process, create a purchase order (PO) with PO type **Stock Transport Order**. Freight units are created automatically and assigned to the STO.

Transportation Management

1. Plan your transportation activities by creating a road freight order based on the freight units created in the previous step.
2. Create an outbound delivery for the freight order.

The system automatically creates an **outbound delivery** in Delivery Management (LE-SHP) that triggers the creation of an outbound delivery in EWM, that is, in the sending warehouse.

3. When you have finished planning your transportation activities, you set the load plan status of the freight order on stop level to **Finalized**.

The system automatically sends a loading instruction to EWM with a `LoadingAppointmentRequest` message.

Extended Warehouse Management

When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates an **outbound transportation unit**.

1. The standard EWM activities (such as truck arrival, picking, packing, and loading) take place in EWM, that is, in the sending warehouse.
2. The warehouse worker posts goods issue for the truck. The LE outbound delivery is automatically updated.

In the STO process, posting goods issue initiates the creation of an LE **inbound delivery** for the STO. The LE inbound delivery is then replicated to the receiving warehouse.

3. The warehouse worker posts the departure of the truck. EWM sends the **LoadingAppointmentNotification** message to TM.

Transportation Management

1. When TM receives this message, it updates the item hierarchy of the freight order with the EWM transportation unit data and updates the associated freight units. The freight units are then reassigned from the LE outbound delivery to the LE inbound delivery. The status of the freight order is set to **Departed** for the stop of the sending location.
2. Sending unloading instructions to the receiving warehouse

Now the combined inbound process continues for the STO: Once the freight order has been posted **Departed**, TM can assume that the LE inbound delivery is available for the STO items that are loaded on the truck and informs the **receiving warehouse**.

This means: A **LoadingAppointmentRequest** message with **unloading instructions** is sent to the receiving warehouse, and an **inbound transportation unit** is created in the receiving EWM-managed warehouse.

If you configure the **Automatic Sending of Unloading Instructions on Departure** in STO Customizing, the system automatically sets the Unload Plan (Stop) status to **Finalized** when departure is posted in EWM, and then automatically sends unloading instructions to EWM. In a multi-pickup scenario, the Unload Plan (Stop) status for the receiving EWM-managed warehouse will be set to **Finalized** only when the last departure of all sending EWM-managed warehouses is posted.

If you don't configure the **Automatic Sending of Unloading Instructions on Departure**, you need to set the Unload Plan (Stop) status manually to **Finalized**.

Inbound Process (Receiving Warehouse)

For more information about the inbound process, see [Example Process: Delivery- and Order-Based Planning \(Inbound\)](#).

Execution in EWM

1. In EWM, you can carry out the steps that are relevant for the warehouse, such as truck arrival, unloading, posting goods receipt, and truck departure.
2. When goods receipt is posted, the LE inbound delivery is updated automatically.
3. When the departure of the truck is posted, EWM sends the **LoadingAppointmentNotification** message to TM.

Update in TM

1. When the **LoadingAppointmentNotification** message is received, TM updates the freight order.

i Note

In the inbound scenario, EWM does not send information on cargo back to the freight order in TM when the truck departure is posted in EWM. Therefore the item hierarchy is not changed in the freight order, even if the items were not all received in EWM.

In the STO process, you cannot activate the sending of **LoadingAppointmentNotification** messages with item details, as described in SAP note [2355948](#) .

2. You complete freight order execution.

Related Information

[Integration of Stock Transport Orders](#)

Unified Package Building in Integration Based on EWM TUs

The integration of Transportation Management (TM) and Extended Warehouse Management (EWM) is based on the EWM transportation unit (TU) and supports unified package building **during freight unit building** and unified package building **based on capacity documents**.

Use

You can have the system create packages as follows:

- When freight units are created

This type of unified package building takes place **early** in the process when the system creates freight units based on order documents (for example sales orders).

- Based on capacity documents

This type of unified package building takes place **late** in the process when capacity documents exist already.

The process steps are optional. In EWM, information about packages is represented by a **handling unit (HU)**.

Prerequisites

- For more information about prerequisites for unified package building, see [Unified Package Building in TM](#).
- If you want to use planned shipping handling units (PSHUs), make sure the delivery item has a process profile for cartonization planning in which the **Create Planned Shipping HUs for Outbound Delivery Orders** checkbox is selected.

For more information, see Customizing for Extended Warehouse Management under **Goods Issue Process** > **Cartonization Planning** > **Define Process Profiles for Cartonization Planning**.

For more information about PSHUs in EWM, see [Planned Shipping Handling Unit](#).

Features

Unified Package Building During Freight Unit Building

Saving a sales order initiates the **unified package building process during freight unit building**. The system creates a complete hierarchy of package and product items that is displayed in the items list of a freight unit. Alternatively, you can configure the system to initiate unified package building once a delivery, created to a sales order in SD or LE, is saved. For more information about the integration options between TM and logistics, see [Logistics Integration Profile for SD, MM, and LE](#).

Unified Package Building Based on Capacity Documents

If you select **Build Packages** in a capacity document or in the transportation cockpit, the system creates a complete hierarchy of package and product items that exist only locally in the document.

Send Loading or Unloading Instruction to EWM

With the [LoadingAppointmentRequest](#) the system automatically sends a loading or unloading instruction to EWM when planning in TM is finished. The system creates PSHUs in EWM based on the sent information about package items. The PSHUs are created during cartonization planning.

PSHUs offer a **packing suggestion** from TM. They can be used as templates to create shipping HUs during picking or packing. Warehouse workers can either pick and pack as suggested, modify or reject the suggestion (and pick and pack differently). The changes will be updated accordingly in TM.

For more information about an example process, see [Example Process: Delivery- and Order-Based Planning \(Outbound\)](#).

Cross-Delivery Handling Units

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on Extended Warehouse Management transportation units (TUs) supports cross-delivery handling units (HUs) for outbound processes. The feature is relevant for shipper scenarios.

Use

You can cross-pack and fill up handling units like cartons or pallets with freight units resulting from two or more deliveries. This helps to improve packing and saves volume if a package or pallet has less than a full load from only one delivery. **Single-stage** freight units are supported.

TM-driven scenario: In a freight order you can build packages for product-items from **different** deliveries in the **same** package and process them in EWM in the corresponding EWM-TU. The EWM-TU has then a cross-delivery HU as suggested packing instruction.

EWM-driven scenario: In EWM you can cross-pack packages or product-items resulting from different deliveries in the same package. As usual, the system transfers the actual item data to TM to update the corresponding freight order.

The system uses **local packages** in the freight order to store the freight units with their document references to the different deliveries. The cross-delivery package is kept local in the freight order, thus the cross-delivery related package is not referenced into the freight unit.

In the EWM transportation unit you can identify HUs that are cross-delivery HUs.

Related Information

[Example Process: Delivery- and Order-Based Planning \(Outbound\)](#)

[Execution-Driven Planning \(Outbound\)](#)

Quantity Deviation

With the integration of Transportation Management (TM) and Extended Warehouse Management (EWM) you can manage situations where you need to deliver more or fewer products than the planned quantity.

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Use

When goods are loaded in EWM, warehouse workers may be able to load only a fraction of the planned quantity onto the truck. Sometimes less than the planned quantity of a product is available in stock. Another reason may be that the truck is different to the one planned and has a lower capacity. In both cases the quantity deviates from the planned quantity.

Quantity deviation in the TM-EWM integration:

- **You Deliver Remaining Amount Later**

You want to deliver the available products now and plan to deliver the remaining products at a later point in time. This generates a delivery split in Logistics Execution (LE), a freight unit split, and also impacts the freight order in TM. For more information, see [Delivery Split](#).

For more information about delivery splits in EWM, see [Delivery Split for Outbound Deliveries](#).

- **You do not Deliver Remaining Amount Later**

You want to deliver the available products now and for whatever reason do not plan to deliver the rest at a later point in time. In that case, the product **amounts can be adjusted** in EWM, indicating that the missing products will not be delivered in the future. The related delivery will be adjusted accordingly.

Delivery Split

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports delivery splits.

Use

When goods are loaded in EWM, warehouse workers may only be able to load a fraction of the planned quantity on to the truck. The quantity deviates from the planned quantity.

You want to deliver the available products now and plan to deliver the remaining products to a later point of time. This generates a delivery split in Logistics Execution (LE), a freight unit split and also impacts the freight order in (TM).

As an example, a freight order with an unpacked product item is sent to EWM. There, the product is loaded, but not all pieces are available. On posting goods issue, EWM initiates the creation of a new delivery. Both the old and the recently created **split delivery** are assigned to the freight order in TM. A **freight unit split** is executed accordingly and freight units (FUs) are assigned as follows:

- The old FU is assigned to the new split delivery.
- The new split FU is assigned to the old delivery.

For more information, see [Quantity Deviation](#).

For more information about delivery splits in EWM, see [Delivery Split for Outbound Deliveries](#).

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

You have set the entry for subsequent delivery split in customizing:

In customizing choose, ► **Logistics Execution** ► **Shipping** ► **Deliveries** ► **Subsequent Delivery Split** ►. Select a **Split Profile** and doubleclick **Per Delivery Type** to assign a profile to the delivery type.

Process

A sample process is described below on the basis of a sales order:

Execution in EWM

1. When loading the truck, warehouse workers can only load some of the overall quantity and have to leave an entire delivery item.
2. The warehouse workers report the change in quantity and post goods issue. A new delivery in LE (split delivery) is created.
3. The warehouse workers post the departure of the truck. EWM sends the **LoadingAppointmentNotification** message to TM. With this message EWM reports the change in quantity or the delivery items that are not included (or both) to TM.

Update and Further Steps in TM

1. In TM, the quantity in the freight unit is automatically reduced. If the warehouse workers could not include an item, TM sets the remaining quantity to zero.

If required, the system also **splits the freight unit**. This means that the system reduces the quantity in the original freight unit and creates an additional freight unit with the remaining quantity. You must then **reschedule** this freight unit. The system assigns the new freight unit to the old delivery and the old freight unit to the new split delivery.

2. Both the old and the recently created split delivery are assigned to the freight order in TM.

Updates in Sales and Distribution (SD)

The sales order is updated and includes links to both freight units and both deliveries.

Batch Split

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports batch splits.

Batches do not have to come from Logistics Execution (LE) but can instead be determined in EWM.

- Warehouse workers load for example the rest of one batch (50 items of a product) on a truck, the other 50 items are loaded from another batch. This causes a line item split in the system when reported. In TM, the batch information for the product items is stored in the freight order as **subitem** of a product item.
- A delivery can also be **split** at the same time as a batch. Warehouse workers load for example 50 items of a product of one batch on a truck, the other 50 items from the **same** batch will be delivered at a different point in time. For more information, see [Delivery Split](#).

It is possible to load one batch per pallet or distributed/split batches per pallet. You can also load distributed batch-managed product items and non-batch-managed product items on a pallet.

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Process

A sample process is described below on the basis of a sales order:

1. Create a freight order in TM with items that do not have any batch numbers.

2. Picking of certain batch numbers is possible. You can also enter the batch number in the warehouse task.
3. When the message `LoadingAppointmentRequest` is received from TM, EWM determines automatically whether there are any batches. If there is more than one batch and these batches need to be split, EWM creates subitems.
4. When EWM reports the data to TM using the message `LoadingAppointmentNotification`, TM automatically creates any required subitems in the freight order and in the related freight units. The batch number is then visible on the **Content Identification** tab page (details of the cargo item).

Seals

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports the usage of seals. You use seals in order to protect vehicle resources and containers from unauthorized access during transportation. Sealing information can be entered in EWM or in TM.

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Entering Seals in EWM

1. **EWM:** In the outbound delivery order, open the transportation unit and then enter the sealing information, for example, a seal number, the partners who seal or unseal, and the date and time for sealing or unsealing. For more convenience, switch to the **Form View**.

EWM reports the data automatically to TM using the message `LoadingAppointmentNotification` when the status **Departure from Checkpoint** is set.

2. **TM:** After processing the message in TM, the item hierarchy and sealing information are updated in the freight order.

Entering Seals in TM

You can enter sealing information for vehicle resources and containers.

1. **TM:** Enter the sealing information on the seals tab page of the freight order, for example, a seal number, the partners who seal or unseal, and the date and time for sealing or unsealing.

TM reports the data automatically to EWM using the message `LoadingAppointmentRequest` when the status **Load Plan (Stop) Finalized** is set.

2. **EWM:** In the outbound delivery order, open the transportation unit and display or update the sealing information that was transferred from TM. For more convenience, switch to the **Form View**.

EWM reports the data automatically to TM using the message `LoadingAppointmentNotification` when the status **Departure from Checkpoint** is set.

3. **TM:** After processing the message in TM, the item hierarchy and sealing information are updated in the freight order.

Related Information

[Use of Seals](#)

Transportation Planning with Trailers

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports planning and execution with trailers.

Data that is received from EWM is confirmed to all affected business documents across several document levels. This means that in addition to the freight order, the data is also confirmed to one or more trailer units and freight units.

Prerequisites

In the resource master data you have defined a vehicle resource for your tractor (truck) and a passive vehicle resource for your trailer.

In customizing for **Transportation Management**, choose ► **Master Data** ► **Resources** ► **Means of Transport and Compartment** ► **Define Means of Transport** ►.

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

Process

A **sample process** is described below for transportation planning with a trailer based on **outbound deliveries**.

Preparation in SD

1. You create a **sales order** in Sales and Distribution (SD).
2. You create **outbound deliveries** in SD and distribute them to TM and EWM.
 - With the delivery, freight units are automatically created in TM, in EWM an outbound delivery order is created.
 - **Only relevant for integration with external logistics systems:** In TM, a delivery-based transportation requirement (DTR) is automatically created along with the associated freight units. In EWM, an outbound delivery request and an outbound delivery order are created automatically. For more information, see [External TM System Integration](#).

Transportation Planning in TM

1. You plan your transportation activities. When you do so, you create a freight order manually or the system creates one automatically. Also create a trailer unit and assign a passive vehicle resource (trailer) to it. Then assign the trailer unit to the freight order.
2. You set the **load plan status** of the freight order on stop level to **Load Plan Finalized**. The system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

Execution in EWM

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates transportation units. The updated delivery has then two transportation units: one transportation unit representing the truck and one representing the trailer.
2. Picking is carried out by the warehouse workers. The system can create handling units automatically.
3. When the truck arrives at the yard, the warehouse workers report its arrival and assign a door to it.
4. The warehouse workers load the truck and the trailer. For more information, see [Loading and Unloading](#).
5. The warehouse workers post the goods issue. The outbound delivery is automatically updated in Logistics Execution (LE).
6. The warehouse workers post the departure of the truck. EWM sends the `LoadingAppointmentNotification` message to TM.

Update and Further Steps in TM

1. Only relevant for integration with external logistics systems: When the data is updated in LE, the delivery-based transportation requirement (DTR) is automatically updated in TM.
2. When the `LoadingAppointmentNotification` message is received, TM updates the freight order and the corresponding freight units and sets the execution status to **In Execution**. The cargo receipt status at item level is automatically set to **Shipped**. If the item in question is to be unloaded at the next transportation stop, the handling execution status is automatically set to **Not Unloaded**.
3. Complete freight order and trailer unit execution.

More Information

[Quantity Deviation](#)

[Delivery Split](#)

[Batch Split](#)

[Seals](#)

Early Picking

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports early picking of products.

In this process, picking in EWM starts early - before a freight order is created and TM therefore before a transportation unit is created in EWM. Without having a transportation unit in the system, warehouse workers already start picking (and possibly packing) on basis of the delivery. In the early picking process, the delivery in Logistics Execution (LE) and the freight units in TM are updated right after picking (for example with batch information).

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

You have defined that executive steps in EWM can be performed without planning for the outbound delivery.

In Customizing choose, ► **SCM Extended Warehouse Management** ► **Extended Warehouse Management** ► **Goods Issue Process** ► **Outbound Delivery** ► **Integration with Transportation** ► **Allow EWM Execution Without Transportation Planning (Outbound)** ►.

Process

A **sample process** for the early picking process is described below.

Preparation in SD

1. You create a **sales order** in Sales and Distribution (SD).
2. You create outbound deliveries in SD and distribute them to TM and EWM.

The system creates an outbound delivery order in EWM and freight units in TM.

Execution in EWM

1. You open the [outbound delivery order](#).
2. You unblock the delivery and set the status for transportation planning to [Fixed Manually](#).
3. The warehouse workers start early picking (and packing) prior to the transportation unit being created in EWM.

i Note

Note that the delivery in Logistics Execution (LE) and the freight order in TM are not yet updated with the package information. The delivery in LE is updated when warehouse workers post goods issue for the delivery in EWM. Freight order and freight units in TM are updated when the departure of the truck is posted in EWM.

Preparation in TM

1. You create a freight order and assign the freight units.
2. You set the **load plan status** of the freight order on stop level to [Load Plan Finalized](#). The system sends the data automatically to EWM using the message `LoadingAppointmentRequest`.

Execution in EWM

1. When the message is received from TM, EWM automatically creates transportation units.

i Note

The freight units and freight order still do not contain the packing information. This information is sent later with the LDAP notification message from EWM.

2. Warehouse workers pack (if not yet done) and load on the transportation units. For more information, see [Loading and Unloading](#).
3. After the warehouse workers post the **goods issue**, the outbound delivery is updated in LE.
4. When the warehouse workers post the **departure** of the truck, EWM sends the `LoadingAppointmentNotification` message to TM.

Update and Further Steps in TM

1. When the message is received, TM updates the freight order with the corresponding freight units.
2. You complete the freight order execution.

Multi-Pickup, Multi-Drop

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) based on EWM transportation units (TUs) enables you to plan for multiple warehouses on one road freight order. This is also known as a multi-pickup and multi-drop scenario. It is relevant for inbound and outbound scenarios.

Prerequisites

In EWM, you have defined a warehouse with a shipping point to enable the multi-pickup and multi-drop scenario for the integration based on EWM TUs. In TM, the warehouse is represented as a **main stop** and can be used as the start or end point of a stage in the freight order.

Shipping points in EWM can be defined for Advanced Shipping and Receiving and for the integration based on EWM TUs. This topic describes the multi-pick, multi-drop scenario in the integration based on EWM TUs. For more information about Advanced

Shipping and Receiving, see [Advanced Shipping and Receiving](#).

Optional: You can combine the use of multiple warehouses in one road freight order with “early updates from EWM”. For more information about prerequisites, see [Early Updates from EWM](#).

Features

By using multiple warehouses for picking up or dropping off handling units, you can improve your freight order planning.

You can use the **Overview** tab in the freight order to update the status values for the multi-pickup and multi-drop scenario. Here you can mark single main stops, and set the status **Load Plan Finalized** or **Unload Plan Finalized** for outbound or inbound scenarios.

This status initiates the sending of the message `LoadingAppointmentRequest` (LDAP request) to the relevant warehouses. This is not restricted to the current main stop. You can also send LDAP requests for all main stops representing warehouses at once. Nevertheless, the stop sequence that was planned in TM must be kept in order when warehouse activities are executed in the multiple warehouses.

If you set the main stop status in the freight order on **top level**, the system still sets the status for the **next** stop only.

Canceling Freight Orders, Canceling Stops (Stages)

You can cancel freight orders or stops in a multi-pick, multi-drop scenario. Canceling might be necessary if a transportation plan changes in the middle of the process. You can cancel whole freight orders, or one or several stops (stages) of a freight order.

- **Canceling Freight Orders or Stops in TM**

You can cancel a freight order if it is not yet in execution.

You cannot cancel a freight order if a truck (an EWM transportation unit) has already arrived at the gate of one of the multiple warehouses, and the handling execution status at the stop level has been changed to **Arrived**. In some cases, you can reverse the arrival of the truck, and then cancel the freight order or stop.

You cannot cancel a freight order if the load plan status on stop level was set to **Load Plan Finalized** or **Unload Plan Finalized**, and the loading or unloading instructions have already been sent to EWM. In this case, the system needs to cancel the EWM-relevant stops first. This is done in the following way: When you choose **Cancel Document** in the freight order, the system sends a sequence of cancellation requests for all stops of the freight order to EWM, for one stop after the other. The system waits for a cancel confirmation from EWM before the next cancellation request is sent to EWM. EWM will not confirm a cancellation if arrival has already been posted.

The freight order is still in process until the cancellations for all stops have been sent and confirmed. When the cancellation is confirmed for all stops, the system sets the **Life Cycle Status** of the freight order to **Canceled**.

You cannot reverse the canceling of a freight order document or stop.

- **Canceling Transportation Units in EWM (Canceling Single Stops in TM)**

It happens that a freight unit is not shipped with the planned freight order, but will be planned to another freight order, for example, if a driver does not pick up any material at one customer stop and drives directly to the next stop. In such a case, EWM cancels the single transportation unit, which triggers the cancellation of the corresponding stop in TM and an update of the stage list. In addition, the system unassigns the corresponding FUs. If the last existing stop in TM is canceled, the system will cancel the whole freight order.

Combinations and Features

Various features are available for the multi-pick, multi-drop scenario, such as:

- For shipper scenarios, you can plan multiple warehouse-related main stops in one road freight order.
- The TU-based integration only supports main stops. Sub-stops are not supported.
- You can plan inbound- and outbound deliveries in one freight order when they are planned for different main stops or shipping points.
- At a warehouse, you can either load or unload items.
- Scenario Examples:
 - **Multi-pickup and single drop (outbound):** You can pick up handling units at multiple warehouses, for example pick up some handling units at warehouse A, pick up more handling units at warehouse B, and deliver (drop) all items at one customer's stop.
 - **Single pickup and multi-drop (inbound):** You can pick up handling units at your supplier's stop, drop handling units at your TU-managed warehouse A, and drop the rest of the handling units at your TU-managed warehouse B.
 - You can combine inbound and outbound scenarios in one freight order.
- You can combine stops from TU-based integration with IM-managed warehouses in one road freight order. IM-managed means warehouses managed by Inventory Management and Physical Inventory (MM-IM).
- You cannot combine stops from TU-based integration with stops related to Advanced Shipping and Receiving.
- Multi-pickup, multi-drop is not supported for logistics service providers (LSP), Transit Warehousing, and Stock Room Management (STRM).
- For information about the supported functions and the restrictions, see SAP Note [3158252](#) 📄.

Example Process

Proceed as follows, to enable a multi-pickup and multi-drop scenario for the TU-based integration.

For more information about the overall process, see:

- [Example Process: Delivery- and Order-Based Planning \(Outbound\)](#)
- [Example Process: Delivery- and Order-Based Planning \(Inbound\)](#)

Transportation Planning in TM

1. You create a freight order (FO).

You create a road freight order with a FO type for an inbound or outbound scenario. Enter a carrier.

2. You assign freight units (FUs).

To assign FUs to the freight order, choose deliveries for different warehouses with TU-based shipping points. As a result, the system creates two or more stages. You can display them on the [Overview](#) tab.

3. You reschedule the freight order.

After inserting deliveries or freight units for different warehouses to a freight order, you need to reschedule the stage data manually. Choose [Backward Scheduling](#) or [Forward Scheduling](#). Save your freight order.

4. You finalize the load plan.

You set the load plan status of the freight order on stop level to [Load Plan Finalized](#) or [Unload Plan Finalized](#).

- Set the status for the next stop only:

If you set the stop status on top level in the freight order, the system sets the status for the next stop only.

- Set the status for one or several stops at once:

If you mark one or several main stops, and set the status on the **Overview** tab, the system sets the status for the selected stops (no matter which stop is next in sequence). This allows the warehouse workers to already start with picking and packing in any of the warehouses.

After you set the status on stop level to **Load Plan Finalized** or **Unload Plan Finalized**, the system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

Execution in EWM

When the message is received from TM, EWM automatically creates a transportation unit, one TU for each stage.

1. Perform EWM tasks including truck departure as usual. With the departure of the truck, EWM sends the `LoadingAppointmentNotification` message to TM.

The handling execution status of the stop in the freight orders is adjusted accordingly.

2. Check whether all messages to TM were sent correctly:
 - Display the PPF actions to check if you need to trigger the resending manually.
 - Enable the **Update Check for Deliveries**. The check verifies that content-related messages are not processed before the delivery has been updated. For more information, see [Reprocessing of Failed Loading Appointment \(LDAP\) Messages](#).

Update and Further Steps in TM

1. When the `LoadingAppointmentNotification` message is received, TM updates the item hierarchy of the freight order with the EWM transportation unit data and updates the associated freight units.
2. You complete freight order execution.

Related Information

[Automatic Status Updates \(Multi-Pickup and Multi-Drop Scenario\)](#)

Execution-Driven Planning (Outbound)

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports execution-driven processes.

In an execution-driven process, an **empty** freight order is sent from TM to EWM. The transportation unit (TU) created in EWM contains no predefined items and the warehouse execution is done without a planning suggestion from TM. The warehouse workers pick, pack, and load the products on the empty transportation unit. The transportation units are assigned to the freight order when it is finally updated after the process is completed.

Prerequisites

For more information, see [Prerequisites for Integration Based on EWM TUs](#).

- You have created an **EWM Integration Profile** where **Send (Un)Loading Instructions Only Once** is defined as **Delta Handling Setting**.

- You have created a **freight order type** with the integration profile created above assigned to it.

In Customizing for **Transportation Management**, choose ► **Freight Order Management** ► **Freight Order** ► **Define Freight Order Types** ►.

- You have defined that processing the outbound delivery in EWM can be performed without planning in TM. This means that you allow the outbound delivery in EWM to be processed before the status **Load Plan Finalized** on stop level is set in the freight order (even before a freight order in TM and a transportation unit in EWM is created).

In Customizing for **SCM Extended Warehouse Management**, choose ► **Extended Warehouse Management** ► **Goods Issue Process** ► **Outbound Delivery** ► **Integration with Transportation** ► ► **Allow EWM Execution Without Transportation Planning (Outbound)** ►.

In customizing for **Transportation Management**, choose ► **Integration** ► **Integration with Extended Warehouse Management** ► **Define EWM Integration Profile** ►.

Process

A **sample process** is described below based on an outbound delivery created as successor document of a sales order.

Preparation in SD

1. You create a **sales order** in Sales and Distribution (SD).
2. You create **outbound deliveries** in SD and send them to TM and EWM.

In EWM, an outbound delivery order is created. In TM, freight units are created automatically. The freight units are not yet assigned to a freight order.

Preparation in TM

1. You create a freight order and enter at least a source location (shipping point), destination location and a delivery date.

You set the **load plan status** of the freight order on stop level to **Load Plan Finalized**. The system automatically sends the **empty** freight order to EWM with the message **LoadingAppointmentRequest**. For your convenience, you can create a stock of such empty freight orders.

2. After sending the message, the freight order is blocked for editing (see prerequisites).

Execution in EWM

1. When the message is received from TM, EWM automatically creates an **empty** transportation unit. The system chooses the relevant EWM warehouse based on the information in the shipping point.
2. The warehouse workers use this empty transportation unit, search manually for the delivery, and assign it to the empty transportation unit.
3. Picking, packing, and loading are done as usual. For more information, see [Loading and Unloading](#).

In addition, you can enable updates from EWM to TM with status **Loading Finished**. For more information, see [Early Updates from EWM](#).

4. The warehouse workers post the **goods issue**. The outbound delivery is automatically updated in Logistics Execution (LE).
5. The warehouse workers post the **departure** of the truck. EWM sends the **LoadingAppointmentNotification** message to TM.

Update and Further Steps in TM

1. When the message is received, TM updates the freight order with the corresponding freight units. The freight order item hierarchy then includes packages and product items from the delivery items that have been packaged in EWM.
2. You complete the freight order execution.

More Information

[Quantity Deviation](#)

[Delivery Split](#)

[Batch Split](#)

[Seals](#)

Reversal of Departure

The integration of Transportation Management (TM) with Extended Warehouse Management (EWM) supports the reversal of the status **Departed from Checkpoint** in EWM.

Reversal of the departure and the sending of a second departure message from EWM to TM is available for inbound and for outbound processes. For information about the supported functions and the restrictions, see SAP Note [3018355](#) .

Prerequisites

You have activated reversal of the departure in the Customizing for [Extended Warehouse Management](#) under [SCM Extended Warehouse Management > Extended Warehouse Management > Interfaces > TM Integration > Make Settings for TM Message Processing](#).

Features

In an outbound process, it is sometimes necessary to reverse the departure of a truck from the checkpoint.

Example 1: Departure Was Set Too Early

Reversing the departure is necessary when, for example, a departure was set for the wrong truck, or the departure was unintentionally posted too early although the truck has not yet left the location.

- In such cases, a warehouse clerk can reverse the departure in EWM, and the system sends an update message for the freight order to TM. In TM, the current departure event is deleted, and the handling execution status is set back to **Loaded**. Everything in the freight order is reverted to the state it had after **Finished Loading**.
- Later, when the second, actual departure event is posted in EWM, the system again sends an update message to TM. In TM, a new departure event with the new date is created, and the handling execution status is set back to **Departed**.

If the truck has not yet arrived at the next stop, the reversal of the departure can be used to update the departure time. If the truck has already arrived at the next stop, the reversal of the departure is mainly used to initiate a correction of the truck load.

Example 2: Transportation Unit or Delivery Needs to be Corrected

Reversing the departure is also necessary when, for example, pallets were loaded or not loaded but were already reported to the system, or when a truck load is too heavy and the load needs to be adjusted. The reversal of the departure is a prerequisite for the changes to a transportation unit or delivery.

When you reverse the departure of the truck, you can correct the data, for example, add or remove deliveries or handling units, or decrease or increase product quantities.

In the case of a multi-pickup scenario from EWM-managed warehouses, reversal of the departure is only possible if the arrival at the next EWM-managed warehouse was not reported yet. If the arrival has already been reported, you must reverse the arrival before you can reverse the departure.

The freight order execution information is used to determine which changes that were made in EWM are reflected in TM. This means which events and statuses are to be updated in the freight order.

Updates to deliveries, handling units, or product quantities are always reflected in TM.

Note the following:

- **Truck has already arrived at the next stop**

The system will **not** update the freight order with any event or status changes for the first stop as it is not relevant anymore. For example, it does not make sense to correct the truck departure when the truck has already arrived at the next stop.

- Event dates for **Loading Complete** and **Departure** remain the same as when they were posted for the first time.
- The handling execution status is not changed.

- **Truck has not yet arrived at the next stop**

If no arrival at the next stop is posted, the event dates and the handling execution status are updated when the departure reversal message and update messages (like the reversal of the loading end or reversal of goods issue) are received from EWM.

- For newly added items, the handling execution status is set to **Loaded**.
- A new departure event with the new date is created, and the handling execution status is set back to **Departed**.
- If the unloading of the receiving stop is reported already, the assumption is that the new items are also unloaded and get the same status.

Transit Warehousing

Use

Transit warehousing enables you to seamlessly integrate the planned flow of goods in Transportation Management (TM) and the physical flow of goods in Extended Warehouse Management (EWM) over the entire transportation chain. The integration is based on execution documents resulting from forwarding orders.

With this process you can store cargo temporarily in transit warehouses or gateways (air freight) or in container freight stations (ocean freight) until it is transported. You can also organize cargo over the entire transportation chain and get transparency over warehouse-specific data. The process of handling cargo at a location can be repeated for several locations.

In EWM, the warehouse-specific processes are based on business documents that are sent from TM using a loading or unloading instruction. You can also configure the warehouse layout according to the requirements of the transit warehousing process. For example, specific storage types are available and a product master is no longer necessary. The handling of cargo is based on package level and not on product level.

For more information about the **processes in EWM**, see SAP Help Portal:

- For EWM, see <https://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management** ► **Transit Warehousing** ►.
- For SAP EWM, see <http://help.sap.com/ewm> under ► **Application Help** ► **SAP Extended Warehouse Management (SAP EWM)** ► **Transit Warehousing** ►.

Prerequisites

- You have defined your transportation network. You have also defined a logical system (EWM system) and a warehouse number for the locations whose warehouses you manage using EWM. You have sent this data to EWM using report /SCMTMS/BUPA_SEND_TRANS_WH.
- You have defined an EWM integration profile and assigned it to your document types (that is, to your freight order types and freight booking types). Among other things, you can specify in the EWM integration profile whether loading and unloading instructions are to be sent manually or automatically and whether they can be sent once or multiple times.

Moreover, you have to specify in this profile that item changes are propagated by the system. This means that you can still change the forwarding order until the goods have physically arrived in the warehouse and the actual quantities and attributes have been reported by EWM. The changes are propagated to the freight order.

For more information, see Customizing for **Transportation Management** under:

- ► **Integration** ► **Integration with Extended Warehouse Management** ► **Define EWM Integration Profile** ►
- ► **Freight Order Management** ► **Freight Order** ► **Define Freight Order Types** ►
- ► **Freight Order Management** ► **Freight Booking** ► **Define Freight Booking Types** ►
- You have defined a discrepancy profile and assigned it to your document types (that is, to your freight order types and freight booking types). For more information, see Customizing for **Transportation Management** under:
 - ► **Freight Order Management** ► **Define Discrepancy Profile** ►
 - ► **Freight Order Management** ► **Freight Order** ► **Define Freight Order Types** ►
 - ► **Freight Order Management** ► **Freight Booking** ► **Define Freight Booking Types** ►
- You have made the following additional settings in your document types (that is, in your freight order types and freight booking types):
 - You have selected the **Transportation Activity Creation in EWM** option in the **Document Creation Relevance** field.
 - This setting is optional if in the EWM integration profile in the **Send. (Un)Load. Instr.** field ("Sending of (Un)Loading Instructions") you have selected the option **Manual Sending of (Un)Loading Instructions** or **Automatic and Manual Sending of (Un)Loading Instructions**.
 - This setting is required if in the EWM integration profile in the **Send. (Un)Load. Instr.** field ("Sending of (Un)Loading Instructions") you have selected the option **Automatic Sending of (Un)Loading Instructions**.
 - You have specified an output profile that contains actions for sending loading/appointment messages, for example, /SCMTMS/TOR.
 - If you are working with freight orders for customer self-delivery and pick-up, you have configured the following settings in your freight order type:
 - In the **Freight Order Can Be Subcontracted** field, you have selected the option **Not Relevant for Subcontracting**.
 - In the **Shipper/Consignee Determination Rule** field, you have selected the option **Shipper/Consignee Not Relevant**.

- For freight orders for customer self-delivery, you have selected the option **Star-Shaped Based on FU Stages (Reversed)** in the **Sequence Type of Stops** field. For freight orders for customer pick-up, you have selected the option **Star-Shaped Based on FU Stages**.
- In the **Self-Delivery/Customer Pick-Up** field, you have selected the option **Freight Order for Customer Self-Delivery or Pick-Up**.
- In the **Web Dynpro Application Config.** field, you have entered the application configuration **/SCMTMS/FRE_ORDER_SDCP**.

The SAP standard system provides the freight order types 4010 for customer self-delivery and 5010 for customer pick-up in the transit warehousing process.

For more information, see Customizing for **Transportation Management** under:

- **Freight Order Management** > **Freight Order** > **Define Freight Order Types**
- **Freight Order Management** > **Freight Booking** > **Define Freight Booking Types**
- SAP provides the following, predefined event codes for cancelations as well as the corresponding event reason codes:

Event Code	Description	Event Reason Code	Description
CANCEL_EXEC_DOC	Cancel Execution Document	CANCELLED_BY_CARRIER	Canceled by Carrier
CANCEL_STATUS_RESET	Cancel and Reset Cargo Execution Status	TM_PLAN_NOT_READY	TM Planning Not Yet Ready
CANCEL_UNASSIGN_FUS	Cancel and Unassign FUs of Current Stop (cancel and unassign the freight units for the current stop)	STOP_CANCELLED	Loading Appointment Canceled (loading cannot be executed in the warehouse as planned)

These event codes are set as the default codes.

You can define your own event codes and event reason codes. For each event type, you must specify at least one of your event codes as the default code.

For more information, see Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **Define Event Reason Codes**.

Features

The following processes are supported:

- Transit Warehousing Process (see [Transit Warehousing Process](#))
- Planning with trailers (see [Planning with Trailers](#))

Handling Discrepancies

Discrepancies and exceptions may occur during the transit warehousing process. For more information, see [Handling Discrepancies in Transit Warehousing Process](#).

Handling Unexpected Package Items

If the confirmation from EWM contains unexpected package items and you have configured your TM system in such a way that no discrepancies are to be created, TM automatically reassigns these package items. For more information, see [Automatic](#)

Splitting Package Items for Distribution to Several Transportation Units

When goods are being transported between transit warehouses or to a gateway they may have to be redistributed to other transportation units, such as trailers. You can do this by splitting the package item of a freight order that has already been assigned to a transportation unit and assigning it to several transportation units. You can also drag and drop package items between transportation units.

Once EWM has posted the goods as received, the system updates automatically the item hierarchy in the freight order with information about the handling units from EWM. Note, however, that this splitting function is integrated with EWM only for trailer units.

For more information, see [Cargo Management](#).

Message Exchange with EWM

Loading and unloading instructions are sent automatically from the freight document. This is triggered by a status change (see [Transit Warehousing Process](#)). You can also trigger the sending of these instructions manually and multiple times provided you have made the required settings in your EWM integration profile (see [Prerequisites](#)). You can also cancel a sent loading or unloading instruction.

You can receive multiple updates in TM from EWM and update the freight documents and transportation units to which the message from EWM refers.

Transmission-to-EWM Status provides information about the status of the exchange of messages. You can get an overview of all the status values relevant for the transit warehousing process, such as load or unload plan status, load or unload plan item status, handling execution status, and cargo execution status, in the freight document item table in the **Status Management** hierarchy view. For more information, see [Explanation of Statuses of Business Documents](#).

i Note

- You can send a loading or unloading instruction for a future stop (for example, a transit warehouse) even if execution of the transportation has not finished at the present stop. If you set the load or unload plan status at header level of the freight order to **Finalized**, the system searches for the next relevant stop, sets the status for this stop, and automatically sends a message to EWM. If you set the status at stop level, the system automatically sends a message to EWM for this stop.

The location master data determines which stops are relevant for loading or unloading. You can use the BAdI **Determination of Warehouse Relevance** to change this standard logic and add locations, for example. For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BAdIs) for Transportation Management** > **Freight Order Management** > **Freight Order Management** > **BAdI: Determination of Warehouse Relevance**.

- You can also send and cancel loading and unloading instructions for transportation units if these transportation units are included in a freight document.

Canceling Freight Documents

In the transit warehousing process, you can still cancel freight orders and freight bookings up until the goods have arrived in the warehouse and have been reported by EWM as **Arrived**. If the status **Unload Plan Finalized** has already been set and you cancel the freight document, the system sends a cancellation request to EWM. This request is checked and processed by the EWM system as follows:

- If the goods have not yet arrived in the warehouse, EWM confirms the cancellation, and the life cycle status of the freight document is changed to **Canceled**.
- If the handling execution status at stop level has already been changed to **Arrived**, the system sets the **Transmission-to-EWM Status** to **Partially Received** or **Received**. Execution of the transportation continues and the freight document cannot be canceled.

The freight document in TM can be canceled only if cancellation has been confirmed by EWM and reported to TM.

Document Flow

In TM you can navigate directly to the associated EWM documents from your freight documents. For example, you can jump directly to the associated EWM freight order from a freight order. However, note that the system does not support the navigation from these EWM documents to other EWM transactions.

Transit Warehousing Process

Use

First, you perform transportation planning in Transportation Management (TM) and then you send your freight documents and the associated transportation units to Extended Warehouse Management (EWM). In EWM, you can then carry out the steps relating to the warehouse (loading or unloading, for example). You can bring the cargo to a staging area in transit warehouses or gateways (air freight) or in container freight stations (ocean freight) where it is stored temporarily until it is transported. Because the two systems are integrated with one another, TM informs you whether the truck or the trailer in EWM has already been loaded or unloaded, for example, and whether there are any discrepancies such as quantity discrepancies.

Prerequisites

For more information, see [Transit Warehousing](#).

Process

A sample process for air freight is described below. For the sake of simplicity, it is assumed that each freight order does not have transportation units, in other words this is a truck without trailer. Only the export side is described. The import side runs in the same way. However, during the import process you send the freight booking first to EWM, even though the freight order for pick-up has to be processed by EWM first.

This process is also supported in the same way for ocean freight.

Creation of Order Documents in TM

1. The customer service agent creates a forwarding order in TM.

i Note

If the customer delivers the goods themselves, you have the following options for this process step:

- Expected customer self-delivery

In this case, you know that the customer is delivering the goods themselves. Therefore, the customer service agent creates a forwarding order for customer self-delivery. From this forwarding order, the transportation planner creates a freight order for customer self-delivery.

- Unexpected customer self-delivery

In this case, the customer is delivering the goods themselves without your knowledge. Therefore, you create the freight order for customer self-delivery directly in TM and send it to EWM.

For more information, see [Handling Discrepancies in Transit Warehousing Process](#) (the last section under "Unexpected Cargo").

For more information about freight orders for customer self-delivery, see [Freight Orders for Customer Self-Delivery and Pick-Up](#).

2. The transportation planner creates an actual route manually or has the system determine this route. The actual route contains one or more locations that are relevant for transit warehousing. A location is relevant for transit warehousing if you have entered a warehouse number and a logical system (EWM system) in the location master data.
3. From the forwarding order, the transportation planner creates a freight order for picking up the goods at the shipper and transporting them to the transit warehouse.
4. The transportation planner sets the execution status of the freight order to **Ready for Transportation Execution**, which means that the transportation activities can now start.
5. The transportation planner sets the unload plan status of the freight order for the transit warehouse to **Unload Plan Finalized**. The system automatically sends an unloading instruction to EWM with the message `LoadingAppointmentRequest`.

If you have specified in the Customizing settings for the EWM integration profile that changes are propagated to the follow-up document and that loading and unloading instructions can be sent multiple times, you can continue to make changes to the forwarding order. The changes are then copied automatically to the freight order. When you set the unload plan status back to **Unload Plan Finalized**, the system sends another unloading instruction to EWM. Once the cargo receipt status has been set to **On Hand**, the system no longer copies any changes made to the forwarding order to the freight order and the freight units.

i Note

You can set the unload plan status for the freight order either at header level or at stop level. If you set the status at header level, the system automatically searches for the next relevant stop, sets the status for this stop, and sends an unloading instruction to EWM. It does this regardless of the extent to which execution has progressed at the present stop and regardless of the value of the handling execution status.

Unloading in the Transit Warehouse (EWM)

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit and the related handling units, as well as an inbound delivery.
2. When the truck arrives at the yard, the warehouse workers report its arrival and assign a door to it. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically with information about the successful arrival.
3. The warehouse workers unload the truck and post the goods receipt. For more information about possible discrepancies and how to handle them, see [Handling Discrepancies in Transit Warehousing Process](#).

If the warehouse workers have received the cargo in its entirety and without discrepancies, EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order and the freight units automatically with information about the received cargo.

4. The warehouse workers post the departure of the truck. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically with information about the successful departure.

5. The warehouse workers bring the cargo to the putaway bin in the transit warehouse or to the staging area that is provided for further transportation of the cargo.

Planning the Pre-Carriage in TM

1. From the forwarding order, the transportation planner creates a freight order for the transportation of goods from the transit warehouse to the gateway.
2. The transportation planner sets the execution status of the freight order to **Ready for Transportation Execution**, which means that the transportation activities can now start.
3. The transportation planner sets the load plan status of the freight order for the transit warehouse to **Load Plan Finalized**. The system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

Loading in the Transit Warehouse (EWM)

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit and an outbound delivery order and assigns the required handling units to the outbound delivery order.
2. When the truck arrives at the yard, the warehouse workers report its arrival and assign a door to it. EWM automatically specifies in the order that handling units, such as dangerous goods, will be brought into the staging area assigned to the door. Furthermore, EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.
3. The warehouse workers bring the handling units into the staging area.
4. The warehouse workers load the truck from the staging area and other storage areas. For more information about possible discrepancies and how to handle them, see [Handling Discrepancies in Transit Warehousing Process](#).

EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order and the freight units automatically with information about the loading process.

5. The warehouse workers post the departure of the truck and post the goods issue. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.

Preparation for Unloading at the Gateway (TM)

1. While the truck is on its way to the gateway, the transportation planner sets the execution status of the freight order to **Ready for Transportation Execution** and the unload plan status of the freight order to **Unload Plan Finalized**. The system automatically sends an unloading instruction to EWM with the message `LoadingAppointmentRequest`.

Unloading at the Gateway (EWM)

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit, planned handling units, and an inbound delivery.
2. When the truck arrives at the yard, the warehouse workers report its arrival and assign a door to it. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.
3. The warehouse workers unload the truck and post the goods receipt. For more information about possible discrepancies and how to handle them, see [Handling Discrepancies in Transit Warehousing Process](#).

EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order and the freight units automatically.

4. The warehouse workers post the departure of the truck. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.
5. The warehouse workers bring the goods into the area in which they will later be consolidated.

Planning the Main Carriage in TM

1. From the forwarding order, the transportation planner creates an air freight booking.
2. The transportation planner sets the load plan status of the air freight booking to **Load Plan Finalized**. The system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

Loading the Unit Load Devices in the Gateway (EWM)

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates an outbound delivery order, planned handling units, and planned handling unit containers. Furthermore, EWM assigns the required handling units to the outbound delivery order.
2. The warehouse workers bring the handling units and the unit load devices to the outbound staging area.
3. The warehouse workers load the handling units into the unit load devices. The system creates handling unit containers. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight booking and the freight units automatically.

i Note

The warehouse workers can distribute partial quantities of items to different unit load devices.

Preparation for the Transportation of Unit Load Devices from the Gateway to the Airport (TM)

1. From the air freight booking, an employee in the shipping office creates a freight order for the pick-up.
2. This employee sets the execution status of the freight order to **Ready for Transportation Execution**, which means that the transportation activities can now start.
3. The employee sets the load plan status of the freight order for the gateway to **Load Plan Finalized**. The system automatically sends a loading instruction to EWM with the message `LoadingAppointmentRequest`.

Loading the Truck at the Gateway (EWM)

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit. If handling unit containers have been reported by TM, EWM assigns these to the transportation unit automatically.
2. When the truck arrives at the yard, the shipping office employee reports its arrival and assigns a door to it. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.
3. The warehouse workers load the unit load devices onto the truck. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order, the air freight booking, and the freight units automatically.
4. The warehouse workers post the departure of the truck and post the goods issue. EWM sends the `LoadingAppointmentNotification` message to TM. When the message is received, TM updates the freight order automatically.

Completion of the Export Side (TM)

1. The transportation planner receives the information from the airline that the aircraft has departed and sets the uplift confirmation status of the air freight booking to **Uplift Confirmed**.

Planning with Trailers

Use

The transit warehousing process supports planning and execution with trailers. Data that is received from Extended Warehouse Management (EWM) is confirmed to all associated business documents across several document levels. This means that in addition to the freight order, the data is also confirmed to one or more trailer units and freight units.

Prerequisites

- In the resource master data you have defined a resource for your tractor (truck) and a passive vehicle resource for your trailer.
- You have made the settings for the transit warehousing process. In particular, you have specified in your EWM integration profile that related transportation units are to be included when loading or unloading instructions are sent. For more information, see [Transit Warehousing](#).
- The trailer must have been coupled before the warehouse-specific steps are executed.

Process

A sample process (inbound) for a transportation stop and a trailer is described below. If your freight order comprises several transportation stops, the process is repeated for each stop if the stop contains unloading locations relevant to EWM. Note, however, that multiple pickup and multiple delivery scenarios are not supported.

Transportation Planning in TM

1. The customer service agent creates a forwarding order in Transportation Management (TM) and creates the freight units.
2. The transportation planner creates a trailer unit and assigns the freight units.
3. The transportation planner creates a freight order and assigns the trailer unit.
4. The transportation planner sets the execution status of the freight order to **Ready for Transportation Execution**, which means that the transportation activities can now start.
5. The transportation planner sets the handling execution status of the freight order to **Departed**.
6. The transportation planner sets the cargo execution status of the freight order to **Cargo Ready for Unloading**. The system automatically sends an unloading instruction to EWM with the message `LoadingAppointmentRequest`. This message includes the assigned trailer unit.

Execution in EWM

1. When the `LoadingAppointmentRequest` message is received from TM, EWM automatically creates a transportation unit and the related handling units.
2. When the truck arrives at the yard, the warehouse workers report its arrival and assign a door to it.
3. The warehouse workers unload the trailer and uncouple it (optional).
4. The warehouse workers post the goods receipt (optional).

5. The warehouse workers post the departure of the truck. EWM sends the `LoadingAppointmentNotification` message to TM.

Update and Further Steps in TM

1. When the `LoadingAppointmentNotification` message is received TM updates the freight order, the trailer unit, and the associated freight units.
2. Complete freight order and trailer unit execution.

i Note

If you receive the message from EWM that the trailer was uncoupled, the corresponding item in the freight order and in the trailer unit is assigned the handling execution status **Uncoupled**. You can also set these values manually.

Handling Discrepancies in Transit Warehousing Process

Use

Discrepancies and exceptions may occur during the transit warehousing process in Transportation Management (TM). For example, during execution in Extended Warehouse Management (EWM) you can determine discrepancies between the actual quantities and the quantities that have been ordered. Furthermore, exceptions can occur such as missing or damaged cargo. This data is sent from EWM to TM and copied automatically into the freight order or freight booking.

The customer service agent or transportation planner must decide together with the ordering party how to proceed with the discrepancies. For example, will missing cargo be delivered at a later date or will the customer stick with the quantity of cargo that has been delivered? Depending on this decision, the customer service agent can adjust the ordered data entered in the forwarding order to the actual data or can plan the subsequent delivery of missing cargo.

Examples of discrepancies and exceptions that may occur in the transit warehousing process are described below. This document also describes how the customer service agent or transportation planner can handle these cases.

Prerequisites

For more information, see [Transit Warehousing](#) and [Resolution of Discrepancies](#). To handle discrepancies you have created a discrepancy profile in Customizing for **Transportation Management** with the required discrepancy types and discrepancy subcategories.

Features

Determining Discrepancies

The system enters any discrepancies between the quantities and attributes of the goods that are to be transported and the data that has been ordered in the freight document and displays this in the forwarding order. The actual quantities compared against the ordered quantities are displayed in the item details of the forwarding order. These details also show whether an attribute such as **Overdimensional Cargo** or **Dangerous Goods** differs from the order data.

You can also search for specific discrepancies. You can display the discrepancies together with all the relevant data in your personal object worklist for discrepancies. Furthermore, you can solve discrepancies directly and access the related documents such as forwarding orders and freight orders.

For more information, see [Handling Discrepancies](#) (forwarding order) and [Discrepancies and Events](#) (freight order).

Handling and Resolving Discrepancies

Missing Cargo

If goods are reported as missing by the EWM system and were entered in the freight document in TM, the system sets the cargo receipt status to **Missing Cargo**. This status is displayed in the forwarding order and freight document. It is also highlighted in the forwarding order that the discrepancy **Missing Cargo** needs to be resolved. You can view details of the discrepancy on the **Discrepancies** tab page in the forwarding order and freight document.

If the quantity reported by the EWM system has changed in the meantime, the system adjusts the actual quantity that is displayed.

Once you have finally decided with the ordering party whether to redeliver the goods or not, you have the following options on the **Discrepancies** tab page:

- Goods are redelivered

In this case, the missing goods are delivered with the next freight order. Even if not all the goods can be loaded, they can be scheduled for the next freight order. Proceed as follows:

1. On the **Discrepancies** tab page or in your personal object worklist for discrepancies, choose  **Resolve**  **Resolve (Retain Missing Cargo)**  for the forwarding order or freight order item.
2. The system splits the freight unit into a freight unit containing the available quantity and a freight unit containing the missing quantity. The freight unit containing the missing quantity can be transported with the next freight order, and the capacity of the means of transport remains reserved.
3. The system resolves the discrepancy and resets the cargo receipt status.

- Goods are not redelivered

In this case, the ordering party accepts that the missing cargo will not be delivered. You can copy the actual quantities to the forwarding order as the ordered data. Proceed as follows:

1. On the **Discrepancies** tab page or in your personal object worklist for discrepancies, choose  **Resolve**  **Resolve (Delete Missing Cargo)**  for the forwarding order or freight order item.
2. The system adjusts the planned quantity to the actual quantity and resolves the discrepancy.

If all of the goods of a forwarding order item are missing (the quantity is 0) and you resolve the discrepancy for missing cargo, the system deletes all associated freight units. To ensure that it does not create new freight units for this item when you save the forwarding order, the system marks them with the processing status **S - Statistical Item; No Logistical Relevance**. This means that the item exists only in the forwarding order.

Note that if cargo is missing you can resolve discrepancies only at main item level, and not at subitem level.

Unexpected Cargo

The discrepancy **Unexpected Cargo** is displayed on the **Discrepancies** tab page in the forwarding order or freight document as well as in your personal object worklist for discrepancies. You can resolve this discrepancy as follows:

- If the discrepancy was reported by the EWM system to TM and there is already an item in the forwarding order for the unexpected cargo, the system creates a cargo item and assigns it to a freight unit. You can resolve this discrepancy by choosing  **Resolve**  **Resolve Discrepancy** .

Unexpected cargo can also be dangerous goods. Dangerous goods information that was entered first in the EWM system is marked accordingly on the user interface. You must check and complete this information before you resolve the discrepancy.

- If the unexpected cargo has not been assigned to an item in the forwarding order yet, the system creates a local item in the freight document. You can resolve this discrepancy by assigning the local cargo item to a freight unit item of an existing forwarding order. To do so, choose **► Forwarding Order ► Assign Forwarding Order Item ►**. On the **Discrepancies** tab page or in your personal object worklist for discrepancies, you can also choose **► Resolve ► Resolve Unexpected Cargo (Assign FWO Item) ►**.

In both cases, a dialog box appears in which you can assign each of your local cargo items to an item of a freight unit. Make sure, however, that a cargo item is assigned to each item of the freight unit. Otherwise, specific criteria must be met. For example, the route and the item category of the two items must be identical. The freight unit must also not yet be split.

- If the unexpected cargo includes not only a single item but all items, for example an entire truck arrives at the warehouse unexpectedly, you create a freight order for customer self-delivery in TM and enter the items. You create the forwarding order from this freight order. You then send the freight order to SAP EWM. The warehouse employees who work with SAP EWM execute the warehouse-specific steps and send the freight order back to TM. Depending on your settings in the discrepancy profile, discrepancies for unexpected cargo are created automatically in TM. You have to resolve this discrepancy as described above.

Damaged Cargo

The discrepancy **Damaged Item** or **Damaged Cargo** is displayed in the item details of the forwarding order and freight document as well as in your personal object worklist for discrepancies. You can resolve this discrepancy as follows:

1. You can split the freight unit manually to create an additional freight unit for the damaged item. You can then assign this freight unit to another freight document to return the goods or plan a later delivery. You do this on the user interface of the freight unit by choosing **Split/Merge**.

If you select **Adopt Stops**, the system copies the stops from the existing freight unit. The system also copies the planning results (routing and assignments to freight orders) from the existing freight unit if it has already been planned.

2. You can resolve the discrepancy by choosing **► Resolve ► Resolve Discrepancy ►** on the **Discrepancies** tab page or in your personal object worklist for discrepancies.

The damage to cargo can also affect a package subitem that was not created in the forwarding order but was created only in the EWM system. The TM system enters these cargo items in the freight document after they have been reported by the EWM system. The corresponding freight units are not assigned to any forwarding order item and are therefore not displayed in the forwarding order. However, the damage is displayed in the forwarding order as a discrepancy for the overlying package. You can also split the freight units manually and assign the freight unit for the damaged item to a different freight document.

You can still split freight units manually after you have started executing the transportation (once the cargo receipt status has been set to **On Hand**). Note, however, that the forwarding order is no longer adjusted accordingly.

You can also apply the process of manually splitting freight units that is described here to other exceptions in the transit warehousing process. For example, other cases are possible:

- Some of the goods are to be sent to a different consignee.
- It is discovered in a transit warehouse that not all the goods can be loaded onto a means of transport and that some of the goods have to be transported using a different means of transport and freight order.

Subsequent Corrections in TM Following Message from SAP EWM

You can make manual corrections in TM after you have received a message from SAP EWM. This may be useful in the following cases:

- After a message is sent to TM, the warehouse workers discover that something has been forgotten and forward the corrections by telephone to the customer service agent or transportation planner in TM.

- The delivery of an automatic message from SAP EWM has failed.

You have the following options:

- Delete items

The system automatically creates a discrepancy called **Missing Cargo**. You have to resolve this discrepancy as described above.

- Reassign items

You can use drag and drop to reassign items easily.

- Add items

The system automatically creates a discrepancy called **Unexpected Cargo**. You have to resolve this discrepancy as described above.

Discrepancies Between Quantities and Attributes

Quantity discrepancies and discrepancies in attributes, such as **Overdimensional Cargo** or **Dangerous Goods**, are entered in the freight document after confirmation from the EWM system. These are then also displayed in the forwarding order. In the details of the forwarding order items you can see whether the quantities and attributes still correspond to the data originally ordered in the forwarding order. If this is no longer the case, you can discuss this with your ordering party. If the ordering party accepts the discrepancies you can proceed as follows:

1. You copy the actual data to the order data. You do this in the item table by choosing **Copy Actual Data > Copy Quantities and Attributes**.
2. You resolve the discrepancies on the **Discrepancies** tab page by choosing **Resolve > Resolve Discrepancy**.

In other cases, such as for missing cargo as described above, you first resolve the discrepancy and then you copy the actual data to the order data. You can also change the order data manually.

Some restrictions apply to the copying or manual changing of some attributes. These include:

- Dangerous goods

Before you can copy the data for dangerous goods you have to define a dangerous goods profile in Customizing for the forwarding order type and a dangerous goods UI profile in Customizing for the item type. For more information, see Customizing for **Transportation Management** under **Forwarding Order Management > Forwarding Order > Define Forwarding Order Types** and **Forwarding Order Management > Define Item Types for Forwarding Order Management**.

Otherwise you have to enter any missing data for dangerous goods to the forwarding order at a later date. If a dangerous goods number (UN number) has already been entered in the transit warehouse, this can be used as advance information. You must then complete this data in the forwarding order either using dangerous goods template data or manually.

If you copy the attribute for dangerous goods without the corresponding data or with incomplete data from the freight document and if you have already resolved the discrepancy, this attribute may be overwritten by the system the next time you run the dangerous goods check.

- Overdimensional cargo

If you want to change this attribute manually, you have to set the method of checking overdimensional cargo to **Manual Check** in Customizing for the item type (see above). Otherwise you can copy attributes only for package items.

- High-value cargo

It is only possible to change this attribute if you have set the **High-Value Cargo** checkbox at subitem level. It cannot be changed at a higher level in the item hierarchy.

- Non-stackable packages

You can copy the attribute only for packages.

- Equipment group and equipment type

You can copy the attribute only for container items and ULD items.

If there are several freight units for a forwarding order item (this is displayed in the item details) and some of the attributes for these freight units differ, the system splits the forwarding order items according to the freight units and copies the attributes to the different forwarding order items.

The **Processing Status of Item** field indicates whether you have already copied actual data to your order data (it contains the value **Actual Data Was Copied**).

i Note

You can also still change data in the forwarding order after execution has started and the cargo receipt status has been set to **On Hand**. However, these changes are not copied to follow-on documents. If a settlement document already exists, first you have to cancel this document, then you change the data, and finally you create a new document.

Canceling Warehouse Activities

When warehouse workers cancel their posted activities in SAP EWM and report these again, the system automatically updates the freight order in TM with this information. In some cases, however, you still have to make manual changes. For example, you may need to resolve new discrepancies. For more information, see [Cancellation of Warehouse Activities in Transit Warehousing Process](#).

More Information

[Handling Discrepancies](#) (forwarding order)

[Discrepancies and Events](#) (freight order)

[Quantity Discrepancies](#) (freight order)

[Other Discrepancies](#) (freight order)

[Resolution of Discrepancies](#)

Cancellation of Warehouse Activities in Transit Warehousing Process

Use

In SAP Extended Warehouse Management (SAP EWM), you can cancel many activities that have already been reported as finished. For example, you can reset loading or unloading processes and report them again. In the transit warehousing process, this impacts the integration with Transportation Management (TM).

SAP EWM reports most activities, such as the arrival of goods in the transit warehouse or the completion of the unloading process, to TM by sending messages. In TM, the freight order and the freight units are then automatically updated with information about

the received cargo. If a warehouse worker cancels these activities and posts them again, the freight order and the freight units are updated automatically again in TM. In some cases, however, the transportation planner has to resolve any discrepancies in TM that may be caused by the updated information (such as missing or unexpected cargo).

The examples below describe the most common cancellations, explain how data is updated in TM, and tell you what you have to do to resolve any blocks and discrepancies.

Prerequisites

For more information, see [Transit Warehousing](#) and [Resolution of Discrepancies](#). To handle discrepancies, you have to create a discrepancy profile in Customizing for Transportation Management with the required discrepancy types and discrepancy subcategories.

Example

The following examples describe cancellations made by warehouse workers in SAP EWM during loading and unloading processes. The cancellations themselves are not reported directly to TM. However, TM receives a message when the warehouse worker updates the unloading data and reports again that the loading or unloading process is finished.

Example 1: Missing Handling Units During Unloading

A warehouse worker accidentally reports only part of the pallets of a freight unit as unloaded and closes the unloading process in SAP EWM. The worker then cancels the completion of the unloading process and reports all the pallets as unloaded.

The process in TM is as follows:

1. The freight unit originally contains five subitems; the three handling units that were first reported as unloaded are displayed with the cargo receipt status **On Hand**. The two unreported handling units are given the cargo receipt status **Missing Cargo**. This freight unit also contains the discrepancy **Missing Cargo**, which refers to the two missing handling units.

The freight order and the two freight units are blocked for further planning.

2. Once the warehouse worker has reported the two missing pallets as unloaded, these two handling units are displayed in the freight unit with the cargo receipt status and the discrepancy **Unexpected Cargo**. The freight unit is still blocked due to the discrepancy.
3. To remove the block and resolve the discrepancy, choose **Resolve > Resolve Discrepancy** on the **Discrepancies** tab page in the item details.

Example 2: Missing Handling Units During Loading

A warehouse worker accidentally loads two pallets fewer than the intended five pallets onto a truck and reports the loading process as completed. The worker then cancels this process and reports another pallet as loaded. Finally, the worker cancels the process again and reports all five pallets as loaded.

The process in TM is as follows:

1. The freight unit originally contains five subitems; the three handling units that were first reported as loaded are displayed with the status **Shipped**. The main item also contains the discrepancy **Missing Cargo** as well as the two unreported handling units with the status **Missing Cargo**.

The freight order and the two freight units are blocked by the system.

2. Once the warehouse worker has reported another pallet as loaded, this handling unit is also displayed in the freight unit as **Shipped**. The freight unit now contains only one handling unit with the status **Missing Cargo**.

3. Once the warehouse worker has reported the last of the five pallets as loaded, all five handling units are displayed in the freight unit as **Shipped**.

The system also automatically deletes the discrepancy **Missing Cargo** and removes the blocks.

Example 3: Exchange of a Missing Handling Unit with Another Handling Unit

During the unloading process, a warehouse worker has reported a pallet as missing. The worker then cancels the process and reports as unloaded a different pallet to that which is missing.

The process in TM is as follows:

1. The system creates a discrepancy called **Missing Cargo**. The missing handling unit is displayed in the freight unit with the status **Missing Cargo**.
2. Once the warehouse worker has canceled the process and reported as unloaded a different handling unit to that which is missing for this freight unit, the system resolves the discrepancy. The main item of the freight unit is displayed with the status **On Hand**. The missing handling unit is displayed at subitem level with the status **Missing Cargo** and the new handling unit is displayed as **On Hand**.

Other Discrepancies

Discrepancies in attributes, such as quantity and weight discrepancies, are automatically updated in TM after cancellations and new notifications in SAP EWM.

More Information

[Handling Discrepancies in Transit Warehousing Process](#)

Automatic Reassignment of Package Items

Use

If the warehouse workers have finished unloading in the transit warehouse, gateway, or in the container freight station and the truck has left the yard, the system automatically sends the message `LoadingAppointmentNotification` from Extended Warehouse Management (EWM) to Transportation Management (TM) (see [Transit Warehousing](#)). This message may contain unexpected package items, in other words items other than those that are planned. If you have configured your TM system in such a way that no discrepancies are to be created, TM automatically reassigns these package items.

Prerequisites

For more information, see [Transit Warehousing](#).

Features

The following describes how the TM system handles unexpected package items contained in the message from EWM.

- Reassignment across freight orders without freight unit split

Initial situation: One or more package items belonging to the same freight unit are assigned to a different freight order than that referred to in the message from EWM.

In this case, TM removes the package items from this freight order and assigns them to the freight order referred to in the message from EWM.

- Assignment of unplanned freight units without freight unit split

Initial situation: One or more package items belong to an unplanned freight unit for the current transportation stage.

In this case, TM assigns the package items to the freight order referred to in the message from EWM.

- Removal of the assignment of freight units

Initial situation: One or more package items are assigned to the freight order referred to in the message from EWM. However, they no longer appear in the message.

In this case, TM removes the package items from the freight order. In other words, it removes the assignment of the freight unit to the freight order for the current transportation stage.

- Assignment of unplanned freight units with freight unit split

Initial situation: One or more package items belong to an unplanned freight unit. However, the freight unit has other package items that are not contained in the message from EWM.

In this case, TM first splits the freight unit. TM assigns the package items contained in the message to the freight order referred to in the message from EWM.

- Reassignment across freight orders with freight unit split

Initial situation: One or more package items are assigned to a different freight order to that referred to in the message from EWM. The corresponding freight unit has other package items that are not contained in the message from EWM.

In this case, TM removes the package items from the other freight order and assigns them to the freight order referred to in the message from EWM. TM then splits the freight unit. This means that after reassignment not all package items in the freight unit are assigned to the freight order contained in the message; only those package items that are contained in the message from EWM are assigned. You have to reschedule the other package items that are not contained in the message from EWM for the current transportation stage.

More Information

For more information about handling discrepancies, see [Handling Discrepancies in Transit Warehousing Process](#).

Warehouse Billing

Use

You can perform the following activities:

- Sell warehouse services to a customer and periodically bill the customer for the services you have provided for a time period.
- Purchase warehouse services from an external service provider and self-bill to pay the service provider for the services you have used for a time period.

Warehouse billing uses the agreements, charge calculation, and settlement features in Transportation Management (TM).

i Note

- You can also integrate TM with an external SAP Extended Warehouse Management.

Agreement Management

You can create a freight agreement with an external service provider for the warehouse services that you want to purchase, or you can create a forwarding agreement with a customer for the warehouse services that you want to sell. TM transfers the agreement information to Extended Warehouse Management (EWM) and automatically creates warehouse billing measurement requests in EWM. For more information, see [Creation of Agreement](#).

Charge and Settlement Management

In EWM, you can measure and determine the quantity of warehouse services used for a time period. EWM transfers the quantity information to TM. TM uses this feature for warehouse services charge calculation and settlement. For more information, see [Charge Calculation and Settlement](#).

Integration

Creation of Forwarding Orders

TM creates a forwarding order for billing a customer by Web services. EWM sends warehouse billing measurements using the service proxy `TransportationRequestRequest_Out`. TM receives the data and creates a forwarding order using the service proxy `TransportationRequestRequest_In`.

To change or extend the data transferred through the service interface `TransportationRequestRequest_In`, use the Business Add-In (BAI) in Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Integration](#) > [Enterprise Services](#) > [Forwarding Order Management](#) > [Forwarding Order](#) > [BAI for TransportationRequestRequest_In](#).

Creation of Service Orders

TM creates a service order for self-billing for a service provider by Web services. EWM sends warehouse billing measurements using the service proxy `TransportationOrderGenericRequest_Out`. TM receives the data and creates a service order using the service proxy `TransportationOrderGenericRequest_In`.

To change or extend the data transferred through the service interface `TransportationOrderGenericRequest_In`, use the BAI in Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Integration](#) > [Enterprise Services](#) > [Freight Order Management](#) > [Freight Order](#) > [BAI for TransportationOrderGenericRequest_In](#).

For more information, see <http://help.sap.com/s4hana> under [Product Assistance](#) > [Enterprise Business Applications](#) > [Supply Chain](#) > [Warehousing](#) > [Extended Warehouse Management \(EWM\)](#) > [Warehouse Billing](#).

More Information

- [Agreement Maintenance](#)
- [Charge Calculation](#)
- [Freight Agreement Maintenance](#)
- [Forwarding Agreement Maintenance](#)
- [Settlement](#)

Creation of Agreement

Use

You can create a forwarding agreement for selling warehouse services to a customer or a freight agreement to purchase warehouse services from a service provider. Subsequently, you can bill or self-bill for the warehouse services used.

You can create an agreement with items that are assigned a calculation sheet for charging services for a specific warehouse.

In the charge line of the calculation sheet, you can specify a measurement remark to describe how to measure the warehouse service. When Transportation Management (TM) sends the agreement information to Extended Warehouse Management (EWM), a warehouse expert can view the measurement remark and decide how the item should be measured.

When you release the agreement, TM automatically transfers the agreement data to EWM for warehouse billing measurement. You cannot edit the item after you have released the agreement. You must change the status of the agreement to **In Process** to make any changes.

Prerequisites

In **Master Data**, you have performed the following actions:

- Specified the EWM logical system and the warehouse number in a location as the warehouse location
- Specified the billing period for the customer or service provider.

For more information, see [Master Data](#)

You have performed the following actions in Customizing:

- You have selected the **SAP EWM Integration** checkbox and specified an output profile in Customizing for **Transportation Management** under **Master Data > Agreements and Service Products** in the following Customizing activities:
 - **Define Freight Agreement Types** for freight agreements
 - **Define FWAs and Service Product Catalog Types** for forwarding agreements

i Note

When you release the agreement, the output profile executes a B2B message that enables the system to transfer agreement data to EWM.

- In Customizing for **Transportation Management** under **Basic Functions**, you have performed the following actions:
 - Defined service types for the warehouse services in the Customizing activity **Define Service Types** under **General Settings**
 - Defined charge types for the warehouse services in the Customizing activity **Define Charge Types** under **Basic Settings**
 - Assigned the charge types to the service types in the Customizing activity **Assign Charge Types to Service Types** under **Basic Settings**

i Note

Ensure that both of the charge type and the service type have the same name.

Process

1. From the launchpad, open one of the following apps to create a forwarding agreement or freight agreement:
 - To create a forwarding agreement, open the [Create Forwarding Agreement](#) app.
 - To create a freight agreement, open the [Create Freight Agreement](#) app.
2. Specify the agreement type and create the agreement. Specify the header information and create the agreement item with the relevant service type. You can specify the warehouse in the [Preconditions](#) tab page for the item. In the [Trade Lane](#) tab page, specify the following:

Field	Value
Orientation	From
Source Type	Location
Source	<Warehouse Location>

3. You specify calculation sheet details for the charge type associated with the service type for the item. You specify rates based on the quantity of the service.
 - You can specify a single rate in the following format in the calculation sheet:

Field	Value
Charge Type	<Charge Type>
Amount	<Rate Amount>
Currency	<Currency>
Calculation Base	QUANTITY_VAL
Quantity	<Quantity>
Unit	AU

- For multiple rates, you can create a rate table. In the rate table, you specify QUANTITY_VAL as the dimension and specify the rate values based on the quantity.

❖ Example

You specify the following:

- A charge type for the storage service according to the daily stock in tons
- A currency of [USD](#) and calculation method type of [Clipping](#)
- You specify the following rate table details:

Scale Type	Scale from
Calculation Base	QUANTITY_VAL
Quantity	1

Unit	AU
------	----

- You define the following scale:

QUANTITY_VAL	Rate
0	1.1
100	1
1000	0.8

Depending on the calculation method of the charge line, if 450 tons of stock are measured from July 12, 2014 to July 19, 2014, one of the following occurs:

- The first 100 tons of stock is charged at 1.1 USD per ton, and the second 350 tons are charged as 1 USD per ton.
- A standard rate of 1.1 USD per ton is applied to all 450 tons.

More Information

- For more information, see <http://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management (EWM)** ► **Warehouse Billing** ► **Request for Warehouse Billing Measurement** ►.
- [Agreement Maintenance](#)
- [Freight Agreement Maintenance](#)
- [Forwarding Agreement Maintenance](#)
- [Settlement](#)
- [Assigning Billing Periods for Business Partners](#)

Charge Calculation and Settlement

Use

You use Transportation Management for automatically calculating charges for the warehouse services used and for settlement in Sales and Distribution (SD).

You can periodically transfer the quantity of warehouse services used from Extended Warehouse Management to Transportation Management (TM). TM creates a forwarding order or a service order when you transfer the quantity. You can use these documents to display the calculated charges and for settlement in SD.

Integration

EWM determines the quantity of services used for a billing period. For more information, see <http://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management (EWM)** ► **Warehouse Billing** ►.

More Information

[Charge Calculation of Warehouse Services](#)

[Settlement of Service Orders and Forwarding Orders](#)

[Creation of Agreement](#)

[Warehouse Billing](#)

Charge Calculation of Warehouse Services

Use

When you distribute a warehouse billing measurement (WBM) from Extended Warehouse Management (EWM) to Transportation Management (TM), TM creates a service order or forwarding order corresponding to the distributed WBM.

The forwarding order or service order automatically calculates charges for the warehouse services used based on the agreement you have defined with the customer or service provider in TM.

Integration

EWM determines the quantity of services used for a billing period. For more information, see <http://help.sap.com/s4hana> under **Product Assistance** > **Enterprise Business Applications** > **Supply Chain** > **Extended Warehouse Management (EWM)** > **Warehouse Billing**.

Prerequisites

- In Customizing for **Transportation Management** under **Master Data** > **Agreements and Service Products**, you have performed one or more of the following activities:
 - Assigned a default service order type to a freight agreement type in the Customizing activity **Define Freight Agreement Types**
 - Assigned a forwarding order type to a forwarding agreement type in the Customizing activity **Define FWA and Service Product Catalog Types**
- In the Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings**, you have performed the following activities:
 - You have specified a **Calculation Date Type** of Order Date in the Customizing activity **Define Calculation Profiles** for a calculation profile.
 - You have assigned this calculation profile to the relevant charge profile in the Customizing activity **Define Charges Profile**.

This enables the system to use the execution end date from the service order or forwarding order as the charge calculation date instead of the system date.

Features

A forwarding order for billing a customer, or a service order for warehouse service procurement for a service provider contains the following warehouse billing relevant information:

- Items with quantity for a time period
- **Service Type** of the warehouse services
- Warehouse numbers
- Billing periods
- Service providers, or the customers as the External Sold-to parties
- The warehouse billing measurement document in the **Document Flow** tab page

Activities

To find forwarding orders for warehouse billing, enter the warehouse billing measurement document number in the field **Buyer's Reference Number** in the quick criteria maintenance screen in the **Forwarding Orders** app. Then choose **Show Quick Criteria Maintenance**.

To find service orders for warehouse billing, enter the warehouse billing measurement document number in the field **Document** in the quick criteria maintenance screen in the **Service Orders** app. Then choose **Show Quick Criteria Maintenance**.

You can perform the following activities and recalculate charge:

- Change the quantity
- Specify new items for warehouse services
- Delete or add new charges

More Information

- [Charge Calculation and Settlement](#)
- [Warehouse Billing](#)
- [Forwarding Order](#)
- [Service Order](#)
- [Charge Calculation](#)

Settlement of Service Orders and Forwarding Orders

Use

You can use the forwarding settlement document for billing a customer. You can use the freight settlement document to credit a service provider. Subsequently, you can send this to Sales and Distribution (SD) for final settlement.

You can create settlement documents from service orders or a forwarding orders relevant to warehouse billing. For more information about finding service orders and forwarding orders relevant for warehouse billing, see [Charge Calculation of Warehouse Services](#)

Integration

Extended Warehouse Management (EWM) determines the quantity of services used for a billing period. For more information, <http://help.sap.com/s4hana> under ► **Product Assistance** ► **Enterprise Business Applications** ► **Supply Chain** ► **Extended Warehouse Management (EWM)** ► **Warehouse Billing** ►.

Prerequisites

- You have enabled invoice Integration of Transportation Management (TM) with SD.
- In Customizing for **Transportation Management** under ► **Settlement** ► **Define Settlement Profiles** ►, you have specified weather you want to create a collective settlement or individual settlements based on the billing period.
- You have activated the business add-in implementation /SCMTMS/CFIR_DATES_WB_IMP for invoice date determination in Customizing for **Transportation Management** under ► **Business Add-Ins (BAdIs) for Transportation Management** ► **Settlement** ► **Forwarding Settlement** ► **BAdI: Determination of Invoice Date for FWSDs** ►.
- You have activated the business add-in implementation /SCMTMS/SFIR_DATES_WB_IMP for settlement date determination in Customizing for **Transportation Management** under ► **Business Add-Ins (BAdIs) for Transportation Management** ► **Settlement** ► **Freight Settlement** ► **BAdI: Determination of Settlement Date for Freight Settlement Documents** ►.
- You can modify warehouse billing forms in the PPF action profile for printing in Customizing for **Cross-Application Components** under ► **Processes and Tools for Enterprise Applications** ► **Reusable Objects and Functions for BOPF Environment** ► **PPF Adapter for Output Management** ► **Maintain PPF Settings** ►.

Features

- The system can take into account the billing period to create either a collective settlement document or individual settlement documents for service orders and forwarding orders.
- You can print warehouse billing forms from the settlement document. You can print the forms using the action profile /SCMTMS/PRINT_PFCFI_WB in the **Output Management** tab page of a settlement document.
- The system uses the service period of the service order or the forwarding order as the invoice date of the settlement document. The system usually uses the end date of the period as invoice date.
- You can post the settlement document in SD for final settlement.
- After the settlement document is posted in SD, you can create additional credit memos or forwarding settlement documents to adapt any changes to the charge and post it in SD for correction.

More Information

- [Settlement](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)
- [Charge Calculation of Warehouse Services](#)
- [Charge Calculation](#)

Assigning Billing Periods for Business Partners

Use

You assign a billing period for a customer or a service provider to specify the periods that the system uses to calculate charge for the warehouse services that are used.

In the Extended Warehouse Management (EWM) system, the system generates a warehouse billing measurement document for a billing period and transfers quantity data to the Transportation Management (TM) system for charge calculation and settlement.

Prerequisites

Integration of Billing Schedule with Extended Warehouse Management (EWM)

When you save a forwarding agreement or freight agreement and the agreement status is Released, TM sends the calendar ID and the forwarding agreement to EWM. You must specify a calendar in EWM that has the exact same settings as the TM calendar that you transfer to EWM. You can specify a factory calendar for EWM in Customizing for [SAP NetWeaver](#) under [General Settings](#) [Maintain Calendar](#).

❖ Example

You have a forwarding agreement or freight agreement in place with a business partner for a warehouse service where you pack a pallet with goods. You charge USD 1 to pack a single pallet. You have a weekly billing schedule in place with the business partner. When the weekly billing schedule reaches its due date, EWM sends the number of pallets it packed during the week for the business partner to TM. You can then calculate the amount due and include the charge in a forwarding settlement document in TM.

Procedure

Specifying Billing Periods for a Customer

1. Go to the [Maintain Business Partner](#) transaction (transaction BP).
2. Enter the business partner.
3. Display the BP role [Sold-to Party](#).
4. In the [Sales Org. Settings](#) screen area on the [Customer Org. Data](#) tab page, specify a sales organization.
5. Enter the factory calendar in the [Billing Schedule](#) field, for example, [Weekly Settlement](#).

Specifying Billing Periods for a Service Provider

1. Go to the [Maintain Business Partner](#) transaction (transaction BP).
2. Enter the business partner.
3. Display the BP role [Carrier](#).
4. In the [Purchase Org. Settings](#) screen area on the [Vendor Org. Data](#) tab page, specify a purchasing organization.
5. Enter the factory calendar in the [Billing Schedule](#) field, for example, [Monthly Settlement](#).

More Information

The EWM system determines the quantity of services used for a billing period. For more information, see http://help.sap.com/s4hana_op_1709 under [Product Assistance](#) [Enterprise Business Applications](#) [Supply Chain](#) [Extended Warehouse Management \(EWM\)](#) [Warehouse Billing](#).

[Creation of Agreement](#)

Integration with SAP Event Management

Use

This function enables you to integrate Transportation Management (TM) with SAP Event Management, and to track and monitor events for business objects in business processes.

Prerequisites

- You have installed SAP Event Management, and made the settings in Customizing for Event Management integration.
- You have made the necessary Customizing settings in TM. For more information, see Customizing for [Transportation Management](#) under [Integration > Tracking and Tracing of Processes and Documents > Prerequisites for Tracking and Tracing](#).
- You have maintained the system connection to TM.

Features

SAP Event Management offers visibility processes to monitor transportation execution in connection with TM.

The transportation execution visibility processes enable users in roles such as transportation dispatcher, shipper, or sold-to party, to track and monitor planned events or to report an actual event. Tracking and tracing is provided for the following business document types in TM:

- Freight units
- Freight orders
- Freight bookings
- Transportation unit
- Resources
- Instruction execution for standard operating procedures

More Information

- [Tracking of Freight Units](#)
- [Tracking of Freight Orders and Freight Bookings](#)
- [Tracking of Transportation Units](#)
- [Tracking of Resources](#)
- [Tracking of Instructions](#)

Tracking of Freight Units

Use

You can use this function as a tracking process to track freight units. It is one of the functions that monitor the execution of transportation processes managed by Transportation Management (TM).

Execution tracking can be done with or without SAP Event Management:

- For information about tracking executions without an external event management system, see [Execution Tracking](#).
- If you connect to SAP Event Management, you can use this visibility process to integrate TM and SAP Event Management. For more information, see [Integration with SAP Event Management](#).

In transportation planning, the shipper or sold-to party asks you to transport goods from one location to another location. On receiving the order, your transportation dispatcher creates a transportation requirement in TM. TM creates freight units to fulfill the transportation requirement. If you have connected SAP Event Management, you can monitor the execution phase of the freight units by tracking the corresponding event handlers in SAP Event Management. SAP Event Management enables the following features:

- Your shipper or sold-to party can monitor the status changes of freight units.
- Your transportation dispatcher, shipper or sold-to party, ship-to party, and carrier can monitor the expected and actual events for freight units, from loading the goods to the proof of delivery at the ship-to party.
- Your ship-to party can report the proof of delivery.
- Your carrier can report events, including unexpected events, for example, a delay.
- Your transportation dispatcher can receive an alert when an unexpected event is reported, for example, damage to an item.

Features

You can track the following expected events for freight units in SAP Event Management:

- [Scheduled](#)
- [Loading Begin](#)
- [Loading End](#)
- [Departure](#)
- [Arrival at Destination](#)
- [Unloading Begin](#)
- [Unloading End](#)

There are two ways of reporting the expected event:

- When your carrier sends an event message from the SAP Event Management Web user interface to TM, TM updates the freight unit with the information in the event message.
- When you update the freight unit directly in TM, TM sends an event message to SAP Event Management. SAP Event Management updates the data for the freight unit.

SAP Event Management can also track the following unexpected events. When these events occur, SAP Event Management sends an alert to TM:

- [Delay](#)
- [Damage](#)

Tracking of Freight Orders and Freight Bookings

Use

You can use this function as a tracking process to track freight orders and freight bookings. It is one of the functions that monitor the execution of transportation processes managed by Transportation Management (TM).

Execution tracking can be done with or without SAP Event Management:

- For information about tracking executions without an external event management system, see [Execution Tracking](#).
- If you connect to SAP Event Management, you can use this visibility process to integrate TM and SAP Event Management. For more information, see [Integration with SAP Event Management](#).

When connected to SAP Event Management, this function enables the following features:

- Your shipper or sold-to party can monitor the expected events for their freight orders and freight bookings.
- Your transportation dispatchers can monitor all events including unexpected events for freight orders and freight bookings in their organizational unit. For example, they can receive an alert if there is a delay.

Features

You can track the following expected events for freight orders and freight bookings in SAP Event Management:

- [Loading Begin](#)
- [Loading End](#)
- [Proof of Pick-up](#)
- [Departure](#)
- [Clear Customs](#)
- [Arrival at Destination](#)
- [Unloading Begin](#)
- [Unloading End](#)
- [Proof of Delivery](#)

There are, for example, the following ways of reporting the expected event:

- When your carrier sends an event message from the SAP Event Management Web user interface to TM, TM updates the freight order or freight booking with the information in this event message.
- When you update the freight order or freight booking directly in TM. SAP Event Management updates the data for the freight order or freight booking.

You can display the source of execution information on the [Execution](#) tab page of a freight order or freight booking in TM.

SAP Event Management can also track unexpected events. You can report these unexpected events for an item of a freight order or freight booking in TM ([Discrepancies and Events](#) tab page). When delays occur, SAP Event Management issues an alert.

Tracking of Transportation Units

Use

You can use this function as a tracking process to track transportation units. It is one of the functions that monitor the execution of transportation processes managed by Transportation Management (TM).

Execution tracking can be done with or without SAP Event Management:

- For information about tracking executions without an external event management system, see [Execution Tracking](#).
- If you connect to SAP Event Management, you can use this visibility process to integrate TM and SAP Event Management. For more information, see [Integration with SAP Event Management](#).

In transportation planning, the shipper or sold-to party asks you to transport goods from one location to another. On receiving the order, your transportation dispatcher creates a transportation requirement in TM. TM creates freight units to fulfill the transportation requirement. If you have connected SAP Event Management, you can monitor the execution phase of the transportation units by tracking the corresponding event handlers in SAP Event Management. SAP Event Management enables the following features:

- Your shipper or sold-to party can monitor the status changes of transportation units.
- Your transportation dispatcher, shipper or sold-to party, ship-to party, and carrier can monitor the expected and actual events for transportation units, from loading the goods to the proof of delivery at the ship-to party.
- Your transportation dispatcher, shipper or sold-to party, ship-to party, and carrier can monitor the expected and actual events for transportation units.
- Your carrier can report events, including unexpected events, for example, a delay.
- Your transportation dispatcher can receive an alert when an unexpected event is reported, for example, delay of the transportation unit.

Features

You can track the following expected events for transportation units in SAP Event Management:

- [Loading Begin](#)
- [Loading End](#)
- [Coupling](#)
- [Departure](#)
- [Arrival at Destination](#)
- [Decoupling](#)
- [Unloading Begin](#)
- [Unloading End](#)

There are two ways of reporting the expected event:

- When your carrier sends an event message from the SAP Event Management Web user interface or any other external interface to SAP Event Management, SAP Event Management updates the data of the corresponding event handler and

sends an update to TM. Then TM updates the transportation unit with the information in the event message and if applicable propagates the event to related freight documents such as freight units.

- When you update the transportation unit directly in TM, TM sends an event message to SAP Event Management. SAP Event Management updates the data for the transportation unit.

SAP Event Management can also track the following unexpected events:

- **Delay**

When these events occur, SAP Event Management can send an alert to TM.

Tracking of Resources

Use

You can use this function to track resources that are defined as master data in Transportation Management (TM). This tracking process is part of the functions to monitor the transportation execution of processes managed by TM. The visibility process integrates TM and SAP Event Management. For more information, see [Integration with SAP Event Management](#).

You can use this resource tracking visibility scenario to track resources over their entire lifecycle:

- Tracking starts when the resource is entered into the TM master data with the SAP Event Management system connected. Further, you must set the resource attribute **Relevant for Event Management** and assign a **Means of Transport** that is not marked as **Multiresource**.
- Tracking ends when the attribute **Relevant for Event Management** is removed.

If you have connected SAP Event Management, you can monitor the execution events that are related to a certain resources by tracking the corresponding event handlers in SAP Event Management. SAP Event Management enables the following features:

- Your transportation dispatcher can check the availability of a certain resource or can get an overview over all available resources.
- Your carrier can report events directly for a resource, for example to update the actual position.
- Your transportation dispatcher can receive an alert when an unexpected event is reported, for example, the damage of a resource

Features

The following resource types are supported:

- Active vehicle resources such as trucks or locomotives
- Passive vehicle resources such as trailers or railcars
- Transportation units such as containers

You can track the following expected events for resources in SAP Event Management:

- **Coupling** (only relevant for vehicle resources)
- **Departure**

- **Arrival at Destination**
- **Decoupling** (only relevant for vehicle resources)

There are three ways of reporting the expected event:

- When your carrier sends an event message for the related freight document via SAP Event Management to TM, TM propagates this event message back to the related resource event handler(s) in SAP Event Management. SAP Event Management updates the data for the resource event handler(s) accordingly.
- When you update the execution information of a freight document directly in TM, TM sends an event message to the related resource event handler(s) in SAP Event Management. SAP Event Management updates the data for the resource event handler(s) accordingly.
- When your carrier sends an event message directly for the resource via SAP Event Management. SAP Event Management updates the data for the resource event handler accordingly. However there is no automatic propagation of the event to the related freight document(s) in TM.

SAP Event Management can also track the following unexpected events:

- **Sighting** (to update the actual position of the resource)
- **Damage**

Tracking of Instructions

Use

You can use this function to track and monitor the execution of instructions that are assigned to forwarding orders or freight units in Transportation Management (TM). Instructions are introduced with TM 9.0 SP02 in what are known as standard operating procedures (SOP) in the transportation industry. The visibility process integrates TM and SAP Event Management. For more information, see [Integration with SAP Event Management](#).

Typically, instructions are pre-defined in forwarding agreements and serve as templates. Once a forwarding order is created from a forwarding agreement, the corresponding instructions are copied into the forwarding order and where applicable, also propagated to the related freight units. If you have connected SAP Event Management, you can monitor the execution of these instructions by tracking the corresponding event handlers in SAP Event Management. SAP Event Management enables the following features:

- Your transportation dispatcher can monitor the status changes of an instruction.
- If the alert framework has been set up, your transportation planner can receive an alert when the alert date of the instruction is reached before the execution of the instruction is completed

Features

You can track the one expected event for instructions in SAP Event Management:

- **Instruction Execution**

For each instruction there is an alert date and a due date, both of which are set in TM. Each instruction has one of the following six statuses in TM:

- Pending (initial value) – can be set manually in TM
- In progress – can be set manually in TM

- Completed – can be set manually in TM
- Completed after Due Date – automatically set by TM if completion date is after due date
- Alert – set by SAP Event Management or
- Overdue – set by SAP Event Management

The expected event date of the expected event is determined as follows:

- Until the alert date is reached, the expected event date is the alert date.
- After the alert date, the expected event date is changed to the due date.

Monitoring of the status of the instruction is done in SAP Event Management. The monitoring sequence is as follows:

- After the alert date until the due date, if the instruction has not been completed, the status of the instruction is set to alert (in TM and SAP Event Management).
- After the due date if the instruction has not been completed, the status of the instruction is set to overdue (in TM and SAP Event Management).

Immediately a change to the status of an instruction is made manually in TM, SAP Event Management is informed, and the corresponding event handler status is updated. Similarly if the status of the instruction changes due to monitoring in SAP Event Management, the corresponding event handler status is updated and the changed status is sent to TM.

i Note

There are no unexpected events supported for instructions.

Integration with SAP Global Trade Services

Integration with SAP Global Trade Services (GTS) allows for customs processing, trade compliance checks, and customs transit discharges declarations for Transportation Management (TM). This integration ensures compliance with embargos, sanction party lists, and customs clearance, and supports outbound and inbound processes for trade compliance.

You can integrate GTS with your TM system in an internal, external, and decentralized setup.

Use

The integration of with the GTS application enables you to do the following:

- Handle customs processing for TM business documents in SAP Global Trade Services
- Perform trade compliance checks for TM business documents in SAP Global Trade Services
- Handle customs transit discharges declarations for TM freight orders in SAP Global Trade Services.

You can find more information about these topics under [Related Information](#).

Related Information

[Integration with SAP Global Trade Services in Advanced Shipping and Receiving](#)

[SAP GTS Integration for the Outbound Process](#)

[SAP GTS Integration for the Inbound Process](#)

[Integration Scenarios for SAP Global Trade Services](#)

Integration Scenarios for SAP Global Trade Services

You can integrate Transportation Management (TM), and SAP Global Trade Services (GTS) by using SAP NetWeaver Process Integration (SAP NetWeaver PI) or a point-to-point connection (Web Services Reliable Messaging). Data is exchanged using enterprise services.

Integration

The following integration scenarios and enterprise services are available in TM:

- [TM_GTSExportDeclarationIntegration](#)

The following enterprise services are available for this scenario in the [Transportation Order Processing](#) process component:

- [Request Export Declaration](#)
- [Request Export Declaration Cancellation](#)
- [Change Transportation Order Based on Export Declaration Confirmation](#)

- [TM_GTSTradeComplianceCheckIntegration](#)

The following enterprise services are available for this scenario in the [Transportation Order Processing](#) process component:

- [Request Trade Compliance Check](#)
- [Request Trade Compliance Check Cancellation](#)
- [Change Transportation Order Based on Trade Compliance Check Confirmation](#)

The following enterprise services are available for this scenario in the [Transportation Request Processing](#) process component:

- [Request Trade Compliance Check](#)
- [Request Trade Compliance Check Cancellation](#)
- [Change Transportation Order Based on Trade Compliance Check Confirmation](#)

- [TM_GTS CustomsTransitDischargeDeclarationIntegration](#)

The following enterprise services are available for this scenario in the [Transportation Order Processing](#) process component:

- [Request Customs Transit Discharge Declaration](#)
- [Change Transportation Order Based on Customs Transit Discharge Declaration Confirmation](#)

- Enterprise Services Without Integration Scenario

The following enterprise services are not assigned to an integration scenario:

- [Request Customs Invoice](#)
- [Change Transportation Order Based on Customs Invoice Confirmation](#)

Related Information

Master Data Replication

You use this function to replicate master data from a source system, such as an ECC system or an SAP S/4HANA system, to the Transportation Management (TM) component of an SAP S/4HANA system (target system).

Data Replication Framework

You can use the [Data Replication Framework](#) (DRF) to replicate master data such as business partners, products, and locations between source systems and target systems.

The DRF uses a Key Mapping Framework to map the IDs of the corresponding objects in the source and the target system when replicating master data. This allows the target system to determine the IDs of local master data entities based on the IDs of the master data objects in the source system. This is to ensure that an object in the target system can be assigned the same value as the corresponding object in the source system, even if the object IDs in the two systems are not identical. Key mapping is essential if a target system receives data from several source systems which may contain the same objects with different IDs or where the usage of the same key for different entities in different source systems might occur.

Related Information

[Set Up Data Replication](#)

[Data Replication](#)

Business Partner

To replicate business partner master data, you set up a replication model in your source system using the DRF customizing transaction DRFIMG, and define the target system as part of the replication model.

You use outbound implementation 986_3 – Outbound Impl. for BP/REL Services for the business object 986 – Business Partner.

The replication uses the business partner service BusinessPartnerSUITEBulkReplicateRequest and the system activates unified key mapping automatically.

To activate key mapping you use transaction IDMIMG. The target system uses confirmation message BusinessPartnerSUITEBulkReplicateConfirmation to transfer key mapping information back to the source system.

For business partners, the key mapping supports the two object identification types 888 – Business Partner ID and 889 – Business Partner UUID.

The following business partner roles are replicated with the Data Replication Framework:

- 000000 Business Partner (Gen.)
- BUP004 Organizational Unit
- CRM010 Carrier
- SLLCOF Customs Office

- TM0001 Driver
- FLVN01 Supplier
- FLVN00 FI Supplier
- FLCU01 Customer
- FLCU00 FI Customer

i Note

If you need locations for business partners in the target system you create these locally in the TM component of your SAP S/4HANA system.

Customer/Vendor Integration

In SAP S/4HANA the business partner is the central object to hold partner data. Customer/Vendor Integration (CVI) creates customer master data or vendor master data, if required, based on a business partner. This means a business partner is always available for handling customer or vendor data in an SAP S/4HANA system.

Product

You replicate product master data for products which do not depend on organizational levels such as plant, storage location, sales organization, distribution channel, warehouse number, or storage type. The master data replicated in TM includes client-dependent data, language-dependent material descriptions, and conversion factors for units of measure.

For the replication of products, you set up a replication model in your source system using the DRF customizing, transaction DRFIMG, and define the target system as part of the replication model.

If the source system is an SAP S/4HANA system, you use the existing outbound implementation 194_3 – Outbound Impl. for Product via Services for the business object 194 – Material.

If the source system is an SAP ECC system, you use outbound implementation 194_1 – Material through IDoc. Note that if you use SAP ECC enhancement package 7 or 8, you first need to implement SAP Note [3313230](#)  **Enable Outbound Implementation 194_1 (Material via iDoc) for DRF**.

You activate the unified key mapping for products in the customizing of the target system, transaction IDMIMG, to enable the identification of local product entities.

The key mapping information is transferred back to the source system with confirmation message ProductMDMSUITEBulkReplicateConfirmation.

The key mapping for products supports the two object identification types 20 – Material ID Internal Format and 924 – Material ID External Format.

Location

Plants and shipping point customizing data exist in SAP S/4HANA and SAP ECC systems but in TM processes this information is represented in Location master data. To replicate plant and shipping point information the source system creates location master data which it then replicated to the target system. However if the source system is an ECC system, no location object is available

and so the plant and shipping point Customizing settings are replicated to TM where the locations can then be created based on this data.

For the replication of locations, you set up a replication model in your source system via the DRF customizing, transaction **DRFIMG**, and define the target system as part of the replication model.

If the source system is an SAP S/4HANA system, you use the outbound implementation **189_1** which is the outbound implementation for locations using service **189 - Location**.

If the source system is an SAP ECC system, you use the following outbound implementations:

- **464_L**

The outbound implementation to replicate plants into locations, based on the business object 464 (Plant)

- **DRF_0045_L**

The outbound implementation to replicate shipping points / receiving points into locations, based on the business object DRF_0045 (Shipping Point / Receiving Point)

The target system uses confirmation message **LocationBulkReplicateConfirmation** to transfer key mapping and status information back to the source system.

i Note

If the source system is an SAP ECC system the key mapping information is not stored in the source system.

You activate the unified key mapping for locations in the customizing of the target system, transaction **IDMIMG**, to enable the identification of local product entities.

The key mapping for locations supports the two object identification types **198 - Location Number** and **580 - Location UUID**. In this case the source system is an SAP S/4HANA system.

i Note

If you need locations for business partners in the target system you create these locally in the TM component of your SAP S/4HANA system.

Analytics

Transportation Management analytics provide apps, called overview pages, that show the key performance indicators (KPIs) that you need to manage your operational business. At a glance, you can see all the vital data you need for your daily decision-making. Each overview page displays information from a different area of TM, for example, freight booking, freight ordering, tendering, and business share.

Analytics Overview Pages

Each overview page consists of a number of cards (similar to tiles) that display information based on one or more KPIs. The cards are read-only; you can't enter any information or perform queries or searches on them. The cards present data in different ways; some show lists, others display tables or charts. Some cards present summary information and you can drill down to see further details, such as individual transactions.

For cards displaying graphical charts (line, bubble), selecting the header takes you to a detailed view showing all the unfiltered, while selecting a data point in a chart takes you to a detailed view, filtered to show only the information relevant to that data point.

All the cards show information in real time, so that you always have the most up-to-date information. This can be invaluable in identifying problems such as delays or bottlenecks, and enable you to take swift action to remedy the situation.

You can re-size the cards and change the order in which they're displayed to suit your own requirements.

Each overview page is based on one or more Core Data Services (CDS) views.

You access the TM overview pages from the Fiori launchpad.

Detailed Views

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line on the overview page takes you to the detailed view, which presents summary information graphically in the form of a combination of donut, bar, and line charts, and also in a table listing relevant Transportation Management documents. From the table you can navigate directly to the relevant document, such as a freight order or a freight booking, in the Transportation Management system and change it if necessary.

Detailed views are also read-only; you can't enter any information or perform queries or searches on them.

Some overview pages have two detailed views, each containing information relevant to specific cards, while others only have one. For example, on the [Cost Analysis](#) overview page, selecting the [Invoicing Block Analysis](#) card takes you to the [Transportation Invoice Blocked Detailed View](#), while selecting any other card takes you to the [Transportation Costs Detailed View](#).

CDS Views

A CDS view is a query view, that is, an SQL query on the database requesting certain data to be returned. One (or more) CDS view feeds this information into each overview page (OVP). These CDS views are top-level views, called consumption views, that contain other contributing CDS views. They consist of key figures and attributes.

You need to have the appropriate authorization objects to access CDS views.

The following table lists the TM analytics overview pages, detailed views, and the CDS views that feed information to them:

Summary

Overview Page	Detailed Views	Contributing CDS Views
Allocation Analysis	Transportation Allocation	Transportation Allocation Compliance Query (C_TranspAllocComplianceQ)
Business Share Analysis	Business Share	Business Share Compliance Query (C_BusinessShareComplianceQ)
Cost Analysis	Transportation Costs Transportation Invoice Blocked	Freight Settlement Cost Query (C_FreightSettlementCostQ) Transportation Order Invoicing Block Status Query (C_TranspOrdInvchgBlockStsQ)
Freight Booking Execution Monitoring	Freight Booking Execution Freight Booking Execution Status	Freight Booking Execution Event Query (C_FrtBkgExecutionQuery)

Overview Page	Detailed Views	Contributing CDS Views
		Freight Booking Execution Status Query (C_FrtBkgExecutionStsQ)
Freight Booking Quantity Analysis	Freight Booking Quantity	Freight Booking Quantity Query (C_FrtBkgQuantityQuery)
Freight Order Execution Monitoring	Freight Order Execution Freight Order Execution Status	Freight Order Execution Event Query (C_FrtOrdExecutionQuery) Freight Order Execution Status Query (C_FrtOrdExecutionStsQ)
Freight Order Quantity Analysis	Freight Order Quantity	Freight Order Quantity Query (C_FrtOrdQuantityQuery)
Tendering Analysis	Tendering	Tendering Query (C_TenderingQuery)

Metadata Extensions to Replace Delivered Time Filters

SAP provides time filters with all the CDS views behind the OVPs and analytical list pages (ALPs) in Transportation Management. The time filters in all cases contain the current and previous month, on a two-month rolling basis. You can change the standard delivered time filters by using metadata extensions. For more information, see SAP Note [2779334](#) .

Allocation Analysis

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You can use the [Allocation Analysis](#) overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. At a glance, you can see all the vital data you need for your daily decision-making.

The KPIs show the weight, volume, and amounts of the transportation allocation, including net, gross, minimum, and maximum values. You can also see the total and average utilization for all units of measure.

The overview page provides you with KPIs for the current and previous month.

Key Features

- **Trade Lane Weight Load Analysis**

You can view the total current, minimum, and maximum weights, listed by orientation (for example, along) for each carrier. This may help you identify the most effective trade lane for goods of a particular size and orientation.

- **Trade Lane Volume Load Analysis**

You can see the aggregated current, minimum, and maximum volumes that each carrier is transporting, listed by orientation. This may help you identify the most effective trade lane for goods of a particular volume and orientation.

- **Trade Lane Container Load Analysis**

You can review the total current, minimum, and maximum container counts, listed by orientation for each carrier. This may help you identify the most effective trade lane for transporting containers.

- **Transportation Mode Category Analysis**

For each transportation mode category (for example, air, sea), you can view the weight, volume, and quantities that have been transported.

- **Overall Weight Utilization Distribution**

You can see the total weight that each carrier is transporting per month (for the current and previous month). You can also analyze any increase from the average minimum threshold for the transported weight, or any decrease from the average maximum threshold.

- **Overall Container Utilization Distribution**

You can see the total container count for each carrier per month (for the current and previous month). You can also analyze any increase from the average minimum container count threshold, or any decrease from the average maximum threshold.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the transportation allocation in the Transportation Management system and change it if necessary.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2673

This overview page uses the C_TranspAllocComplianceQ CDS view.

You must activate the C_TRANSPALLOCCOMPLIANCEQ_CDS OData service in the Gateway system.

More Information

Subject	More Information
Carrier selection overview for a business person	Carrier Selection
Allocation analysis detailed view for a business person	Transportation Allocation Detailed View
Underlying technical object for a technical person	C_TranspAllocComplianceQ CDS view Transportation Allocation Compliance Query

Transportation Allocation Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze your transportation allocation and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered transport allocation data in different ways.

You can quickly see the total gross weight of goods you transported in the current and previous month represented visually by a donut chart. The total gross weight of goods you transported, filtered by transportation mode category, is depicted by a line chart. Bar charts show the total gross volume per means of transport, and the number of containers by orientation.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the total gross weight of goods you transported by road and per orientation used.

- Field view

You can change to a field filter (compact filter), where you can enter the following filter criteria: weight, volume, and quantity unit, minimum and maximum tolerance (percentage), transportation mode, means or transport, and orientation.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by current value of gross weight, volume, and number of containers in the current and previous month. You can change the chart to a table or a combined table and chart. Use the [View By](#) filter to view more specific information such as means of transport, destination location, and trade lane. For example, you can quickly see the number of containers being transported by road to a specific destination location in the current month.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as allocation, orientation, source and destination location, and gross weight and volume. You can add or remove columns in [Settings](#) to give you the information you want to see, for example, the volume of goods transported from a source location in the previous month and the gross volume utilization.

Selecting an allocation in the table and pressing [Change Transportation Allocation](#) takes you directly to the transportation allocation in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to find out the destination locations to which you transported the highest gross weight by road last month, and the number of containers used. You see that one location transported a low weight but used a relatively high number of containers. You navigate directly to the transportation allocation to find out which carrier transported the goods and view further details on that location.

You can now investigate any issues with that destination location and whether the carrier could be more efficient.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2745

This detailed view uses the C_TranspAllocComplianceQ CDS view.

You must activate the C_TRANSPALLOCCOMPLIANCEQ_CDS OData service in the Gateway system.

More Information

[Allocation Analysis](#)

[Transportation Allocation Compliance Query](#)

Transportation Allocation Compliance Query

Technical Name	View Type
C_TRANSPALLOCCOMPLIANCEQ	Consumption view

Purpose

This view contains key performance indicators (KPIs) based on the weight, volume, and amounts of the transportation allocation, including net, gross, minimum, and maximum values. All the KPIs are available on the database level.

The view also calculates the total and average utilization from these KPIs for all units of measure.

Prerequisites

Authorization is based on both standard Transportation Management and transportation allocation authorization objects, as well as business partner and trade lane authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN) with activity [H1](#) checked, or the [Transp. Management: Allocation](#) authorization object (T_TAL) with activity [03](#) checked.

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all the business partners you access in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all the locations you access in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or T_TAL authorization object.		
		ACTVT	H1 (Deactivate)

Authorization Object	Check Condition	Field	Value
T_TAL	Check if the user has either T_ADMIN or T_TAL authorization object.		
		TM_TALTYPE	TranspAllocType
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier is set to Completed (value of view field CarrierBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03. The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code> . The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.Authorizat</code> and <code>_DestinationLocation.Autho</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Allocation business object.

Main Input Parameters

The main input parameters are as follows:

- P_Language

Specifies the display language of the texts. The system automatically uses the system language as the P_Language.

- P_TodayDate

Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.

- P_DisplayWeightUnit

Specifies the units of weight, for example, **Gross Weight** in kilograms.

- P_DisplayVolumeUnit

Specifies the units of volume, for example, **Item Volume** in meters cubed.

- P_DisplayQuantityUnit

Specifies the units of quantity, for example, **Number of Containers** in pieces.

- P_TrAlcMinToleranceInPercent

Sets another tolerance value for the minimum value.

- P_TrAlcMaxToleranceInPercent

Sets another tolerance value for the maximum value.

The weight and volume unit parameters are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- TransportationAllocation

Transportation Allocation Attributes

- YearMonth
- TranspAllocCategory
- TranspAllocType
- TranspAllocValdtyStartDateTime
- TranspAllocValdtyEndDateTime
- TranspAllocBucketType
- TrAlcBucketValdtyStrtDate
- TrAlcBucketValdtyStrtDteTme
- TrAlcBucketValdtyEndDteTime
- TranspAllocValdtyEndDateTime
- TrAlcBucketValdtyStrtDteYrMnth
- TrAlcBucketValdtyEndDteYrMnth

- TrAlcCtnUnitOfMeasureDimn
- TrAlcGrossVolUnitOfMsrDimn
- TrAlcGrossWgtUnitOfMsrDimn
- TradeLane
- TradeLaneOrientation
- SourceLocation
- DestinationLocation
- TransportationMode
- MeansOfTransport
- TransportationModeCategory

Transportation Allocation Measures

- NumberOfTranspAllocBuckets
- TrAlcMinGrossWgtInDspUnit
- TrAlcMaxGrossWgtInDspUnit
- TrAlcCurGrossWeightInDspUnit
- TrAlcMinGrossWgtTolInDspUnit
- TrAlcMaxGrossWgtTolInDspUnit
- TrAlcGrossWgtTolHigh
- TrAlcGrossWgtDvtnHigh
- TrAlcGrossWgtTolLow
- TrAlcGrossWgtDvtnLow
- TrAlcUtilznGrossWeightRatio
- TrAlcRltvDeviationGrossWeight
- TrAlcMinGrossVolInDspUnit
- TrAlcMaxGrossVolInDspUnit
- TrAlcCurGrossVolumeInDspUnit
- TrAlcMinGrossVolTolInDspUnit
- TrAlcMaxGrossVolTolInDspUnit
- TrAlcGrossVolTolHigh
- TrAlcGrossVolDvtnHigh
- TrAlcGrossVolTolLow
- TrAlcGrossVolDvtnLow
- TrAlcUtilznGrossVolumeRatio

- TrAlcRltvDeviationGrossVolume
- TrAlcMinNmbrOfCtnsInDspUnit
- TrAlcMaxNmbrOfCtnsInDspUnit
- TrAlcCurNmbrOfCtnsInDspUnit
- TrAlcMinNmbrOfCtnsTolInDspUnit
- TrAlcMaxNmbrOfCtnsTolInDspUnit
- TrAlcNmbrOfCtnsTolHigh
- TrAlcNmbrOfCtnsDvtnHigh
- TrAlcNmbrOfCtnsTolLow
- TrAlcNmbrOfCtnsDvtnLow
- TrAlcUtilznContainerRatio
- TrAlcRltvDeviationContainer

Units

- DisplayQuantityUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Business Share Analysis

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You can use the [Business Share Analysis](#) overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. The KPIs show the current amount and target quantity of each carrier's transports, the current value to target quantity ratio, and business share quantities per month. At a glance, you can see all the vital data you need for your daily decision-making. The overview page provides you with KPIs for the current and previous month.

Key Features

- **Weight Fulfillment Ratio**

You can see how carriers have performed in respect of total weight transported, that is, how much of their allocated share they have achieved. You can also see this as a percentage. For example, one carrier may have shipped 328,750kg out of their allocation of 493,125kg (66.67%).

- **Volume Fulfillment Ratio**

You can see how carriers have performed in respect of total volumes transported, that is how much of their allocated share they have achieved. You can also see this as a percentage. For example, one carrier may have shipped 5000m3 out of their allocation of 10000m3 (50%) .

- **Trade Lane Fulfillment**

You can view how each trade lane is performing for all units of measure (such as weight and volume), that is, how much of their allocated share they have actually transported. You can also see this as a percentage. For example, one trade lane may have transported 987,500kg out of their allocation of 987,500kg (100%).

- **Road Fulfillment Distribution (Weight)**

You can review the volume of goods transported by road by seeing the actual booked carrier quantity against the target quantity. This is illustrated as a graph.

- **Sea Fulfillment Distribution**

You can see the volume of goods transported by sea by viewing the actual booked carrier quantity against the target quantity. This is illustrated as a graph.

- **Air Fulfillment Distribution**

You can review the volume of goods transported by air by seeing the actual booked carrier quantity against the target quantity. This is illustrated as a graph.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the business share document in the Transportation Management system and change it if necessary.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2674

This overview page uses the C_BusinessShareComplianceQ CDS view.

You must activate the C_BUSINESSSSHARECOMPLIANCEQ_CDS OData service in the Gateway system.

More Information

Subject	More Information
Business share overview for a business person	Business Share
Business share detailed view for a business person	Business Share Detailed View
Underlying technical object for a technical person	C_BusinessShareComplianceQ Business Share Compliance Query

Business Share Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze and change your business share quantities and change them if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered business share data in different ways.

You can quickly see how your bookings for the month to date compare with the bookings for the previous month represented visually by a donut chart. A line chart shows fulfilled quantities per transportation mode category, such as road, sea, and air, while fulfilled quantities per carrier are depicted by a bar chart.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for booked quantities in the current and previous month and the fulfillment ratio per transportation mode category.

- Field view

You can change to a field filter (compact filter), where you can enter the following filter criteria: weight and volume unit, dimensions (mass or volume), current or previous month, transportation mode, and carrier.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by target against booked quantities per transportation mode category for the current and previous month. For example, you can quickly see whether booked quantities aligned with target quantities last month. Use the **View By** filter to view more specific information such as carrier, means of transport, and trade lane. For example, you can quickly see the quantity of goods being transported within Germany by road in refrigerated trucks by a specific carrier in the current month.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as fulfillment ratio, booked quantities, and target quantities in the current and previous month. You can add or remove columns in **Settings** to give you the information you want to see. This enables you to see how a business share is performing against how you planned that it would perform.

Selecting an allocation in the table and pressing **Change Business Share** takes you directly to the business share in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

In a specific trade lane for the month to date, you planned that a carrier would transport 70% of your shipments and another carrier would transport 30% of your shipments. However, in detailed analysis, you notice that the first carrier has actually transported 90% of your shipments. The second carrier hasn't transported any shipments.

You know that you have already incurred penalties because you have passed the 70% threshold for the first carrier, and you will not reach the 30% threshold for the second carrier in the given time period.

You research why the first carrier got so much of your business, and you change the business share information to favor the second carrier. You are more confident now that the month will come in closer to what you had originally planned.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2746

This detailed view uses the C_BusinessShareComplianceQ CDS view.

You must activate the C_BUSINESSSHARECOMPLIANCEQ_CDS OData service in the Gateway system.

More Information

[Business Share Analysis](#)

[Business Share Compliance Query](#)

Business Share Compliance Query

Technical Name	View Type
C_BUSINESSSHARECOMPLIANCEQ	Consumption view

Purpose

This view contains business share key performance indicators (KPIs) based on the current amount and target quantity of the carrier's transports. The KPIs show relative and absolute tolerance values, the current value to target quantity ratio, and business share quantities per month.

Prerequisites

Authorization is based on both standard Transportation Management and business share authorization objects, as well as business partner and trade lane authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN) with activity **H1** checked, or the [Transp. Management: Business Share](#) authorization object (T_BUSSHARE) with activity **03** checked.

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or T_BUSSHARE authorization object.		
		ACTVT	H1 (Deactivate)
T_BUSSHARE	Check if the user has either T_ADMIN or T_BUSSHARE authorization object.		
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier is set to Completed (value of view field CarrierBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03 . The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code> . The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management. The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGr</code> and <code>_DestinationLocation.Authorizat</code>
		ACTVT	03 (Display(

Structure

Business Objects

This view is built on the Business Share business object.

Main Input Parameters

The main input parameters are as follows:

- **P_Language**
Specifies the display language of the texts. The system automatically uses the system language as the P_Language.
- **P_TodayDate**
Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- **P_DisplayWeightUnit**
Specifies the units of weight, for example, **Gross Weight** in kilograms.
- **P_DisplayVolumeUnit**
Specifies the units of volume, for example, **Item Volume** in meters cubed.

The weight and volume unit parameters are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- BusinessShare

Business Share Attributes

- Carrier
- CalendarYear
- YearMonth
- BusShrValidityStartDate
- BusShrValidityEndDate
- BusShrValidityStartDateTime
- BusShrValidityEndDateTime
- TradeLaneOrientation
- TradeLane
- SourceLocationUUID
- DestinationLocationUUID
- TransportationMode
- TransportationModeCategory
- MeansOfTransport

- UnitOfMeasureDimension

Business Share Measures

- BookedBusinessShrQtyInDspUnit
- BookedCarrBusShrQtyInDspUnit
- BusShrTargetQuantityInDspUnit
- BusShrPositiveToleranceValue
- BusShrNegativeToleranceValue
- BusShrPositiveToleranceInPct
- BusShrNegativeToleranceInPct
- BusinessShareRelativeDeviation
- BusinessShareAbsoluteDeviation
- BusinessShareRatio

Units

- RatioUnit
- DisplayUnit

Constraints

None.

Cost Analysis

As a transportation manager or person responsible for costs, you need to track your costs and be aware of any issues with freight orders or freight bookings that are blocked for invoicing. You can use the [Cost Analysis](#) overview page to provide you with the key performance indicators (KPIs) that you need to analyze costs and take the appropriate action if required.

The KPIs show cost amounts analyzed in different ways: by invoicing week, transportation mode category (with weights and volumes), carrier, charge type, purchasing organization, source and destination country or region, and number of transportation orders (freight orders and freight bookings) blocked for invoicing, with blocking reasons. At a glance, you can see all the vital data you need for your daily decision-making.

The overview page provides you with KPIs for the current and previous month.

This overview page is based on two CDS views (data sources), each generating different cards.

Key Features

The following features are available with data source 1 (C_FreightSettlementCostQ CDS view):

- **Cost Distribution**

A chart displays the increase or decrease in invoiced amounts, credit amounts, and cost amounts in each invoicing week. The total cost appears at the top, for example, 5.3M euros.

- **Cost and Quantity Analysis**

A bubble chart shows the total cost for each transportation mode category, together with the gross weight and volume transported. The overall total cost appears at the top, for example, 1.6B euros.

- **Logistic Network Analysis**

You can see a table displaying the total cost for goods transported from source to destination countries or regions, with the highest cost at the top. For example, Germany to USA has a total cost of 2.2M euros. This lets you see at a glance where you are spending most on transportation costs.

- **Carrier Cost Analysis**

You can view a table showing carriers with the highest costs (highest at the top). The total cost per carrier is shown alongside their name. This gives you a quick visual indication of the carriers with whom you are spending the most.

- **Cost and Quantity Analysis**

You can see the total cost per transportation mode category, together with the gross weight and volume transported in each category.

- **Charge Type Analysis**

You can review charge types with the highest costs (highest at the top). The total cost, with gross weight and volume, is displayed for each charge type (for example, basic freight charges, pick up charges), enabling you to see where you are spending the most.

- **Purchasing Organization Cost Analysis**

You can see purchasing organizations with the highest costs (highest at the top). The total cost for each purchasing organization is shown alongside their name, enabling you to see at a glance the purchasing organizations with which you are doing the most business.

The following feature is available with data source 2 (C_Transp0rdInvcgBlockStsQ CDS view):

- **Invoicing Block Analysis**

You can see how many transportation orders (freight orders or freight bookings) are blocked for invoicing. The numbers are shown per blocking reason, such as time conflict, compliance, incomplete for planning. This may prompt you to investigate further, perhaps to find out why specific invoices are blocked, and to take immediate action if necessary.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight booking or freight order in the Transportation Management system and change it if necessary.

Selecting the [Invoicing Block Analysis](#) card takes you to the [Transportation Invoice Blocked Detailed View](#). Selecting any other card takes you to the [Transportation Costs Detailed View](#).

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2672

This overview page uses the the following CDS views:

- C_FreightSettlementCostQ (data source 1)
-
- C_TranspOrdInvcgBlockStsQ (data source 2)

You must activate the following OData services in the Gateway system:

- C_FREIGHTSETTLEMENTCOSTQ_CDS (data source 1)
- C_TRANSPORDINVCGBLOCKSTSQ_CDS (data source 2)

More Information

Subject	More Information
Cost analysis overview for a business person	Settlement
Cost analysis detailed views for a business person	Transportation Costs Detailed View Transportation Invoice Blocked Detailed View
Underlying technical objects for a technical person	Freight Settlement Cost Query C_FreightSettlementCostQ Freight Settlement Cost Query Transportation Order Invoicing Block Status Query C_TranspOrdInvcgBlockStsQ Transportation Order Invoicing Block Status Query

Transportation Costs Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyse your costs and change them if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your costs filtered in different ways.

For example, you can quickly see the total cost for goods you transported by road and the proportion of your business this represented, represented visually by donut and line charts as follows:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for invoice year week and transportation mode category.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as weight unit, volume unit, and currency.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as carrier, charge type, and source and destination countries or regions. For example, you can quickly see the cost of goods transported by road to the USA by a carrier in a specific invoicing week.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as carrier, charge type, purchasing organization, and gross weight and volume. It also lists the relevant freight settlement documents. You can add or remove columns in **Settings** to give you the information you want to see, for example, the volume of goods transported to a country or region and by which carrier in a specific invoicing week.

Selecting a row in the table takes you directly to the freight settlement document in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to see how much you spent on transporting goods by air from the USA to Japan last month, with a view to making savings. You see that you spent a high amount in one invoice week, while no costs were involved in other weeks. You navigate directly to the freight settlement document to find out which carrier transported the goods and view the charges involved. You notice that goods were transported by air five times during that week. You also navigate to the air freight booking to see more details.

You now have the facts to further investigate the reasons for transporting the goods by air that week; perhaps it had become an urgent shipment and if more notice had been given, you could have used a less expensive mode of transport.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2754

This detailed view uses the C_FreightSettlementCostQ CDS view.

You must activate the C_FREIGHTSETTLEMENTCOSTQ_CDS OData service in the Gateway system.

More Information

[Cost Analysis](#)

Transportation Invoice Blocked Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to investigate transportation orders blocked for invoicing and take action if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view transportation orders (freight orders and freight bookings) that are blocked for invoicing filtered in different ways.

You can quickly see the total cost of transportation orders blocked for invoicing per transportation order life cycle status year week and per transportation mode category, represented visually by donut and line charts as follows:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for cost amount by life cycle status year week and transportation mode category.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as weight unit, volume unit, and currency.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as carrier, block category, and block reason. For example, you can quickly see the total cost of transportation orders blocked for invoicing involving goods transported by sea for a specific carrier, and the reasons why they are blocked.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as block category, block reason, carrier, source and destination country or region, and document category. It also lists the relevant Transportation Management documents. You can add or remove columns in **Settings** to give you the information you want to see. This enables you to see, for example, which carrier was used and the destination of the goods being transported in a specific transportation order life cycle status year week.

Selecting a row in the table takes you directly to the freight booking or freight order in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to know how many freight orders are currently blocked for invoicing for goods being transported by sea and find out why, so that you can take remedial action. You see that three orders are blocked with a block category of time conflict and a block reason of penalty charges. You go directly to the freight orders to view the charges and gain more insight into what happened. You find out that the carrier arrived late for their customs slot on two occasions and can see all the relevant details.

Armed with the facts, you can now contact the carrier to resolve any dispute over payment of the penalty charges and ensure that they keep to their schedule in future.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2755

This detailed view uses the C_Transp0rdInvcgBlockStsQ CDS view.

You must activate the C_TRANSPORDINVCGBLOCKSTSQ_CDS OData service in the Gateway system.

More Information

[Cost Analysis](#)

Freight Settlement Cost Query

Technical Name	View Type
C_FREIGHTSETTLEMENTCOSTQ	Consumption view

Purpose

This view contains key performance indicators (KPIs) to enable you to see if you've reached your business goals for individual charge types and timeframe.

The view also contains calculations showing you how much you've spent on transportation for each carrier and how much you've spent on each transportation mode.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the **Transp. Management Administrative Settings** authorization object (T_ADMIN), or all of the following authorization objects:

- **Transp. Management Order Management & Planning (Purchase)** (T_TOR_PUR)
- **Transp. Management Order Management & Planning (Execution)** authorization object (T_TOR_EXE)
- **Transp. Management Freight Settlement** authorization object (T_SD_FSD)

In addition, you'll need the **Business Partner: Authorization Groups** authorization object (B_BUPA_GRP) for all business partners accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Field	Value	Description
T_ADMIN			Transp. Management: Administrative Settings authorization object
	ACTVT	H1 (Deactivate)	Activity

Authorization Object	Field	Value	Description
T_TOR_PUR			Transp. Management: Order Management & Planning (Purchase) authorization object
	TM_OC	TransportationOrderCategory	Field in query view
	TM_OT	TransportationOrderType	Field in query view
	TM_XPORG	PurchasingOrgExternalID	Field in query view
	TM_XPGRP	PurchasingGroupExternalID	Field in query view
	TM_RP	Transp0rdResponsiblePerson	Field in query view
	TM_MOT	TransportationMode	Field in query view
	ACTVT	09 (Display prices)	Activity
T_TOR_EXE			Transp. Management: Order Management & Planning (Execution) authorization object
	TM_OC	TransportationOrderCategory	Field in query view
	TM_OT	TransportationOrderType	Field in query view
	TM_XEORG	PlngAndExecOrgExternalID	Field in query view
	TM_XEGRP	PlngAndExecGroupExternalID	Field in query view
	TM_RP	Transp0rdResponsiblePerson	Field in query view
	TM_MOT	TransportationMode	Field in query view
	ACTVT	09 (Display prices)	Activity
T_SD_FSD			Transp. Management: Freight Settlement authorization object
	TM_SF_CAT	SuplrFrtInvcReqCategory	Field in query view
	TM_SF_TYPE	SuplrFrtInvcReqType	Field in query view
	ACTVT	03 (Display)	Activity
B_BUPA_GRP			Business Partner: Authorization Groups authorization object
	BEGRU	CarrierAuthorizationGroup ShipperAuthorizationGroup ConsigneeAuthorizationGroup	Fields in query view
	ACTVT	03 (Display)	Activity

Structure

Business Objects

This view is built on the following business business objects:

- Freight Settlement Documents
- Transportation Order
- Transportation Charges

Main Input Parameters

The main input parameters are as follows:

- **P_Language**
Specifies the display language of the texts. The system automatically uses the system language as the P_Language.
- **P_TodayDate**
Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- **P_DisplayTimeUnit**
Specifies the units of time, for example, **Event Delay** in hours or minutes.
- **P_DisplayWeightUnit**
Specifies the units of weight, for example, **Gross Weight** in kilograms.
- **P_DisplayVolumeUnit**
Specifies the units of volume, for example, **Item Volume** in meters cubed.
- **P_DisplayQuantityUnit**
Specifies the units of quantity, for example, **Number of Containers** in pieces.
- **P_DisplayLengthUnit**
Specifies the units of length, for example, **Package Length** in meters.
- **P_DisplayCurrency**
Specifies the currency, for example, **Invoice Amount** in euros.

The unit parameters (time, weight, volume, quantity, length, and currency) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- SupplierFreightInvoiceRequest
- TransportationOrder
- SuplrFrftInvReqUUID

- TransportationOrderUUID
- TransportationChargesObjUUID
- TranspChargeItemUUID
- TranspChargeItemElementUUID

Freight Settlement Document Attributes

- SuplrFrtInvcReqCategory
- SuplrFrtInvcReqType
- SuplrFrtInvcReqLifeCycSts
- SuplrFrtInvcReqConfStatus
- SuplrFrtInvcReqConstcySts
- SuplrFrtInvcReqArchStatus
- InvoicingStatus
- PurchasingOrganization
- PurchasingGroup
- BillFromParty
- Carrier
- InvoiceDate
- InvoiceDateYearWeek
- InvoiceDateYearMonth
- InvoiceDateYearQuarter
- InvoiceDateYear

Freight Order/Freight Booking Attributes

- TransportationOrderCategory
- TransportationOrderType
- TranspOrdShippingType
- TransportationMode
- TransportationModeCategory
- MeansOfTransport
- TranspOrdTrafficDirection
- TranspOrdLifeCycleStatus
- TranspOrdPlanningStatus
- PlngAndExecOrganization
- PlngAndExecGroup

- Shipper
- Consignee
- SourceLocation
- SourceLocationType
- SourceLocationCountry
- SourceLocationRegion
- DestinationLocation
- DestinationLocationType
- DestinationLocationCountry
- DestinationLocationRegion

Transportation Charges Attributes

- TranspChargeCategory
- TranspChargeSubCategory
- TranspChargeType
- TranspChargeInstrnType
- TranspChargeCalcStatus
- CarrierInvoicingStatus
- TranspChargeItmInvcgStatus
- TranspChargeCalcLevel
- TranspChargeCalcSheetLineNmbr
- TranspChargeCalcReslnBase

Freight Order/Freight Booking Measures

- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit
- TranspOrdNumberOfCtnsInDspUnit
- TranspOrdDistanceInDspUnit
- TranspOrdNetDurationInDspUnit

Transportation Charges Measures

- InvoicedAmountInDspCurrency
- CreditAmountInDisplayCurrency
- CostAmountInDisplayCurrency

Units

- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit
- DisplayQuantityUnit
- DisplayLengthUnit
- DisplayCurrency

Constraints

None.

Transportation Order Invoicing Block Status Query

Technical Name	View Type
C_TRANSPORDINVCGBLOCKSTSQ	Consumption view

Purpose

This view contains key performance indicators (KPIs) to help you find out if any freight orders or booking orders are blocked for invoicing (for a specific timeframe).

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the **Transp. Management Administrative Settings** authorization object (T_ADMIN), or all of the following authorization objects:

- **Transp. Management Order Management & Planning (Purchase)** (T_TOR_PUR)
- **Transp. Management Order Management & Planning (Execution)** authorization object (T_TOR_EXE)
- **Transp. Management Freight Settlement** authorization object (T_SD_FSD)

In addition, you'll need the **Business Partner: Authorization Groups** authorization object (B_BUPA_GRP) for all business partners accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Field	Value	Description
T_ADMIN			Transp. Management: Administrative Settings authorization object
	ACTVT	H1 (Deactivate)	Activity

Authorization Object	Field	Value	Description
T_TOR_PUR			Transp. Management: Order Management & Planning (Purchase) authorization object
	TM_OC	TransportationOrderCategory	Field in query view
	TM_OT	TransportationOrderType	Field in query view
	TM_XPORG	PurchasingOrgExternalID	Field in query view
	TM_XPGRP	PurchasingGroupExternalID	Field in query view
	TM_RP	TranspOrdResponsiblePerson	Field in query view
	TM_MOT	TransportationMode	Field in query view
	ACTVT	09 (Display prices)	Activity
T_TOR_EXE			Transp. Management: Order Management & Planning (Execution) authorization object
	TM_OC	TransportationOrderCategory	Field in query view
	TM_OT	TransportationOrderType	Field in query view
	TM_XEORG	PlngAndExecOrgExternalID	Field in query view
	TM_XEGRP	PlngAndExecGroupExternalID	Field in query view
	TM_RP	TranspOrdResponsiblePerson	Field in query view
	TM_MOT	TransportationMode	Field in query view
	ACTVT	09 (Display prices)	Activity
T_SD_FSD			Transp. Management: Freight Settlement authorization object
	TM_SF_CAT	SuplrFrtInvcReqCategory	Field in query view
	TM_SF_TYPE	SuplrFrtInvcReqType	Field in query view
	ACTVT	03 (Display)	Activity
B_BUPA_GRP			Business Partner: Authorization Groups authorization object
	BEGRU	BillFromPtyAuthorizationGroup CarrierAuthorizationGroup ShipperAuthorizationGroup ConsigneeAuthorizationGroup	Fields in query view
	ACTVT	03 (Display)	Activity

Business Objects

This view is built on the following business business objects:

- Transportation Order
- Freight Settlement Documents
- Transportation Charges

Main Input Parameters

The main input parameters are as follows:

- **P_Language**
Specifies the display language of the texts. The system automatically uses the system language as the **P_Language**.
- **P_TodayDate**
Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the **P_TodayDate**. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- **P_DisplayWeightUnit**
Specifies the units of weight, for example, **Gross Weight** in kilograms.
- **P_DisplayVolumeUnit**
Specifies the units of volume, for example, **Item Volume** in meters cubed.
- **P_DisplayQuantityUnit**
Specifies the units of quantity, for example, **Number of Containers** in pieces.
- **P_DisplayCurrency**
Specifies the currency, for example, **Invoice Amount** in euros.

The unit and currency parameters (weight, volume, quantity, and currency) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- SupplierFreightInvoiceRequest
- TransportationOrder

Freight Order/Freight Booking Attributes

- TransportationOrderCategory
- TransportationOrderType

- TranspOrdShippingType
- TransportationMode
- TransportationModeCategory
- MeansOfTransport
- TranspOrdTrafficDirection
- TranspOrdLifeCycleStatus
- TranspOrdPlanningStatus
- TransportationOrderExecSts
- PurchasingOrganization
- PurchasingGroup
- PlngAndExecOrganization
- PlngAndExecGroup
- Shipper
- Consignee
- Carrier
- SourceLocation
- SourceLocationType
- SourceLocationCountry
- SourceLocationRegion
- DestinationLocation
- DestinationLocationType
- DestinationLocationCountry
- DestinationLocationRegion
- DangerousGoodsAreContained
- TranspOrdPlanningBlock
- TranspOrdExecutionIsBlocked
- InvoicingIsBlocked
- InvoicingBlockReason
- CarrierOriginOfEntry
- TranspOrdLfcycStsChgDteTime
- TranspOrdLfcycStsChgDte
- TranspOrdLfcycStsChgDteYrWeek
- TranspOrdLfcycStsChgDteYrMnth
- TranspOrdLfcycStsChgDteYearQtr

- TranspOrdLfcycStsChgDteYear

Freight Settlement Document Attributes

- SuplrFrtInvcReqCategory
- SuplrFrtInvcReqType
- SuplrFrtInvcReqLifeCycSts
- SuplrFrtInvcReqBlockReason
- SuplrFrtInvcReqBlockStatus
- SuplrFrtInvcReqConfStatus
- SuplrFrtInvcReqConstcySts
- SuplrFrtInvcReqArchStatus
- InvoicingStatus
- BillFromParty
- InvoiceDate
- InvoiceDateYearWeek
- InvoiceDateYearMonth
- InvoiceDateYearQuarter
- InvoiceDateYear

Transportation Charges Attributes

- TranspChargeCalcStatus
- CarrierInvoicingStatus
- TranspChargeItmInvcgStatus
- TranspChargeCalcLevel

Freight Order/Freight Booking Measures

- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit
- TranspOrdNumberOfCtnsInDspUnit
- NumberOfTransportationOrders
- NumberOfInvcgBlkdTranspOrds

Transportation Charges Measures

- InvoicedAmountInDspCurrency
- CreditAmountInDisplayCurrency

- CostAmountInDisplayCurrency

Units

- DisplayWeightUnit
- DisplayVolumeUnit
- DisplayQuantityUnit
- DisplayCurrency

Constraints

None.

Freight Booking Execution Monitoring

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You need to monitor your freight bookings with carriers to make sure they are being executed on time.

A freight booking contains the logistical plan for the transportation of goods, for example, fixed departure times of a ship. You use a freight booking to reserve freight space on a ship or in an airplane, and it also contains information on how the plan is actually carried out.

You can use the **Freight Booking Execution Monitoring** overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. At a glance, you can see all the vital data you need for your daily decision-making.

The overview page provides you with KPIs for the current and previous month.

Key Features

This overview page is based on two CDS views (data sources), each generating different cards.

The following features are available with data source 1 (C_FRTBKGEXECUTIONSTSQ_CDS CDS view):

- **Life Cycle Status Analysis**

You can view the number of execution-relevant freight bookings, grouped by freight booking life cycle status. Execution-relevant freight bookings have the execution status **In Execution (03)** and have a departure event but no arrival event at the final destination.

You can see the number of freight bookings for each life cycle status (**Draft (00)**, **New (01)**, or **In Process (02)**), sorted in descending order.

- **Execution Block Analysis**

You can review all freight bookings with an execution block, grouped by execution block reason. This helps you to determine the most common causes of execution blocking and take the appropriate action. The list is sorted in descending order.

You specify execution block reasons in the TM Freight Booking transaction; if you haven't done this, the system automatically creates them, based on the blocking category.

- **Planning Block Analysis**

You can review all freight bookings with a planning block, grouped by planning block reason. This helps you to determine the most common causes of planning blocking and take the appropriate action. The list is sorted in descending order.

You specify execution block reasons in the TM Freight Booking transaction; if you haven't done this, the system automatically creates them, based on the blocking category.

- **Delayed Freight Bookings in Transit**

You can view freight bookings that are delayed in transit, that is, if their latest, time-relevant event is delayed (both the expected and actual date and time fields are available and the event status is **Reported**). The delay is the time difference between the actual date and the expected date.

You can see the total number of freight bookings and then drill down to view details such as **Freight Booking ID**, **Means of Transport**, **Gross Weight**, and **Delay**. You specify the units as parameters in the relevant CDS query view.

You can also navigate directly to the related freight booking transaction by selecting the first line or header line of the details view.

The following features are available with data source 2 (C_FRTBKGEXECUTIONQUERY_CDS CDS view):

- **Blocking Discrepancies**

You can review the discrepant freight booking events that led to a block, sorted by event code and discrepancy type (such as delivery of item, late, or delay). The discrepancy type with the most blocking events is at the top.

- **Non-Blocking Discrepancies**

You can see the discrepant freight booking events that didn't result in a block, sorted by event code and discrepancy type (such as arrival at destination, departure, or loading end). The discrepancy type with the most non-blocking events is at the top.

- **Carrier On Time Performance**

You can view the percentage of freight bookings that each carrier delivered on time. You can also see this expressed in numbers, for example, 6 / 14 were on time. A freight booking is on time if all its events are classed as on time, otherwise it is classed as delayed. This helps you identify the most efficient carriers.

The calculation for all the on time performance cards takes into account all time-relevant events where both expected and actual date and time fields are available, and where the event status is **Reported**.

- **Location On Time Performance**

This shows the percentage of on time freight booking events for each location, sorted by the transportation order events on time ratio. The lowest percentage is at the top. You can also see this expressed in numbers, for example, 6 / 8 events were on time. This helps you understand how individual locations are performing.

- **Activity On Time Performance**

You can review the percentage of on time freight booking events for each event code (such as loading or unloading), with the lowest number at the top. You can also see this expressed in numbers, for example, 12 / 27 events took place on time. You can use this information to identify potential bottlenecks.

- **Unexpected Events**

You can monitor unexpected events that took place within the freight booking, grouped by event code (such as accident or delay), with the highest number at the top. This information may prove vital in taking preventative measures to avoid such events in the future.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight booking worklist in the Transportation Management system and change it if necessary.

Selecting the blocking and non-blocking discrepancies, carrier, location, and activity on time performance, and unexpected events cards takes you to the [Freight Booking Execution Detailed View](#). Selecting the life cycle status analysis, execution and planning block analysis, and freight bookings delayed in transit cards takes you to the [Freight Booking Execution Status Detailed View](#)

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2677

This overview page uses the following CDS views:

- C_FrtBkgExecutionStsQ (data source 1)
- C_FrtBkgExecutionQuery (data source 2)

You must activate the following OData services in the Gateway system:

- C_FRTBKGEXECUTIONSTSQ_CDS (data source 1)
- C_FRTBKGEXECUTIONQUERY_CDS (data source 2)

More Information

Subject	More Information
Freight booking overview for a business person	Freight Booking
Freight booking execution detailed views for a business person	Freight Booking Execution Detailed View Freight Booking Execution Status Detailed View
Underlying technical objects for a technical person	C_FrtBkgExecutionStsQ CDS view Freight Booking Execution Status Query C_FrtBkgExecutionQuery CDS view Freight Booking Execution Event Query

Freight Booking Execution Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to monitor and analyze your freight booking execution and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

This detailed view provides further information on the following cards on the [Freight Booking Execution Monitoring](#) overview page:

- Blocking and non-blocking discrepancies
- Carrier, location, and activity on time performance
- Unexpected events

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight booking data in different ways.

You can quickly see the event on time ratio by location and carrier, represented visually by a bar chart and a line chart respectively. Donut charts display the ratio of on time events per event (such as arrival at destination, departure, or loading end), expressed as a percentage, and the number of events broken down into discrepancy type leading to a block and discrepancy type not leading to a block.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the on time ratio of arrival events, or for a specific location.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as event status and the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by number of freight bookings per transportation mode category and traffic direction. You can change the chart to a table or a combined table and chart. Use the [View By](#) filter to view more specific information such as discrepancy, discrepancy type, and event. For example, you can quickly see the number of freight bookings transported by road that arrived at their destination with no discrepant events.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as destination country or region, event, event status, and number of delayed freight bookings. You can add or remove columns in [Settings](#) to give you the information you want to see, for example, ship-to party and expected end date.

Selecting a freight booking in the table and pressing [Change Freight Booking](#) takes you directly to the freight booking in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to review the discrepant freight booking events that led to a block and see which discrepancy types are the most common, for example, delivery of item, late, or delay. You see that late is the most frequently occurring discrepancy type. You find out that several late events involved one location and the same means of transport. You navigate directly to the freight booking to see further details. You now have the information to investigate the issue with that location with the carrier and prevent future problems.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2747

This detailed view uses the C_FrtBkgExecutionQuery CDS view.

You must activate the C_FRTBKGEXECUTIONQUERY_CDS OData service in the Gateway system.

More Information

[Freight Booking Execution Monitoring](#)

[Freight Booking Execution Event Query](#)

Freight Booking Execution Status Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyse your freight booking execution status and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

This detailed view provides further information on the following cards on the [Freight Booking Execution Monitoring](#) overview page:

- Life cycle status analysis
- Execution and planning block analysis
- Freight orders delayed in transit

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight booking status data in different ways.

You can quickly see the number of freight bookings per life cycle status represented visually by a donut chart, for example, 1055 in process and 27 completed. Bar charts display the planning blocked freight bookings by transportation mode category and the number of execution blocked freight bookings by traffic direction. The number of execution blocked freight bookings by carrier and the number of freight bookings for destination country or region are depicted by line charts.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the number of execution blocked freight bookings per carrier and per traffic direction and the number of new freight bookings.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as the execution block and the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by number of freight bookings per transportation mode category and traffic direction. You can change the chart to a table or a combined table and chart. Use the [View By](#) filter to view more specific information such as as ship-to party, source country or region, and whether dangerous goods were transported. For example, you can quickly see the number of freight bookings transported by air with planning block, and the category for that block.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as carrier, ship-to party, and life cycle status. You can add or remove columns in [Settings](#) to give you the information you want to see, for example, execution status and life cycle change date.

Selecting a freight booking in the table and pressing [Change Freight Booking](#) takes you directly to the freight booking in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to review how many freight bookings sent by air have an execution block, and the execution block reason. You want to determine the most common causes of execution blocking for goods transported by air. You navigate directly to the freight booking to find out more details and take the appropriate action.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2748

This detailed view uses the `C_FrtBkgExecutionStsQ` CDS view.

You must activate the `C_FRTBKGEXECUTIONSTSQ_CDS` OData service in the Gateway system.

More Information

[Freight Booking Execution Monitoring](#)

[Freight Booking Execution Status Query](#)

Freight Booking Execution Event Query

Technical Name	View Type
C_FRTBKGEXECUTIONQUERY	Consumption view

Purpose

This view contains key performance indicators (KPIs) to help you monitor your freight bookings with carriers to make sure they are being executed on time, for example, KPIs on critical process statuses that lead to blocking situations in your operational business that you need to solve urgently. It also contains each carrier's on-time performance to aid when allocating future business.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	03 (Display)
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID
		TM_XPGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)

Authorization Object	Check Condition	Field	Value
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XEGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03 . The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the event, source location, or destination location is set to Completed (<code>_Location.IsBusinessPurposeCompleted = X</code> or <code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_Location.AuthorizationGroup</code> or <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code> . The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		

Authorization Object	Check Condition	Field	Value
		BRGRU	_Location.AuthorizationGroup and _SourceLocation.Authorizatio and _DestinationLocation.Authori
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with TransportationOrderCategory = **BO**.

Main Input Parameters

The main input parameters are as follows:

- **P_Language**
Specifies the display language of the texts. The system automatically uses the system language as the P_Language.
- **P_TodayDate**
Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- **P_DisplayTimeUnit**
Specifies the units of time, for example, **Event Delay** in hours or minutes.
- **P_DisplayWeightUnit**
Specifies the units of weight, for example, **Gross Weight** in kilograms.
- **P_DisplayVolumeUnit**
Specifies the units of volume, for example, **Item Volume** in meters cubed.
- **P_DisplayQuantityUnit**
Specifies the units of quantity, for example, **Number of Containers** in pieces.
- **P_DisplayLengthUnit**
Specifies the units of length, for example, **Package Length** in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- FreightBooking
- Transp0rdExecution
- TransportationOrderUUID
- TransportationOrderEventUUID

Freight Booking Attributes

- TransportationOrderType
- MeansOfTransport
- TransportationMode
- PlngAndExecOrganization
- PurchasingOrganization
- Transp0rdLifeCycleStatus
- TransportationOrderExecSts
- Carrier
- Shipper
- Consignee
- Transp0rdIsDiscrepant
- DiscrepancyType
- Transp0rdHasQtyDiscrepancy
- SourceLocationName
- SourceLocationType
- SourceLocationCountry
- DestinationLocationName
- DestinationLocationType
- DestinationLocationCountry

Freight Booking Execution Attributes

- Transp0rdEventCode
- Transp0rdEventStatus
- Transp0rdEventReasonCode
- Location
- LocationType
- Country

- GeographicalLocationLatitude
- GeographicalLocationLongitude
- TranspOrdEvtActualDateTime
- TranspOrdEvtExpdEndDateTime
- TranspOrdEvtActualDateTimeZone
- TranspOrdEvtEstimatedDateTime
- TranspOrdEventIsOnTime
- TranspOrdEventIsDelayed
- TranspOrdEventIsTimeRelevant

Freight Booking Measures

- NumberOfTransportationOrders
- NumberOfTimeRlvtTranspOrds
- NumberOfOnTimeTranspOrds
- NumberOfDelayedTranspOrds
- TransportationOrderOnTimeRatio

Freight Booking Execution Measures

- TranspOrdEventDelayInDspUnit
- AverageTranspOrdEventDelay
- NumberOfTranspOrdEvents
- NumberOfTimeRlvtTranspOrdEvts
- NumberOfOnTimeTranspOrdEvents
- NumberOfDelayedTranspOrdEvents
- TranspOrdEventOnTimeRatio

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Freight Booking Execution Status Query

Technical Name	View Type
C_ FRTBKGEXECUTIONSTSQ	Consumption view

Purpose

This view contains the key performance indicators (KPIs) for your daily operational business, such as transports that are critically blocked either in planning or execution, grouped by block reason. This helps you to determine the most common causes of blocking and take the appropriate action. It also contains transports that are currently in transit and delayed so that you can investigate and remedy the situation.

The view also has a workload analysis showing the number of transports in draft, new, in process, and in execution.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	03 (Display)

Authorization Object	Check Condition	Field	Value
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID
		TM_XPGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XEGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03 .		

Authorization Object	Check Condition	Field	Value
	The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code>. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGroup</code> and <code>_DestinationLocation.AuthorizationGroup</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with `TransportationOrderCategory = BO`.

Main Input Parameters

The main input parameters are as follows:

- P_Language**
 Specifies the display language of the texts. The system automatically uses the system language as the `P_Language`.
- P_TodayDate**
 Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the `P_TodayDate`. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- P_DisplayTimeUnit**
 Specifies the units of time, for example, **Event Delay** in hours or minutes.
- P_DisplayWeightUnit**
 Specifies the units of weight, for example, **Gross Weight** in kilograms.
- P_DisplayVolumeUnit**

Specifies the units of volume, for example, **Item Volume** in meters cubed.

- **P_DisplayQuantityUnit**

Specifies the units of quantity, for example, **Number of Containers** in pieces.

- **P_DisplayLengthUnit**

Specifies the units of length, for example, **Package Length** in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- FreightBooking
- TranspOrdExecution
- TransportationOrderUUID
- TransportationOrderEventUUID

Freight Booking Attributes

- TransportationOrderType
- MeansOfTransport
- TransportationMode
- PlngAndExecOrganization
- PurchasingOrganization
- TranspOrdLifeCycleStatus
- TransportationOrderExecSts
- Carrier
- Shipper
- Consignee
- TranspOrdIsInTransit
- TranspOrdIsDlydInTransit
- TranspOrdPlanningBlock
- TranspOrdPlanningBlockRsn
- TranspOrdExecutionIsBlocked
- TranspOrdExecBlockReason
- InvoicingIsBlocked
- SourceLocationName

- SourceLocationType
- SourceLocationCountry
- DestinationLocationName
- DestinationLocationType
- DestinationLocationCountry

Freight Booking Measures

- NumberOfTransportationOrders
- NumberOfTranspOrdsInTransit
- NmbrOfDelayedTranspOrdsInTrnst
- TranspOrdDelayInDspUnit
- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit
- TranspOrdNumberOfCtnsInDspUnit
- NumberOfPlngBlkdTranspOrds
- NumberOfExecBlkdTranspOrds

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Freight Booking Quantity Analysis

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You need to monitor your freight bookings with carriers to make sure they are being executed on time. A freight booking contains the logistical plan for the transportation of goods, for example, fixed departure times of a ship. You use a freight booking to reserve freight space on a ship or in an airplane, and it also contains information on how the plan is actually carried out.

You can use the **Freight Booking Quantity Analysis** overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. At a glance, you can see all the vital data you need for your daily decision-making. The overview page provides you with KPIs for the current and previous month.

Key Features

- **Gross Weight Distribution**

You can view a chart showing how gross weight distribution for completed freight bookings (that is, with a freight booking life cycle status of **Completed**) has increased or decreased in each completion week (current and previous month only). The total gross weight distribution also appears at the top, for example, 190K.

- **Gross Volume Distribution**

You can view a chart showing how gross volume distribution for completed freight bookings has increased or decreased in each completion week (current and previous month only). The total gross volume distribution also appears at the top, for example, 200K.

- **Quantity Distribution**

You can view a chart showing how quantity distribution for completed freight bookings has increased or decreased in each completion week (current and previous month only). The total gross quantity distribution also appears at the top, for example, 175K.

- **Transportation Mode Category Analysis**

You can see a chart with gross weight and gross volume as axes, plus the total number of freight bookings, for each transportation mode category for completed freight bookings. This can help you identify the most effective method of transportation.

- **Logistic Network Analysis**

You can review a table showing source and destination countries or regions, and total gross weight for completed freight bookings. This identifies where you have transported most goods in the current and previous months. The table is sorted in descending order of gross weight.

- **Carrier Quantity Analysis**

You can view the total gross weight, gross volume, and container count per carrier for completed freight bookings. This gives you information on which carriers you are using the most. The highest weight appears at the top.

- **Shipper Quantity Analysis**

You can view the total gross weight, gross volume, and container count per shipper for completed freight bookings. This gives you information on which shippers you are using the most. The highest weight appears at the top.

- **Ship-To Party Quantity Analysis**

You can view the total gross weight, gross volume, and container count per ship-to party for completed freight bookings. This gives you information on which ship-to parties you are using the most. The highest weight appears at the top.

- **Utilization Distribution**

You can see the lowest average utilization per carrier (expressed as a percentage) for completed freight bookings. This helps you assess carriers' performance and capacity. The lowest utilization percentage appears at the top.

- **Quantity Analysis**

You can view a table showing the gross volume and gross weight transported for each transportation mode category (for example, road and air) for completed freight bookings. You can use this information when deciding on the most efficient method of transportation. The table is sorted in descending order of gross weight.

- **Distance Analysis**

You can see the total net duration and total distance for transported goods for each transportation mode category (for example, sea and air) for completed freight bookings. You can use this information when deciding on the most efficient

method of transportation. The greatest distance appears at the top.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight booking in the Transportation Management system and change it if necessary.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2678

This overview page uses the C_FrtBkgQuantityQuery CDS view.

You must activate the C_FRTBKGQUANTITYQUERY_CDS OData service in the Gateway system.

More Information

Subject	More Information
Freight booking overview for a business person	Freight Booking
Freight booking quantity analysis detailed view for a business person	Freight Booking Quantity Detailed View
Underlying technical object for a technical person	C_FrtBkgQuantityQuery CDS view Freight Booking Quantity Query

Freight Booking Quantity Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze the execution of your freight bookings and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight booking data in different ways.

You can quickly see the total gross weight of goods transported per carrier in the current and previous month represented visually by a line chart, and the number of freight bookings per transportation mode category depicted by a bar chart.

The following views are available:

- o Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for gross weight per carrier in the current and previous month and the number of freight bookings per transportation mode category.

- o Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by gross weight per transportation mode category for completion weeks in the current and previous month. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as carrier, destination country or region, and life cycle status. For example, you can quickly see the total gross weight of goods transported by sea in each completion week of the current and previous month, and by which carriers.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as carrier, number of containers, and source and destination countries or regions. You can add or remove columns in **Settings** to give you the information you want to see, for example, completion date, destination location, and ship-to party.

Selecting a freight booking in the table and pressing **Change Freight Booking** takes you directly to the freight booking in Transportation Management. The document opens in edit mode.

Example

You want to assess the most efficient mode of transport to a specific location in China. Looking at the current and previous months, you can see the total gross weight of goods transported by air, road, and sea. You check the destination location and see that most goods were transported by sea. Navigating directly to the freight booking, you confirm that sea was indeed the most efficient mode of transport, with all bookings being completed on time.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2749

This detailed view uses the `C_FrtBkgQuantityQuery` CDS view.

You must activate the `C_ FRTBKGQUANTITYQUERY_ CDS` OData service in the Gateway system.

More Information

[Freight Booking Quantity Analysis](#)

[Freight Booking Quantity Detailed View](#)

Freight Booking Quantity Query

Technical Name	View Type
C_FRTBKGQUANTITYQUERY	Consumption view

Purpose

This view contains key performance indicators (KPIs) to help you monitor your freight bookings with carriers. It contains the current weights, volumes, and numbers being transported on a weekly basis by each carrier, shipper, and ship-to party, as well as by transportation mode, and source and destination country or region. The view also calculates each carrier's utilization.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the **Transp. Management Administrative Settings** authorization object (T_ADMIN), or both the **Transp. Management Order Management & Planning (Purchase)** authorization object (T_TOR_PUR) and the **Transp. Management Order Management & Planning (Execution)** authorization object (T_TOR_EXE).

You'll need the **Business Partner: Purpose Completed** authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the **Location: Access Based on Authorization Group** authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID

Authorization Object	Check Condition	Field	Value
		TM_XPGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory BO (Freight Booking)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XEGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03. The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (_SourceLocation.IsBusinessPurposeCompleted = X or _DestinationLocation.IsBusinessPurposeCompleted = X), check field BRGRU against the value of view field		

Authorization Object	Check Condition	Field	Value
	<code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code>. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGroup</code> and <code>_DestinationLocation.AuthorizationGroup</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with `TransportationOrderCategory = BO`.

Main Input Parameters

The main input parameters are as follows:

- P_Language**
 Specifies the display language of the texts. The system automatically uses the system language as the P_Language.
- P_TodayDate**
 Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- P_DisplayTimeUnit**
 Specifies the units of time, for example, [Event Delay](#) in hours or minutes.
- P_DisplayWeightUnit**
 Specifies the units of weight, for example, [Gross Weight](#) in kilograms.
- P_DisplayVolumeUnit**
 Specifies the units of volume, for example, [Item Volume](#) in meters cubed.
- P_DisplayQuantityUnit**
 Specifies the units of quantity, for example, [Number of Containers](#) in pieces.
- P_DisplayLengthUnit**
 Specifies the units of length, for example, [Package Length](#) in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- TransportationOrder
- TransportationOrderStage
- TransportationOrderUUID
- SourceLocationUUID
- DestinationLocationUUID

Freight Booking Attributes

- TransportationOrderCategory
- TransportationOrderType
- DangerousGoodsAreContained
- TranspOrdShippingType
- TranspOrdTrafficDirection
- MeansOfTransport
- TransportationMode
- TransportationModeCategory
- TranspOrdLifeCycleStatus
- PlngAndExecOrganization
- PlngAndExecGroup
- PurchasingOrganization
- PurchasingGroup
- TranspOrdCompletionDateTime
- TranspOrdCompletionDate
- TranspOrdCompletionYearWeek
- TranspOrdCompletionYearMonth
- TranspOrdCompletionYearQuarter
- TranspOrdCompletionYear
- ChangedDateTime
- Carrier
- CarrierOriginOfEntry

- Shipper
- Consignee
- SourceLocation
- SourceLocationType
- SourceLocationCountry
- SourceLocationRegion
- DestinationLocation
- DestinationLocationType
- DestinationLocationCountry
- DestinationLocationRegion

Freight Booking Stage Attributes

- TranspOrdStageSrceLocation
- TranspOrdStageSrceLocationType
- TranspOrdStageSrceLocCntry
- TranspOrdStageSrceLocRgn
- TranspOrdStageDestLocation
- TranspOrdStageDestLocationType
- TranspOrdStageDestLocCntry
- TranspOrdStageDestLocRgn

Freight Booking Measures

- NumberOfTransportationOrders
- TranspOrdMaxUtilznRatio
- TranspOrdMaxUtilznMassRatio
- TranspOrdMaxUtilznVolumeRatio
- TranspOrdNumberOfCtnsInDspUnit
- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit

Freight Booking Stage Measures

- NumberOfTranspOrdStages
- TranspOrdStageMaxUtilznRatio
- TranspOrdStageUtilznVolRatio

- TranspOrdStageUtilznMassRatio
- TranspOrdStageDstncInDspUnit
- TranspOrdStageNetDurnInDspUnit

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Freight Order Execution Monitoring

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You can use the [Freight Order Execution Monitoring](#) overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. At a glance, you can see all the vital data you need for your daily decision-making. For example, you can see KPIs on critical process statuses that lead to blocking situations in your operational business that may need your urgent attention. A carrier's on-time performance KPI also helps you make decisions when allocating future business.

The overview page provides you with KPIs for the current and previous month.

Key Features

This overview page is based on two CDS views (data sources), each generating different cards.

The following features are available with data source 1 (C_FrtOrdExecutionStsQ CDS view):

- **Life Cycle Status Analysis**

You can view the number of execution-relevant freight orders, grouped by freight order life cycle status. Execution-relevant freight orders have the execution status **In Execution (03)** and have a departure event but no arrival event at the final destination.

You can see the number of freight bookings for each life cycle status (**Draft (00)**, **New (01)**, or **In Process (02)**), sorted in descending order.

- **Execution Block Analysis**

You can review all freight orders with an execution block, grouped by execution block reason. This helps you to determine the most common causes of execution blocking and take the appropriate action.

You specify execution block reasons in the TM Freight Order transaction; if you haven't done this, the system automatically creates them, based on the blocking category.

The list is sorted in descending order.

- **Planning Block Analysis**

You can review all freight orders with a planning block, grouped by planning block reason. This helps you to determine the most common causes of planning blocking and take the appropriate action.

You specify execution block reasons in the TM Freight Order transaction ; if you haven't done this, the system automatically creates them, based on the blocking category.

The list is sorted in descending order.

- **Delayed Freight Orders in Transit**

You can view freight orders that are delayed in transit, that is, if their latest, time-relevant event is delayed (both the expected and actual date and time fields are available and the event status is **Reported**). The delay is the time difference between the actual date and the expected date.

You can see the total number of freight orders and then drill down to view details such as **Freight Order ID**, **Means of Transport**, **Gross Weight**, and **Delay**. You specify the units as parameters in the relevant CDS query view.

You can also navigate directly to the related freight order transaction by selecting the first line or the header line on the details view.

The following features are available with data source 2 (C_FrtOrdExecutionQuery CDS view):

- **Blocking Discrepancies**

You can review the discrepant freight order events that led to a block, sorted by event code and discrepancy type (such as delivery of item, late, or delay). The discrepancy type with the most blocking events is at the top.

- **Non-Blocking Discrepancies**

You can see the discrepant freight order events that didn't result in a block, sorted by event code and discrepancy type (such as arrival at destination, departure, or loading end). The discrepancy type with the most non-blocking events is at the top.

- **Carrier On Time Performance**

You can view the percentage of freight orders that each carrier delivered on time. You can also see this expressed in numbers, for example, 6 / 14 were on time. A freight order is on time if all its events are classed as on time, otherwise it is classed as delayed. This helps you identify the most efficient carriers.

The calculation for all the on time performance cards takes into account all time-relevant events where both expected and actual date and time fields are available, and where the event status is **Reported**.

- **Location On Time Performance**

This shows the percentage of on time freight order events for each location, sorted by the transportation order events on time ratio. You can also see this expressed in numbers, for example, 6 / 8 events were on time. This helps you understand how individual locations are performing. The lowest percentage is at the top.

- **Activity On Time Performance**

You can review the percentage of on time freight order events for each event code (such as loading or unloading), with the lowest number at the top. You can also see this expressed in numbers, for example, 12 / 27 events took place on time. You can use this information to identify potential bottlenecks.

- **Unexpected Events**

You can monitor unexpected events that took place within the freight order, grouped by event code (such as accident or delay), with the highest number at the top. This information may prove vital in taking preventative measures to avoid such events in the future.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight order in the Transportation Management system and change it if necessary.

Selecting the blocking and non-blocking discrepancies, carrier, location, and activity on time performance, and unexpected events cards takes you to the [Freight Order Execution Detailed View](#). Selecting the life cycle status analysis, execution and planning block analysis, and freight bookings delayed in transit cards takes you to the [Freight Order Execution Status Detailed View](#)

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2675

This overview page uses the following CDS views:

- C_FrtOrdExecutionStsQ (data source 1)
- C_FrtOrdExecutionQuery (data source 2)

You must activate the following OData services in the Gateway system:

- C_FRTORDEXECUTIONSTSQ_CDS (data source 1)
- C_FRTORDEXECUTIONQUERY_CDS (data source 2)

More Information

Subject	More Information
Freight order management overview for a business person	Freight Order Management
Freight order execution monitoring detailed views for a business person	Freight Order Execution Detailed View Freight Order Execution Status Detailed View
Underlying technical objects for a technical person	C_FrtOrdExecutionStsQ CDS view Freight Order Execution Status Query C_FrtOrdExecutionQuery CDS view Freight Order Execution Event Query

Freight Order Execution Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze your freight order execution and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

This detailed view provides further information on the following cards on the [Freight Order Execution Monitoring](#) overview page:

- Blocking and non-blocking discrepancies
- Carrier, location, and activity on time performance
- Unexpected events

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight order data in different ways.

You can quickly see the event on time ratio by location and carrier, represented visually by a bar chart and a line chart respectively. Donut charts display the ratio of on time events per event (such as arrival at destination, departure, or loading end), expressed as a percentage, and the number of events broken down into discrepancy type leading to a block and discrepancy type not leading to a block.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the on time ratio of arrival events, or for a specific location.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as event status and the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by number of freight orders per transportation mode category and traffic direction. You can change the chart to a table or a combined table and chart. Use the [View By](#) filter to view more specific information such as discrepancy, discrepancy type, and event. For example, you can quickly see the number of freight orders transported by road that arrived at their destination with no discrepant events.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as destination country or region, event, event status, and number of delayed freight orders. You can add or remove columns in [Settings](#) to give you the information you want to see, for example, ship-to party and expected end date.

Selecting a freight order in the table and pressing [Change Freight Order](#) takes you directly to the freight order in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to review the performance of carriers and see whether they delivered goods on time. This helps you identify the most efficient carriers. You see that two carriers had a much lower ratio than the others. You look at the events, event statuses, and any

discrepant events. You then navigate directly to the freight order to see if there were any extenuating circumstances. This information helps you make decisions when allocating future business.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2750

This detailed view uses the C_FrtOrdExecutionQuery CDS view.

You must activate the C_FRTORDEXECUTIONQUERY_CDS OData service in the Gateway system.

More Information

[Freight Order Execution Monitoring](#)

[Freight Order Execution Event Query](#)

Freight Order Execution Status Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze your freight order execution status and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

This detailed view provides further information on the following cards on the [Freight Order Execution Monitoring](#) overview page:

- Life cycle status analysis
- Execution and planning block analysis
- Freight orders delayed in transit

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight order status data in different ways.

You can quickly see the number of freight orders per life cycle status represented visually by a donut chart, for example, 1055 in process and 27 completed. Bar charts display the number of planning blocked freight orders by transportation mode category and the number of execution blocked freight orders by traffic direction. The number of execution blocked freight orders by carrier and the number of freight orders per destination country or region are depicted by line charts.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the number of execution blocked freight orders per carrier and per traffic direction and the number of new freight orders.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as the execution block and the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by number of freight orders per transportation mode category and traffic direction. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as ship-to party, source country or region, and whether dangerous goods were transported. For example, you can quickly see the number of freight orders transported by air with planning block, and the category for that block.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as carrier, ship-to party, and life cycle status. You can add or remove columns in **Settings** to give you the information you want to see, for example, execution status and life cycle change date.

Selecting a freight order in the table and pressing **Change Freight Order** takes you directly to the freight order in Transportation Management (it opens in edit mode), where you can change the information to get a more desirable outcome.

Example

You want to review how many freight orders sent by air have an execution block, and the execution block reason. You want to determine the most common causes of execution blocking for goods transported by air. You navigate directly to the freight order to find out more details and take the appropriate action.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2751

This detailed view uses the C_FrtOrdExecutionStsQ CDS view.

You must activate the C_FRTORDEXECUTIONSTSQ_CDS OData service in the Gateway system.

More Information

[Freight Order Execution Monitoring](#)

[Freight Order Execution Status Query](#)

Freight Order Execution Event Query

Technical Name	View Type
C_ FRTORDEXECUTIONQUERY	Consumption view

Purpose

This view contains key performance indicators (KPIs) to help you monitor your freight orders with carriers to make sure they are being executed on time, for example, KPIs on critical process statuses that lead to blocking situations in your operational business

that you need to solve urgently. It also contains each carrier's on-time performance to aid when allocating future business.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	03 (Display)
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID
		TM_XPGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson

Authorization Object	Check Condition	Field	Value
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XEGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03. The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the event, source location, or destination location is set to Completed (_Location.IsBusinessPurposeCompleted = X or _SourceLocation.IsBusinessPurposeCompleted = X or _DestinationLocation.IsBusinessPurposeCompleted = X), check field BRGRU against the value of view field _Location.AuthorizationGroup or _SourceLocation.AuthorizationGroup or _DestinationLocation.AuthorizationGroup. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management		

Authorization Object	Check Condition	Field	Value
	The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	_Location.AuthorizationGroup and _SourceLocation.Authorizatio and _DestinationLocation.Authori
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with TransportationOrderCategory = **TO**.

Main Input Parameters

The main input parameters are as follows:

- **P_Language**
Specifies the display language of the texts. The system automatically uses the system language as the P_Language.
- **P_TodayDate**
Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- **P_DisplayTimeUnit**
Specifies the units of time, for example, [Event Delay](#) in hours or minutes.
- **P_DisplayWeightUnit**
Specifies the units of weight, for example, [Gross Weight](#) in kilograms.
- **P_DisplayVolumeUnit**
Specifies the units of volume, for example, [Item Volume](#) in meters cubed.
- **P_DisplayQuantityUnit**
Specifies the units of quantity, for example, [Number of Containers](#) in pieces.
- **P_DisplayLengthUnit**
Specifies the units of length, for example, [Package Length](#) in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- FreightOrder
- TranspOrdExecution
- TransportationOrderUUID
- TransportationOrderEventUUID

Freight Order Attributes

- TransportationOrderType
- MeansOfTransport
- TransportationMode
- PlngAndExecOrganization
- PurchasingOrganization
- TranspOrdLifeCycleStatus
- TransportationOrderExecSts
- Carrier
- Shipper
- Consignee
- TranspOrdIsDiscrepant
- DiscrepancyType
- TranspOrdHasQtyDiscrepancy
- SourceLocationName
- SourceLocationType
- SourceLocationCountry
- DestinationLocationName
- DestinationLocationType
- DestinationLocationCountry

Freight Order Execution Attributes

- TranspOrdEventCode
- TranspOrdEventStatus
- TranspOrdEventReasonCode
- Location

- LocationType
- Country
- GeographicalLocationLatitude
- GeographicalLocationLongitude
- Transp0rdEvtActualDateTime
- Transp0rdEvtExpdEndDateTime
- Transp0rdEvtActualDateTimeZone
- Transp0rdEvtEstimatedDateTime
- Transp0rdEventIsOnTime
- Transp0rdEventIsDelayed
- Transp0rdEventIsTimeRelevant

Freight Order Measures

- NumberOfTransportationOrders
- NumberOfTimeRlvtTransp0rds
- NumberOfOnTimeTransp0rds
- NumberOfDelayedTransp0rds
- TransportationOrderOnTimeRatio

Freight Order Execution Measures

- Transp0rdEventDelayInDspUnit
- AverageTransp0rdEventDelay
- NumberOfTransp0rdEvents
- NumberOfTimeRlvtTransp0rdEvts
- NumberOfOnTimeTransp0rdEvents
- NumberOfDelayedTransp0rdEvents
- Transp0rdEventOnTimeRatio

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

Freight Order Execution Status Query

Technical Name	View Type
C_ FRTORDEXECUTIONSTSQ	Consumption view

Purpose

This view contains the key performance indicators (KPIs) for your daily operational business, such as transports that are critically blocked either in planning or execution, grouped by block reason. This helps you to determine the most common causes of blocking and take the appropriate action. It also contains transports that are currently in transit and delayed so that you can investigate and sort out the situation.

The view also has a workload analysis showing the number of transports in draft, new, in process, and in execution.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		

Authorization Object	Check Condition	Field	Value
		ACTVT	03 (Display)
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID
		TM_XGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03.		

Authorization Object	Check Condition	Field	Value
	The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code>. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGroup</code> and <code>_DestinationLocation.AuthorizationGroup</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with `TransportationOrderCategory = TO`.

Main Input Parameters

The main input parameters are as follows:

- **P_Language**

Specifies the display language of the texts. The system automatically uses the system language as the `P_Language`.

- **P_TodayDate**

Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the `P_TodayDate`. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.

- **P_DisplayTimeUnit**

Specifies the units of time, for example, **Event Delay** in hours or minutes.

- **P_DisplayWeightUnit**

Specifies the units of weight, for example, **Gross Weight** in kilograms.

- P_DisplayVolumeUnit

Specifies the units of volume, for example, **Item Volume** in meters cubed.

- P_DisplayQuantityUnit

Specifies the units of quantity, for example, **Number of Containers** in pieces.

- P_DisplayLengthUnit

Specifies the units of length, for example, **Package Length** in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- FreightOrder
- TranspOrdExecution
- TransportationOrderUUID
- TransportationOrderEventUUID

Freight Order Attributes

- TransportationOrderType
- MeansOfTransport
- TransportationMode
- PlngAndExecOrganization
- PurchasingOrganization
- TranspOrdLifeCycleStatus
- TransportationOrderExecSts
- Carrier
- Shipper
- Consignee
- TranspOrdIsInTransit
- TranspOrdIsDlydInTransit
- TranspOrdPlanningBlock
- TranspOrdPlanningBlockRsn
- TranspOrdExecutionIsBlocked
- TranspOrdExecBlockReason
- InvoicingIsBlocked

- SourceLocationName
- SourceLocationType
- SourceLocationCountry
- DestinationLocationName
- DestinationLocationType
- DestinationLocationCountry

Freight Order Measures

- NumberOfTransportationOrders
- NumberOfTranspOrdsInTransit
- NmbrOfDelayedTranspOrdsInTrnst
- TranspOrdDelayInDspUnit
- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit
- TranspOrdNumberOfCtnsInDspUnit
- NumberOfPlngBlkdTranspOrds
- NumberOfExecBlkdTranspOrds

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Freight Order Quantity Analysis

As a transportation manager for a shipper, you work with carriers to transport your products to your customers. You can use the [Freight Order Quantity Analysis](#) overview page to provide you with the key performance indicators (KPIs) that you need to manage your execution business. At a glance, you can see all the vital data you need for your daily decision-making.

You can see KPIs for the current weight, volume, and numbers being transported on a weekly basis by each carrier, shipper, and ship-to party, as well as by transportation mode, and source and destination country or region. You can also view each carrier's utilization performance.

The overview page provides you with KPIs for the current and previous month.

Key Features

- **Gross Weight Distribution**

You can view a chart showing how gross weight distribution for completed freight orders (that is, with a freight order life cycle status of **Completed**) has increased or decreased in each completion week (current and previous month). The total gross weight distribution also appears at the top, for example, 5K.

- **Gross Volume Distribution**

You can view a chart showing how gross volume distribution for completed freight orders has increased or decreased in each completion week (current and previous month). The total gross volume distribution also appears at the top, for example, 194K.

- **Quantity Distribution**

You can view a chart showing how quantity distribution for completed freight orders has increased or decreased in each completion week (current and previous month). The total gross quantity distribution also appears at the top, for example, 55K.

- **Transportation Mode Category Analysis**

You can see a chart with gross weight and gross volume as axes, plus the total number of freight orders, for each transportation mode category for completed freight orders. This can help you identify the most effective method of transportation.

- **Logistic Network Analysis**

You can review a table showing source and destination countries or regions, and total gross weight for completed freight orders. This identifies where you have transported most goods in the current and previous months. The table is sorted in descending order of gross weight.

- **Carrier Quantity Analysis**

You can view the total gross weight, gross volume, and container count per carrier for completed freight orders. This gives you information on which carriers you are using the most. The highest weight appears at the top.

- **Shipper Quantity Analysis**

You can view the total gross weight, gross volume, and container count per shipper for completed freight orders. This gives you information on which shippers you are using the most. The highest gross weight appears at the top.

- **Ship-To Party Quantity Analysis**

You can view the total gross weight, gross volume, and container count per ship-to parties for completed freight orders. The highest gross weight appears at the top.

- **Utilization Distribution**

You can see the lowest average utilization per carrier (expressed as a percentage) for completed freight orders. The lowest utilization percentage appears at the top. This helps you assess carriers' performance and capacity.

- **Quantity Analysis**

You can view a table showing the gross volume and gross weight transported for each transportation mode category (for example, road and air) for completed freight orders. You can use this information when deciding on the most efficient method of transportation. The table is sorted in descending order of gross weight.

- **Distance Analysis**

You can see the total net duration and total distance for transported goods for each transportation mode category (for example, sea and air) for completed freight orders. You can use this information when deciding on the most efficient method of transportation. The greatest distance appears at the top.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight order in the Transportation Management system and change it if necessary.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2676

This overview page uses the C_FrtOrdQuantityQuery CDS view.

You must activate the C_FRTORDQUANTITYQUERY_CDS OData service in the Gateway system.

More Information

Subject	More Information
Freight order management for a business person	Freight Order Management
Freight order quantity analysis detailed view for a business person	Freight Order Quantity Detailed View
Underlying technical object for a technical person	C_FrtOrdQuantityQuery CDS view Freight Order Quantity Query

Freight Order Quantity Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyse your freight order execution and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered freight order data in different ways.

You can quickly see the total gross weight of goods transported per carrier in the current and previous month represented visually by a line chart, and the total gross weight of freight orders per transportation mode category depicted by a bar chart.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for gross weight per carrier in the current and previous month and the number of freight orders per transportation mode category.

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs by gross weight per transportation mode category for completion weeks in the current and previous month. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as carrier, destination country or region, and life cycle status. For example, you can quickly see the total gross weight of goods being transported by road in a specific completion week.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as carrier, number of containers, and source and destination countries or regions. You can add or remove columns in **Settings** to give you the information you want to see, for example, completion date and whether dangerous goods are being transported.

Selecting a freight order in the table and pressing **Change Freight Order** takes you directly to the freight order in Transportation Management. The document opens in edit mode.

Example

You want to see which carriers transported goods by road to the USA in the previous month, and the total gross weight transported by each. You see that you used three carriers, but one transported much more than the others. You navigate directly to the freight order to find out more details.

You see that the carrier that transported the most is performing in line with expectations, but the other carriers are not. You can now investigate the reasons why, and take action if necessary.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2752

This detailed view uses the C_FrtOrdQuantityQuery CDS view.

You must activate the C_FRTORDQUANTITYQUERY_CDS OData service in the Gateway system.

More Information

[Freight Order Quantity Analysis](#)

[Freight Order Quantity Query](#)

Freight Order Quantity Query

Technical Name	View Type
C_FRTORDQUANTITYQUERY	Consumption view

Purpose

This view contains key performance indicators (KPIs) to enable you to monitor your freight orders with carriers. It contains the current weights, volumes, and numbers being transported on a weekly basis by each carrier, shipper, and ship-to party, as well as by transportation mode, and source and destination country or region. The view also calculates each carrier's utilization.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType

Authorization Object	Check Condition	Field	Value
		TM_XPORG	PurchasingOrgExternalID
		TM_XGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the carrier, shipper, or ship-to party is set to Completed (value of view field CarrierBusPurposeIsCompleted = X or ShipperBusPurposeIsCompleted = X or ConsigneeBusPurposeIsCompleted = X), check if authorization object B_BUP_PCPT has set field ACTVT to 03. The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code>		

Authorization Object	Check Condition	Field	Value
	or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code>. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGroup</code> and <code>_DestinationLocation.AuthorizationGroup</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with `TransportationOrderCategory = TO`.

Main Input Parameters

The main input parameters are as follows:

- P_Language**
 Specifies the display language of the texts. The system automatically uses the system language as the `P_Language`.
- P_TodayDate**
 Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the `P_TodayDate`. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.
- P_DisplayTimeUnit**
 Specifies the units of time, for example, [Event Delay](#) in hours or minutes.
- P_DisplayWeightUnit**
 Specifies the units of weight, for example, [Gross Weight](#) in kilograms.
- P_DisplayVolumeUnit**
 Specifies the units of volume, for example, [Item Volume](#) in meters cubed.
- P_DisplayQuantityUnit**
 Specifies the units of quantity, for example, [Number of Containers](#) in pieces.
- P_DisplayLengthUnit**

Specifies the units of length, for example, **Package Length** in meters.

The unit parameters (time, weight, volume, quantity, and length) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- TransportationOrder
- TransportationOrderStage
- TransportationOrderUUID
- SourceLocationUUID
- DestinationLocationUUID

Freight Order Attributes

- TransportationOrderCategory
- TransportationOrderType
- DangerousGoodsAreContained
- TranspOrdShippingType
- TranspOrdTrafficDirection
- MeansOfTransport
- TransportationMode
- TransportationModeCategory
- TranspOrdLifeCycleStatus
- PlngAndExecOrganization
- PlngAndExecGroup
- PurchasingOrganization
- PurchasingGroup
- TranspOrdCompletionDateTime
- TranspOrdCompletionDate
- TranspOrdCompletionYearWeek
- TranspOrdCompletionYearMonth
- TranspOrdCompletionYearQuarter
- TranspOrdCompletionYear
- ChangedDateTime
- Carrier

- CarrierOriginOfEntry
- Shipper
- Consignee
- SourceLocation
- SourceLocationType
- SourceLocationCountry
- SourceLocationRegion
- DestinationLocation
- DestinationLocationType
- DestinationLocationCountry
- DestinationLocationRegion

Freight Order Stage Attributes

- TranspOrdStageSrceLocation
- TranspOrdStageSrceLocationType
- TranspOrdStageSrceLocCntry
- TranspOrdStageSrceLocRgn
- TranspOrdStageDestLocation
- TranspOrdStageDestLocationType
- TranspOrdStageDestLocCntry
- TranspOrdStageDestLocRgn

Freight Order Measures

- NumberOfTransportationOrders
- TranspOrdMaxUtilznRatio
- TranspOrdMaxUtilznMassRatio
- TranspOrdMaxUtilznVolumeRatio
- TranspOrdNumberOfCtnsInDspUnit
- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit

Freight Order Stage Measures

- NumberOfTranspOrdStages
- TranspOrdStageMaxUtilznRatio

- TranspOrdStageUtilznVolRatio
- TranspOrdStageUtilznMassRatio
- TranspOrdStageDstncInDspUnit
- TranspOrdStageNetDurnInDspUnit

Units

- DisplayLengthUnit
- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit

Constraints

None.

Tendering Analysis

As a tendering manager or transportation manager for a shipper, you work with carriers to transport your products to your customers. You can use the **Tendering Analysis** overview page to provide you with the key performance indicators (KPIs) that you need to manage your tendering business. At a glance, you can see all the vital data you need for your daily decision-making. The overview page provides you with KPIs for the current and previous month.

You can identify and work with the carriers who perform the best. You can assess the performance of carriers with peer-to-peer requests for quotation (freight RFQs). You can also assess the performance of carriers with all your tendering types.

You can investigate the best tendering type for an individual carrier. You can investigate why a carrier that the system proposed was manually replaced with another carrier.

Key Features

• Open Requests

You have sent freight RFQs to your carriers. You want your carriers to give you freight quotations for providing you with transportation services. This is essentially your current worklist, where you can review all the freight RFQs you have sent out that are still open, and that haven't yet reached their deadlines. You can drill down to see summaries of the open requests and, from each summary, you can open the request in the Transportation Management system.

• Overdue Requests

You can review all the freight RFQs that you have sent out to your carriers that have exceeded their deadlines.

You can work with your critical requests where you may still have time to answer a request.

• Acceptance Analysis

You can review the proportion of freight RFQs to which an individual carrier has responded. This enables you to determine which carriers perform well, and which will most likely support you with an urgent request. You can also see the number of requests accepted out of total requests made, for example, two out of four requests accepted (2 / 4).

- **Tendering Peer-to-Peer Cycle Analysis**

You can review the proportion of peer-to-peer requests to which an individual carrier has responded with a quotation. This enables you to determine which carriers perform well on your peer-to-peer freight RFQs, and which will most likely support you with an urgent peer-to-peer freight RFQ.

You can also review the response times from your carriers. This enables you to see who responds to your freight RFQs immediately, who delays a response, and also who delays a response and submits an increased amount in a quotation.

- **Tendering Cycle Time Analysis**

You use a combination of tendering types with your carriers, for example, peer-to-peer and broadcast tendering types. You can review the average cycle time in hours and the number of cycles taken by a carrier to respond to freight RFQs. This enables you to determine which carriers perform well on your overall tendering business, and which will most likely support you with urgent requests.

- **Tendering Ratio**

You can review the most successful tendering type for individual carriers. For example, you want to see which broadcast tendering type got the best response from a particular carrier. You can answer the questions: which tendering type gets the most responses and which tendering type gets the quickest response?

- **Original Carrier Changed**

You can review the percentage and number of times the carrier that the system proposed was manually replaced with another carrier who fulfilled the transportation service. You can review the reasons why a carrier was manually replaced.

You can use the information on this card to compare the business you actually awarded against the business the system proposed.

Detailed View

You can use visual features to analyze the KPI information at a more detailed level. Selecting a card header or information line takes you to the detailed view, which presents summary information graphically and also in a table, from which you can navigate directly to the freight order in the Transportation Management system and change the tendering if necessary.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Technical Basis

Fiori ID: F2679

This overview page uses the C_TenderingQuery CDS view.

You must activate the C_TENDERINGQUERY_CDS OData service in the Gateway system.

More Information

Subject	More Information
Freight tendering overview for a business person	Freight Tendering
Tendering detailed view for a business person	Tendering Detailed View
Underlying technical object for a technical person	C_TenderingQuery CDS view Tendering Query

Tendering Detailed View

You can get a detailed analysis of each key performance indicator (KPI). You can have instant and flexible access to the information you need to analyze and manage your tendering business and change it if necessary by navigating directly to the relevant document in the Transportation Management (TM) system.

The system displays the information on a two-month rolling basis (current and previous month).

Screen Areas

- Information filter

You can use the information filter on the top of the screen to select the information that the system displays in the chart and table underneath. This enables you to view your filtered tendering data in different ways.

You can quickly see the cycle time by requested carrier, represented visually by a bar chart.

The following views are available:

- Graphics view

By default, the system displays the graphics filter. For example, you can select KPIs for the cycle time in which an individual carrier has responded to freight requests for quotation (RFQs).

- Field view

You can change to a field filter (compact filter), where you can enter filter criteria such as freight order tendering completed, and the weight, volume, and time units in which you want to see the information.

- Bar chart

The chart in the middle of the screen gives you a comparison view of your KPIs for tendering time by requested carrier. You can change the chart to a table or a combined table and chart. Use the **View By** filter to view more specific information such as freight RFQ status, life cycle status, and negative responses. For example, you can quickly see the number of negative responses from the requested carrier.

- Table

The table at the bottom of the screen gives you more information on your KPIs, such as response submission date, number of completed tendering cycles, and positive responses. You can add or remove columns in **Settings** to give you the information you want to see, for example, request time left and number of successful tenderings.

Selecting a tendering in the table and pressing **Change Tendering** takes you directly to the freight order in Transportation Management (it opens in edit mode), where you can change the tendering if necessary.

Example

You want to identify and work with the carriers who perform the best. You look at the completed tendering cycles for each carrier, their positive and negative responses to RFQs, and any rejection reasons.

You identify two carriers who have performed the best, so you investigate which carrier will most likely support you with an urgent request. You then navigate directly to the freight order and look at the tendering to find out further details about the carriers. You then have all the information you need to proceed.

Technical Basis

Your system administrator or another technical user can use the following technical information to change the detailed view.

Fiori ID: F2753

This detailed view uses the C_TenderingQuery CDS view.

You must activate the C_TENDERINGQUERY_CDS OData service in the Gateway system.

More Information

[Tendering Analysis](#)

[Tendering Query](#)

Tendering Query

Technical Name	View Type
C_ TENDERINGQUERY	Consumption view

Purpose

This view contains tendering key performance indicators (KPIs) such as open, overdue, and accepted tendering requests. It also contains calculations for different cycle time definitions and carrier-based tendering ratios.

This information helps you identify which carriers perform the best by assessing their performance with peer-to-peer requests for quotation (freight RFQs) and all the different tendering types.

Prerequisites

Authorization is based on the standard Transportation Management authorization objects.

You'll need either the [Transp. Management Administrative Settings](#) authorization object (T_ADMIN), or both the [Transp. Management Order Management & Planning \(Purchase\)](#) authorization object (T_TOR_PUR) and the [Transp. Management Order Management & Planning \(Execution\)](#) authorization object (T_TOR_EXE).

You'll need the [Business Partner: Purpose Completed](#) authorization object (B_BUP_PCPT) for all business partners accessed in this query.

In addition, you'll need the [Location: Access Based on Authorization Group](#) authorization object (/SCMB/LOCB) for all locations accessed in this query.

The following table lists the authorization objects with required fields and values:

Authorization Object	Check Condition	Field	Value
T_ADMIN	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		ACTVT	H1 (Deactivate)
T_TOR_PUR	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType
		TM_XPORG	PurchasingOrgExternalID
		TM_XPGRP	PurchasingGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
T_TOR_EXE	Check if the user has either T_ADMIN or both T_TOR_PUR and T_TOR_EXE authorization objects.		
		TM_OC	TransportationOrderCategory TO (Freight Order)
		TM_OT	TransportationOrderType

Authorization Object	Check Condition	Field	Value
		TM_XEORG	PlngAndExecOrgExternalID
		TM_XEGRP	PlngAndExecGroupExternalID
		TM_RP	Transp0rdResponsiblePerson
		TM_MOT	TransportationMode
		ACTVT	03 (Display)
B_BUP_PCPT	Only when the business purpose of the Carrier, Requested Carrier, or Selected Carrier is set to Completed (value of view field <code>CarrierBusPurposeIsCompleted = X</code> or <code>TndrgReqdCarrBusPrpsIsCmpltd = X</code> or <code>TndrgSeldCarrBusPrpsIsCmpltd = X</code>), check if authorization object B_BUP_PCPT has set field ACTVT to 03. The system uses B_BUP_PCPT in the data destruction process for Information Lifecycle Management.		
		ACTVT	03 (Display)
/SCMB/LOCB	Only when the business purpose of the source location or destination location is set to Completed (<code>_SourceLocation.IsBusinessPurposeCompleted = X</code> or <code>_DestinationLocation.IsBusinessPurposeCompleted = X</code>), check field BRGRU against the value of view field <code>_SourceLocation.AuthorizationGroup</code> or <code>_DestinationLocation.AuthorizationGroup</code>. The system uses /SCMB/LOCB in the data destruction process for Information Lifecycle Management The field BRGRU enables display access to blocked locations based on the settings you specify in the Define Authorization Group Indicating Blocked Master Data Customizing activity.		
		BRGRU	<code>_SourceLocation.AuthorizationGroup</code> and <code>_DestinationLocation.AuthorizationGroup</code>
		ACTVT	03 (Display)

Structure

Business Objects

This view is built on the Transportation Order (TOR) business object with `TransportationOrderCategory = TO`.

Main Input Parameters

The main input parameters are as follows:

- **P_Language**

Specifies the display language of the texts. The system automatically uses the system language as the P_Language.

- **P_TodayDate**

Specifies the timeframe of the data that you want to display. The system automatically uses the system date as the P_TodayDate. This date is usually your local date. The system retrieves information from the beginning of previous month up to the system date.

- **P_DisplayTimeUnit**

Specifies the units of time, for example, **Event Delay** in hours or minutes.

- **P_DisplayWeightUnit**

Specifies the units of weight, for example, **Gross Weight** in kilograms.

- **P_DisplayVolumeUnit**

Specifies the units of volume, for example, **Item Volume** in meters cubed.

- **P_DisplayQuantityUnit**

Specifies the units of quantity, for example, **Number of Containers** in pieces.

- **P_DisplayCurrency**

Specifies the currency, for example, **Tendering Response Submitted Amount** in euros.

The unit parameters (time, weight, volume, quantity, and currency) are mandatory.

Measures and Attributes

The most important measures and attributes are listed below.

Keys and Identifiers

- TransportationOrder
- TenderingSequenceNumber
- TenderingStepSequenceNumber
- TenderingRequestSequenceNumber
- TenderingResponseSqncNumber
- TransportationOrderUUID
- SourceLocationUUID
- DestinationLocationUUID
- TenderingUUID
- TenderingStepUUID
- TenderingRequestUUID
- TenderingResponseUUID

Freight Order Attributes

- TransportationOrderCategory
- TransportationOrderType
- TransportationMode
- TransportationModeCategory
- MeansOfTransport
- ChangedDateTime
- TranspOrdLifeCycleStatus
- TransportationOrderExecSts
- PlngAndExecOrganization
- PlngAndExecGroup
- PurchasingOrganization
- PurchasingGroup
- Carrier
- CarrierOriginOfEntry
- SourceLocation
- DestinationLocation

Tendering Attributes

- TenderingType
- TenderingProcessResult
- TenderingIsSuccessful
- TenderingHasCompletedCycle
- PeerToPeerTndrgIsSuccssfl
- CarrierIsFromTendering
- FrtOrdHasCompletedTndrg
- TenderingLifeCycleStatus
- TenderingStartDateTime
- TenderingStartDate
- TenderingStartDateYear
- TenderingStartDateYearWeek
- TenderingStartDateYearMonth
- TenderingStartDateYearQuarter
- TenderingResponseDate

- TenderingCreationUser
- TenderingChangeUser

Tendering Step Attributes

- TenderingProcessMode
- CarrierAssignmentMethod
- TrAlcAndBusShrIsUpdtdByRFQ
- TndrgStepLifeCycleStatus

Tendering Request Attributes

- TenderingRequestIsOpen
- TenderingRequestIsOverdue
- TenderingRequestIsCompleted
- TndrgReqLifeCycleStatus
- TenderingRequestedCarrier
- FreightQuotationDueDateTime

Tendering Response Attributes

- TenderingResponseCode
- TndrgResponseIsPositive
- TndrgResponseIsNegative
- TenderingRejectionReason
- TndrgQuotationEvalResult
- IsProposedFreightQuotation
- QuotationReviewIsRequired
- TenderingAwardStatus
- FreightQuotationStatus
- TenderingSelectedCarrier
- TndrgCarrierIsDiscrepant

Freight Order Measures

- NumberOfTransportationOrders
- TranspOrdGrossWeightInDspUnit
- TranspOrdGrossVolumeInDspUnit
- TranspOrdQuantityInDisplayUnit
- TranspOrdNumberOfCtnsInDspUnit

Tendering Measures

- NumberOfTenderings
- NumberOfSuccessfulTenderings
- NumberOfCarriersFromTendering
- NumberOfFrtOrdsWithCmpltdTndrg
- NumberOfCmpltdTenderingCycles
- NumberOfPeerToPeerTndrgCycles
- NmbrOfSuccssflPeerToPeerTndrgs
- AvgNumberOfPeerToPeerCycles
- TenderingCycleTimeInDspUnit
- TenderingAvgCycleTimeInDspUnit
- TenderingSuccessRatio
- FrtOrdsWithTenderingRatio
- CarrierFromTenderingRatio

Tendering Step Measures

- NumberOfTenderingSteps
- TndrgStepAbsltPrcLmtInDspCrcy
- AvgRelativePriceLimitInPercent
- MaxResponseDurationInDspUnit

Tendering Request Measures

- NumberOfTenderingRequests
- NumberOfOpenTndrgRequests
- NumberOfOverdueTndrgRequests
- NumberOfCompletedTndrgRequests
- RmngOpenTndrgReqTimeInDspUnit
- TndrgRspSubmdAmtInDspCrcy

Tendering Response Measures

- NumberOfTenderingResponses
- NumberOfPositiveTndrgResponses
- NumberOfNegativeTndrgResponses
- NumberOfDiscrepantCarrierSel
- PositiveTenderingResponseRatio

- NegativeTenderingResponseRatio
- TndrgReqAbsltPrcLmtInDspCrcy

Units

- DisplayQuantityUnit
- DisplayTimeUnit
- DisplayWeightUnit
- DisplayVolumeUnit
- DisplayCurrency

Constraints

None.